Dossier de Candidature
Concours MCF, section 27

Polytech Montpellier

Équipe AlGCo, LIRMM

picasarri-arrieta lucas

10 mars 2026

Ce document présente ma formation et mon parcours professionnel et expose mes activités en tant que doctorant puis chercheur postdoctoral. Il présente également mon projet d'intégration au sein de Polytech Montpellier, ainsi qu'au sein du LIRMM.

Des informations complémentaires à ce document (rapports de thèse, manuscrit de thèse, attestations d'enseignement, supports de cours) sont disponibles à l'adresse suivante : https://lucaspicasarri.github.io/dossier_Polytech_UM.html.

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Curriculum Vitæ

État Civil

PICASARRI-ARRIETA Lucas

Né le 23 janvier 1998, Français.

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Position Actuelle

Depuis septembre 2024, je suis **chercheur postdoctoral** au **National Institute of Informatics** à Tokyo, au sein de l'équipe *Global Research Center for Big Data Mathematics*, dirigée par **Ken-ichi Kawarabayashi**.

Cursus Universitaire

Octobre 2021 – Août 2024 : Doctorat en Informatique

Thèse : *Coloration de Graphes Dirigés*.

Directeur : Frédéric HAVET, Directeur de recherches, CNRS.

Co-encadrant : Stéphane BESSY, Professeur des universités, Université de Montpellier.

Laboratoire : Laboratoire I3S (UMR 7271) et Centre Inria d'Université Côte d'Azur (Équipe COATI).

Affiliation : Université Côte d'Azur.

Soutenance : 18 juin 2024, devant le jury composé de :

- Stéphan THOMASSÉ, Professeur des universités, ENS Lyon, président du jury,
- Louis ESPERET, Directeur de recherches, CNRS, rapporteur,
- Alexander SCOTT, Professor, University of Oxford, rapporteur,
- Anders YEO, Professor, University of Southern Denmark, rapporteur,
- Marthe BONAMY, Chargée de recherches, CNRS, examinatrice,
- Ararat HARUTYUNYAN, Maître de conférences, Université Paris-Dauphine, examinateur.

Manuscrit : https://lucaspicasarri.github.io/phd/manuscript.pdf.

2019 – 2021 : Master Informatique

Spécialité Informatique Théorique, Université de Montpellier.

Mémoire : *Arc-partitions de graphes orientés*

Encadrants : Stéphane BESSY et Daniel GONÇALVES

Classé 1/15.

2017 – 2019 : Licence Informatique

Université de Montpellier.

Classé 1/91.

Présentation analytique

Enseignements

Durant mon doctorat, j'ai enseigné au département informatique de l'**IUT Nice Côte d'Azur**, pour un total de **192 heures d'enseignement**, en tant que doctorant contractuel avec mission d'enseignement **(DCME)**. Les enseignements que j'y ai effectués sont récapitulés ci-dessous.

Intitulé du cours	Fonction	Volume	Niveau	Année
Communications et fonctionnement bas-niveau	***Responsable***	26 h	BUT1	2023/24
Introduction aux services réseaux	***Responsable***	22 h	BUT1	2023/24
Méthodes d'optimisation pour l'aide à la décision	*Chargé de TD*	16 h	BUT3	2023/24
Automates et langages	*Chargé de TD*	12 h	BUT2	2022/23
Introduction à la programmation	*Chargé de TD*	52 h	BUT1	2022/23
Bases de données avancées	*Chargé de TD*	28 h	BUT1	2021/22
Qualité de développement	***Responsable***	36 h	BUT1	2021/22

Encadrement

— **Samuel Coulomb**

Colouring digraphs with bounded maximum degree.

Stage pré-thèse, Mai – Septembre 2025, *ENS de Paris.*

Co-encadré avec Ken-ichi Kawarabayashi.

— **Martin Jouve-Genty**

Erdős-Pósa properties for small cycles in digraphs.

Stage de Licence 3, Juin – Juillet 2024, *ENS de Lyon.*

Co-encadré avec Clément Rambaud.

— **Teiki Rigaud**

A scheduling problem for LoRa technology.

Stage de Licence 3, Juin – Juillet 2023, *ENS de Paris.*

Co-encadré avec Christelle Caillouet et Frédéric Havet.

Médiation scientifique

Pendant mon doctorat, j'ai eu la chance de participer à **Terra Numerica**, un projet de diffusion de la culture des sciences du numérique. J'ai ainsi pu intervenir au sein d'écoles primaires, de collèges et de lycées, afin de partager à des élèves de tous âges mon goût pour l'informatique et les mathématiques. À plusieurs reprises, j'ai également animé différentes activités à la Fête de la Science, ce qui était l'occasion de m'adresser à une audience plus large. Au total, ma participation au projet Terra Numerica est évaluée à **104 heures**.

Prix

- Accessit au prix **Charles Delorme** 2025.
- Second prix de la meilleure thèse de l'**École Doctorale STIC** en informatique, 2024.

Exposés

Conférences internationales

- **An analogue of Reed's conjecture for digraphs**
  ACM-SIAM Symposium on Discrete Algorithms (SODA25), New Orleans, 2025.
- **On the minimum number of arcs in 4-dicritical oriented graphs**
  49th International Workshop on Graph-Theoretic Concepts in Computer Science (WG), Fribourg, 2023.
- **Strengthening the Directed Brooks' Theorem for oriented graphs and consequences on digraph redicolouring**
  12th European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB'23), Prague, 2023.

Conférences nationales

- **Chromatic discrepancy of locally s-colourable graphs**
  Journées Graphes et Algorithmes, Paris, 2025.
- **Digraph redicolouring**
  Journées Graphes et Algorithmes, Paris, 2022.
- **Complexity of some arc-partition problems for digraphs**
  Journées Graphes et Algorithmes, Montpellier, 2021.

Séminaires d'équipe ou de projet

- **Edge-colouring and orientations : applications to degree- and χ-boundedness**
  – *LIB's seminar*, Dijon, 2026.
  – *IRIF's graph theory seminar*, Paris, 2026.
  – *ACRO's seminar*, Marseille, 2026.
- **Chromatic discrepancy of locally s-colourable graphs**
  – *National Institute of Informatics*, Tokyo, 2025.
- **$(\Delta - 1)$-dicolouring of digraphs**
  – *Imada's seminar*, Odense, 2025.
- **An Analogue of Reed's Conjecture for Digraphs**
  – *LAMSADE's Discrete Mathematics seminar*, Paris, 2025.
  – *GOAL and MC2 common seminar*, Lyon, 2025.
  – *GT Graphs and Optimization*, Bordeaux, 2024.
  – *National Institute of Informatics*, Tokyo, 2024.
- **Blow-ups and extensions of trees in tournaments**
  National Institute of Informatics, Tokyo, 2024.

- **Constrained Flows in Networks**
  COATI seminar, Sophia-Antipolis, 2024.
- **Recolouring digraphs of bounded cycle-degeneracy**
  Third Workshop Complexity and Algorithm (CoA), Paris, 2023.
- **On the minimum number of arcs in 4-dicritical oriented graphs**
  – *Fifth ANR Digraph Meeting*, Sète, 2023.
  – *COATI seminar*, 2023, Sophia-Antipolis.
- **Recolouring digraphs of bounded maximum degree**
  – *Group for Discrete Mathematics and Algebra*, Ilmenau, 2024.
  – *Fourth ANR Digraph Meeting*, Lyon, 2023.
  – *GT Graphes et Optimisation*, Bordeaux, 2023.
  – *PARGO seminar*, Fortaleza, 2023.
- **Digraph redicolouring**
  COATI seminar, Sophia-Antipolis, 2022.
- **Complexity of some arc-partition problems for digraphs**
  COATI seminar, Sophia-Antipolis, 2021.

Workshops et conférences

J'ai participé aux workshops et conférences suivants.

— Journées Graphes et Algorithmes (JGA), Paris, novembre 2025.
— ACM-SIAM Symposium on Discrete Algorithms (SODA), New Orleans, janvier 2025.
— Fourth Workshop on Graphs and Geometric Algorithms, Kyoto, décembre 2024.
— Second Fortaleza Workshop in Combinatorics, Fortaleza, novembre 2023.
— Third Workshop Complexity and Algorithms (CoA), Paris, septembre 2023.
— EuroComb'23, Prague, août 2023.
— WG 2023, Fribourg, juin 2023.
— Fifth ANR Digraph meeting, Sète, juin 2023.
— Fourth ANR Digraph meeting, Lyon, janvier 2023.
— Third ANR Digraph meeting, Les Plantiers, juillet 2022.
— Journées Graphes et Algorithmes (JGA), Paris, novembre 2022.
— International Colloquium on Graph Theory and Combinatorics, Montpellier, juillet 2022.
— School on Graph Theory (SGT), Murol, juin 2022.
— Journées Graphes et Algorithmes (JGA), Montpellier, novembre 2021.

Relecture

J'ai été rapporteur pour les revues et conférences à comité de lecture suivantes.
Total : 22 (journaux ×18, conférences ×4).

— Algorithmica
— Graphs and Combinatorics

— Discrete Applied Mathematics (×2)
— Discrete Mathematics
— Discrete Mathematics & Theoretical Computer Science
— Electronic Journal of Combinatorics (×3)
— European Journal of Combinatorics (×2)
— Filomat
— Journal of Graph Theory (×2)
— LAGOS 2023, 2025 (×2)
— Mathematics in Computer Science
— SIAM Journal on Discrete Mathematics
— SODA 2026
— Theoretical Computer Science

Séjours de recherche

Laboratoires visités lors de séjours de recherche

— IRIF, Paris, février 2026, 10 jours.

> Sujet : *Heroes for the acyclic dichromatic number.*
> Article en cours d'écriture avec P. Aboulker, P. Charbit, S. Coulomb et K. Nurse.

— LIS, Marseille, février 2026, 7 jours.

> Sujet : *Enumerating subtournaments of a tournament.*

— CISPA, Saarbrücken, décembre 2025, 10 jours.

> Sujet : *Minimum number of bounded inversions to make a digraph strong.*
> Article en cours d'écriture avec Florian Hörsch.

— Universität Ulm, Ulm, décembre 2025, 10 jours.

> Sujet : *On the number of maximal independent sets in K_4-free graphs.*
> Article soumis, co-écrit avec T. Hartel et D. Rautenbach [28].

— LAMSADE, Paris, novembre 2025, 10 jours.

> Sujet : *On the list version of a conjecture of Erdős and Neumann-Lara.*
> Article soumis, co-écrit avec Ararat Harutyunyan et G. Puig i Surroca [32].

— Technische Universität Berlin, Berlin, septembre 2025, 10 jours.

> Sujet : *Bounding the dichromatic number by a fraction of the maximum out-degree.*

— University of Southern Denmark, Odense, septembre 2025, 10 jours.

> Sujet : *Acyclic dichromatic number of digraphs.*
> Article soumis, co-écrit avec A. Yeo et J. Bang-Jensen [9].

— LAMSADE, Paris, février 2025, 10 jours.

> Sujet : $(\Delta - 1)$-*dicolouring of digraphs.*
> Article soumis, co-écrit avec Ararat Harutyunyan, K. Kawarabayashi et G. Puig i Surroca [29].

— LaBRI, Bordeaux, février 2025, 10 jours.

> Sujet : *On cuts of small chromatic number in sparse graphs.*
> Article publié, co-écrit avec G. Aubian, M. Bonamy, R. Bourneuf et O. Fontaine [6].

— Technische Universität Ilmenau, Ilmenau, mai 2024, 7 jours.

> Sujet : *Reuniting bivariable degeneracy and DP-coloring*
> Article soumis, co-écrit avec M. Stiebitz [50].

— National Institute of Informatics, Tokyo, mars 2024, 23 jours.

> Sujet : *An analogue of Reed's conjecture for digraphs.*
> Article publié dans les actes de SODA 2025, co-écrit avec K. Kawarabayashi [38].

— Universidade Federal do Ceará, Fortaleza, novembre 2023, 30 jours.

> Sujet : *Backbone colouring of chordal graphs*
> Article publié dans les actes de LAGOS XIII, co-écrit avec J. Araújo et N. Nisse [5].

— University of Southern Denmark, Odense, septembre 2023, 7 jours.

> Sujet : *Constrained flows in networks.*
> Article publié dans Theoretical Computer Science, co-écrit avec J. Bang-Jensen et S. Bessy [8].

Collaborateurs venus effectuer un séjour de recherche

— Ararat Harutyunyan, Tokyo (NII), mars 2026, 13 jours.

> Sujet : *Erdős–Neumann-Lara's conjecture for claw-free graphs.*

— Amadeus Reinald, Tokyo (NII), juin 2025, 19 jours.

> Sujet : χ-*bounding oriented forests.*

— Florian Hörsch, Sophia-Antipolis (I3S), août 2023, 7 jours.

> Sujet : *Characterising the* 3-*dicritical semi-complete digraphs.*
> Article publié dans Electronic Journal of Combinatorics, co-écrit avec F. Havet et F. Hörsch [33].

— Jørgen Bang-Jensen, Sophia-Antipolis (I3S), mai 2023, 7 jours.

> Sujet : *Constrained flows in networks.*
> Travail poursuivi à Odense en septembre 2023, voir ci-dessus.

Liste de publications

Mes publications sont disponibles sur ma page web : lucaspicasarri.github.io/.
Dans la liste ci-dessous, cliquer sur le titre d'un article permet également d'y accéder.

Les auteurs sont classés par ordre alphabétique, ce qui est la norme dans ma communauté de recherche.

Articles soumis ou publiés : 26.

Articles publiés dans des actes de conférences : 5

Articles publiés en journal : 15.

Nombre de co-auteurs : 35.

Pré-publications soumises à un journal ou une conférence à comité de lecture

[P9] **On the list version of a conjecture of Erdős and Neumann-Lara**
A. Harutyunyan, L. Picasarri-Arrieta, G. Puig i Surroca. *ArXiv 2603.01020*, 2026.

[P8] **On the number of maximal independent sets and maximal induced bipartite subgraphs in K_4-free graphs**
T. Hartel, L. Picasarri-Arrieta, D. Rautenbach. *ArXiv 2512.19356*, 2025.

[P7] **Acyclic dichromatic number of oriented graphs**
J. Bang-Jensen, L. Picasarri-Arrieta, A. Yeo. *ArXiv 2511.20246*, 2025.

[P6] **Generalized DP-colorings of digraphs**
L. Picasarri-Arrieta, M. Stiebitz. *ArXiv 2509.17471*, 2025.

[P5] **The complexity of finding and counting subtournaments**
S. Döring, S. Houdaigoui, L. Picasarri-Arrieta, P. Wellnitz. *ArXiv 2509.14807*, 2025.

[P4] **Chromatic discrepancy of locally s-colourable graphs**
T. Corsini, L. Picasarri-Arrieta, T. Pierron, F. Pirot, E. Robinson. *ArXiv 2508.02985*, 2025.

[P3] **$(\Delta - 1)$-dicolouring of digraphs**
A. Harutyunyan, K. Kawarabayashi, L. Picasarri-Arrieta, G. Puig i Surroca. *ArXiv 2507.10266*, 2025.

[P2] **Edge-colouring and orientations: applications to degree-boundedness and χ-boundedness**
A. Char, K. Kawarabayashi, L. Picasarri-Arrieta. *ArXiv 2506.23054*, 2025.

[P1] **Blow-ups and extensions of trees in tournaments**
P. Aboulker, F. Havet, W. Lochet, R. Lopes, L. Picasarri-Arrieta, C. Rambaud. *ArXiv 2410.23566*, 2024.

Publications en conférence

[C5] **Backbone colouring of chordal graphs**

J. Araújo, N. Nisse, L. Picasarri-Arrieta. In the proceedings of the *XIII Latin American Algorithms, Graphs, and Optimization Symposium (LAGOS)*, 2025.

[C4] **An analogue of Reed's conjecture for digraphs**

K. Kawarabayashi, L. Picasarri-Arrieta. In the proceedings of the *2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.

[C3] **On the minimum number of arcs in 4-dicritical oriented graphs**

F. Havet, L. Picasarri-Arrieta, C. Rambaud. In the proceedings of *Graph-Theoretic Concepts in Computer Science (WG)*, 376–387, 2023.

[C2] **Strengthening the Directed Brooks' Theorem for oriented graphs and consequences on digraph redicolouring**

L. Picasarri-Arrieta. In the proceedings of the *12th European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB)*, 2023.

[C1] **Factorially many maximum matchings close to the Erdős-Gallai bound**

S. Bessy, J. Pardey, L. Picasarri-Arrieta, D. Rautenbach. In the proceedings of the *11th International Colloquium on Graph Theory and combinatorics (ICGT)*, 2022.

Publications en journal

[J15] **On cuts of small chromatic number in sparse graphs**

G. Aubian, M. Bonamy, R. Bourneuf, O. Fontaine, L. Picasarri-Arrieta. *Discrete Mathematics*, 349(6):115052, 2026.

[J14] **Brooks-type colourings of digraphs in linear time**

D. Gonçalves, L. Picasarri-Arrieta, A. Reinald. *Journal of Graph Theory*, 110(4):496–513, 2025.

[J13] **Dichromatic number of chordal graphs**

S. Bessy, F. Havet, L. Picasarri-Arrieta. *Graphs and Combinatorics*, 41:81, 2025.

[J12] **The 3-dicritical semi-complete digraphs**

F. Havet, F. Hörsch, L. Picasarri-Arrieta. *Electronic Journal of Combinatorics*, 32(1):P1.1, 2025.

[J11] **Complexity results on the decomposition of a digraph into directed linear forests and out-stars**

F. Hörsch, L. Picasarri-Arrieta. *Electronic Journal of Combinatorics*, 31(4):P4.26, 2024.

[J10] **On the minimum number of arcs in 4-dicritical oriented graphs**

F. Havet, L. Picasarri-Arrieta, C. Rambaud. *Journal of Graph Theory*, 107(4):778–809, 2024.

[J9] **Subdivisions in dicritical digraphs with large order or digirth**

L. Picasarri-Arrieta, C. Rambaud. *European Journal of Combinatorics*, 122:104022, 2024.

[J8] **Constrained flows in networks**

J. Bang-Jensen, S. Bessy, L. Picasarri-Arrieta. *Theoretical Computer Science*, 1010:114702, 2024.

[J7] **Redicolouring digraphs: directed treewidth and cycle-degeneracy**

N. Nisse, L. Picasarri-Arrieta, I. Sau. *Discrete Applied Mathematics*, 356 :191–208, 2024.

[J6] **Minimum number of arcs in k-critical digraphs with order at most $2k-1$**

L. Picasarri-Arrieta, M. Stiebitz. *Discrete Mathematics*, 347(9):114072, 2024.

[J5] **Digraph redicolouring**

N. Bousquet, F. Havet, N. Nisse, L. Picasarri-Arrieta, A. Reinald. *European Journal of Combinatorics*, 116:103876, 2024.

[J4] **Strengthening the Directed Brooks' Theorem for oriented graphs and consequences on digraph redicolouring**

L. Picasarri-Arrieta. *Journal of Graph Theory*, 106(1):5–22, 2024.

[J3] **Unbalanced spanning subgraphs in edge labeled complete graphs**

S. Bessy, J. Pardey, L. Picasarri-Arrieta, D. Rautenbach. *Electronic Journal of Combinatorics*, 30(1):P1.35, 2023.

[J2] **Factorially many maximum matchings close to the Erdős-Gallai bound**

S. Bessy, J. Pardey, L. Picasarri-Arrieta, D. Rautenbach. *Electronic Journal of Combinatorics*, 29(2):P2.52, 2022.

[J1] **Complexity of some arc-partition problems for digraphs**

J. Bang-Jensen, S. Bessy, D. Gonçalves, L. Picasarri-Arrieta. *Theoretical Computer Science*, 928:167–182, 2022.

Enseignement

Expérience d'enseignement

Durant mon doctorat, j'ai enseigné au département informatique de l'**IUT Nice Côte d'Azur**, pour un total de **192 heures d'enseignement**, en tant que doctorant contractuel avec mission d'enseignement **(DCME)**.

Les enseignements que j'y ai effectués sont récapitulés dans le tableau ci-dessous, puis détaillés spécifiquement par la suite.

Intitulé du cours	**Fonction**	**Volume**	**Niveau**	**Année**
Communications et fonctionnement bas-niveau	***Responsable***	26 h	BUT1	2023/24
Introduction aux services réseaux	***Responsable***	22 h	BUT1	2023/24
Méthodes d'optimisation pour l'aide à la décision	*Chargé de TD*	16 h	BUT3	2023/24
Automates et langages	*Chargé de TD*	12 h	BUT2	2022/23
Introduction à la programmation	*Chargé de TD*	52 h	BUT1	2022/23
Bases de données avancées	*Chargé de TD*	28 h	BUT1	2021/22
Qualité de développement	***Responsable***	36 h	BUT1	2021/22

Le lecteur pourra retrouver certains de mes supports de cours ici : supports de cours.

Suivi de stage. Durant l'année scolaire 2023 – 2024, j'ai également participé au suivi des stages effectués en entreprise par les étudiants de deuxième année de BUT. Dans ce contexte, j'ai visité les stagiaires au sein des entreprises, évalué certains rapports de stage, et présidé certains jurys de soutenance.

Communications et fonctionnement bas-niveau.

Responsable du cours, 26h, BUT1 alternance, 2023–2024.

Ce cours d'introduction aux réseaux était à destination des étudiants de BUT1 en alternance. J'étais responsable de l'unité d'enseignement, incluant la conception et l'adaptation des supports, la rédaction des sujets d'examen et l'évaluation des étudiants. Le cours présente de manière générale le modèle en couches OSI, ainsi que le fonctionnement détaillé des couches de plus bas niveau, de la couche physique à la couche transport. Une attention particulière est portée aux protocoles ARP, IP, ICMP, TCP et UDP.

Pour les séances de TD/TP, j'utilisais principalement deux outils pédagogiques : *Packet Tracer* et *Wireshark*. *Packet Tracer* est un simulateur de matériel réseau. Il permet donc de construire un réseau virtuel, et de voir en temps réel quels messages sont échangés entre les différentes machines du réseau. *Wireshark* est un outil qui permet d'enregistrer, au niveau de la carte réseau, tous les messages échangés depuis ou vers notre machine.

Introduction aux services réseaux.

Responsable du cours, 22h, BUT1 alternance, 2023–2024.

Ce cours est dans la continuité du cours “Communications et fonctionnement bas-niveau” ci-dessus, et j’y ai assuré les mêmes responsabilités. On y présente le fonctionnement des principaux protocoles et services de haut niveau, tels que DHCP, HTTP, DNS, FTP et SMTP.

Méthodes d’optimisation pour l’aide à la décision.

Chargé de TD/TP, 16h, BUT3, 2023–2024.

L’objectif de l’UE est d’introduire les étudiants aux principales méthodes d’optimisation et à leur mise en œuvre concrète. Le cours s’attache en particulier à la modélisation de problèmes classiques sous forme de programmes linéaires, en insistant sur la traduction d’un problème réel en variables, contraintes et fonction objectif. L’algorithme du simplexe est présenté en cours afin de donner aux étudiants une compréhension des principes théoriques de la résolution des programmes linéaires, et il est appliqué sur de petits exemples traités à la main en travaux dirigés. En complément, l’UE aborde l’utilisation de solveurs pour résoudre des instances plus complexes : les étudiants utilisent notamment le solveur `CPLEX` en `Java` et l’appliquent à la modélisation et à la résolution de problèmes classiques tels que les problèmes de flots ou le problème du sac à dos.

Le cours était assuré par Christelle Caillouet.

Automates et langages. *Chargé de TD, 12h, BUT2, 2022–2023.*

Dans cette UE, nous introduisons les outils fondamentaux de l’étude des langages formels. Elle aborde notamment les automates (déterministes et non déterministes), les expressions régulières, les grammaires, ainsi que les automates à piles. En parallèle, nous étudions les classes de langages — rationnels et algébriques — reconnues par ces différents modèles de calcul, et présentons des résultats structurants tels que le lemme de l’étoile et le lemme d’itération.

Les étudiants de BUT rencontrant fréquemment des difficultés avec les aspects les plus théoriques de cette UE, j’ai choisi d’introduire, lors des séances de travaux dirigés, une approche plus appliquée de ces notions. Celle-ci vise à illustrer l’usage concret des outils étudiés, par exemple à travers des problèmes de recherche de chaînes de caractères dans un document.

Le cours était assuré par Christelle Caillouet.

Introduction à la programmation. *Chargé de TD/TP, 52h, BUT1, 2022–2023.*

Ce cours, à destination des étudiants de première année, constitue une introduction à la programmation en `C++`. Il s’agissait d’un enseignement nouvellement mis en place à l’IUT lors de l’année concernée, j’ai donc aidé à la conception des supports pédagogiques pour les séances de TD et de TP, que j’ai construit principalement à partir de l’ouvrage *Exercices en langage C++* de Claude Delannoy.

Ce cours correspondait également au tout premier enseignement de programmation suivi par ces étudiants. Il a donc été nécessaire d’y introduire, en parallèle de l’apprentissage du langage `C++`, les bases de l’algorithmique : variables et affectations, structures conditionnelles, boucles, décomposition d’un problème en sous-problèmes, écriture de fonctions et premiers exemples d’appels récursifs.

Pour la partie programmation, le contenu du cours portait sur les fondements du langage (types fondamentaux, structures de contrôle, fonctions, tableaux et chaînes de caractères), ainsi que sur une première approche de la gestion de la mémoire, incluant la distinction entre les différents segments mémoire (pile, tas, données statiques), leur cycle de vie et leurs usages respectifs. L’enseignement s’arrêtait avant l’introduction de la programmation orientée objet.

Le cours magistral était assuré par Thierry Vert.

Bases de données avancées *Chargé de TD/TP, 28h, BUT1, 2021–2022.*

Ce cours fait suite à un premier cours d'introduction aux bases de données et vise à approfondir les connaissances acquises précédemment. L'objectif est tout d'abord de consolider les notions élémentaires de manipulation et de définition des données à l'aide du langage SQL, en mettant l'accent sur la structuration des requêtes et la gestion des relations entre tables.

Ensuite, le cours présente le langage PL/SQL d'Oracle, qui permet d'aller au-delà des simples requêtes SQL en introduisant la programmation procédurale au sein de la base de données. Nous y abordons en particulier les concepts de curseurs pour le traitement des ensembles de données, de procédures et fonctions stockées pour automatiser des tâches répétitives, de gestion d'exceptions pour assurer la robustesse des programmes, ainsi que les déclencheurs (triggers) qui permettent de réagir automatiquement à certains événements dans la base de données.

Le cours était assuré par Lise Brenac.

Qualité de développement. *Responsable du cours, 36 h, BUT1, 2021–2022.*

Ce cours a pour objectif d'initier les étudiants de première année aux bonnes pratiques du développement logiciel, en complément des enseignements plus techniques de programmation. Il vise à leur donner, dès le début de leur formation, une vision structurée et professionnelle de la manière de concevoir, développer et maintenir un logiciel.

Les étudiants y sont notamment sensibilisés à l'utilisation d'outils de gestion de versions, en particulier le logiciel `git`, afin de comprendre les principes du travail collaboratif, du suivi des modifications, de la gestion de l'historique d'un projet et de l'utilisation des branches pour organiser le développement. Le cours aborde également les bases des méthodes agiles, en mettant l'accent sur l'organisation du travail en équipe. Enfin, une introduction aux tests unitaires est proposée, avec notamment l'utilisation de `JUnit` en `Java` pour vérifier le bon fonctionnement du code.

Projet d'intégration au sein de Polytech Montpellier

En cas de recrutement, je serai ravi d'enseigner au sein des formations

- Développement informatique et exploitation Opérationnelle (DO) ; et
- Data science, Management, Software architecture (DaMS) ;

de Polytech Montpellier. Je suis disposé à m'investir dans l'enseignement d'un large éventail de matières en informatique disposés dans celles-ci. Selon les besoins existants, je suis bien sûr prêt à intervenir dans les cours des premières années du cycle ingénieur, ainsi que dans d'autres départements de l'école. Je suis également disposé à enseigner en anglais.

D'une façon générale, je pourrai, dès ma prise de fonctions, intervenir comme chargé de TD dans chacune des matières enseignées au sein des 3ème et 4ème année de ces formations en informatique.

Mon expérience d'enseignement à l'IUT de Nice Côte d'Azur témoigne également de ma capacité à assurer les cours, au sein de la formation DO, des unités d'enseignement suivantes dès ma prise de poste :

— Bases de données relationnelles (XADO501B),
— Gestion de projets (XADO515A, XADO517A),
— Réseaux (XADO625),
— Languages et Automates (XADO705A).

Bien que je n'aie pas encore eu l'occasion de les enseigner, mon cursus universitaire, mes activités de recherche et mon appétence personnelle me permettraient également d'assurer le cours des unités d'enseignement suivantes dès ma prise de poste :

— Initiation à Python (XADO514B, XADO520),
— Algorithmique et structure de données (XADO518A, XADO601A),
— Conception et Programmation Orientée Objet (XADO613A),
— Graphes et Algorithmes (XADO700),
— Programmation Fonctionnelle (XADO820).

Je suis également prêt à assurer les enseignements correspondant au sein de la formation DO. Selon les besoins au sein de ces formations, je suis également prêt à me former afin d'intervenir au plus tôt dans tout autre enseignement en informatique.

De plus, je serais ravi, si le besoin existe, de pouvoir à terme proposer un enseignement en lien direct avec mon domaine de recherche. Cet enseignement pourrait porter sur la théorie des graphes ou sur l'utilisation de méthodes probabilistes en combinatoire et en algorithmique. Ceci pourrait par exemple compléter l'enseignement XADO700 mentionné ci-dessus.

Enfin, j'ai pleinement conscience que le métier d'enseignant-chercheur ne se limite pas aux activités d'enseignement et de recherche, mais comporte également une dimension collective et administrative essentielle au bon fonctionnement du département. Je suis prêt à m'investir de manière générale dans ces responsabilités, qu'il s'agisse de la prise en charge d'unités d'enseignement, de la participation à diverses missions transverses, ou, à plus long terme, de l'exercice d'une responsabilité d'année ou de formation.

Recherche

Thématiques de recherche

Mes recherches portent sur la théorie des graphes, et plus spécifiquement sur la théorie des graphes dirigés (ou digraphes). Les graphes sont connus pour être un outil naturel modélisant efficacement de nombreux problèmes issus du monde réel. Dans de nombreuses situations, les graphes obtenus sont par essence dirigés. C'est particulièrement le cas lorsque l'objectif est de modéliser des relations, entre des objets quelconques, qui ne sont pas nécessairement symétriques. On peut citer, par exemple, le graphe du web, les graphes de certains réseaux sociaux, ou les graphes de planification de tâches. Il apparaît donc nécessaire de comprendre, tant d'un point de vue structurel qu'algorithmique, les similitudes et différences entre graphes et digraphes. Mes recherches sont alignées sur cet objectif, et visent ainsi à enrichir la théorie des graphes dirigés, tant du point de vue structurel que combinatoire et algorithmique.

Un axe central de mes travaux concerne la coloration de graphes dirigés. Cette thématique constitue le fil conducteur d'une part importante de mon activité de recherche. Néanmoins, mes travaux ne se limitent pas à cet axe. J'ai également contribué à d'autres thématiques de la théorie des graphes et des digraphes, notamment la coloration de graphes non dirigés, des questions d'algorithmique sur les graphes dirigés, de reconfiguration, ainsi que des problèmes relevant de la théorie de Ramsey et de l'étude de sous-structures inévitables dans les graphes et digraphes. Par soucis de concision, je ne développe pas mes contributions à ces domaines ici, mais le lecteur pourra retrouver un rapport plus long sur mes activités de recherche ici : `rapport long sur les activités de recherche`.

Ci-dessous, je détaille une partie des résultats obtenus dans les trois articles joints à ce dossier, qui illustrent de manière représentative mes contributions autour de la coloration des graphes dirigés. Le choix de ces publications s'est fait sur la qualité des résultats obtenus, l'originalité des méthodes utilisées, ainsi que leur lien direct avec le programme de recherche présenté plus loin.

Présentation des articles joints.

La notion de coloration de graphes est un problème central très étudié en théorie des graphes. Une *k-coloration propre* d'un graphe G est une fonction qui associe à chaque sommet de G une couleur de $\{1, \ldots, k\}$, de telle sorte que deux sommets adjacents reçoivent des couleurs distinctes. Le plus petit entier k tel que G admette une k-coloration propre est le *nombre chromatique* de G, que l'on note $\chi(G)$. Ces deux notions ont été généralisées aux graphes dirigés à la fin des années 1970 par Erdős et Neumann-Lara [21]. Une *k-dicoloration* d'un digraphe D est une coloration des sommets de D avec k couleurs $\{1, \ldots, k\}$, telle que tout cycle dirigé de D utilise au moins deux couleurs. Le plus petit entier k tel que D admette une k-dicoloration est le *nombre dichromatique* de D, que l'on note $\vec{\chi}(D)$. De façon équivalente, une coloration est une dicoloration si chaque classe de couleur induit un sous-digraphe acyclique. Ces deux concepts pour les graphes dirigés ont été redécouverts et popularisés par Mohar [44, 15] dans les années 2000.

La notion de dicoloration généralise celle de la coloration propre, car une coloration propre d'un graphe G est exactement une dicoloration du digraphe obtenu à partir de G en remplaçant chaque arête par deux arcs de directions opposées. Dans ma thèse, j'ai démontré que de

nombreux résultats classiques portant sur la coloration de graphes peuvent en effet être généralisés à la dicoloration de digraphes. Ceci tend à confirmer que la notion de dicoloration est une généralisation appropriée de la coloration propre.

Le degré maximum $\Delta(G)$ d'un graphe G est le nombre maximum d'arêtes de G incidentes à un même sommet. Une procédure gloutonne triviale montre que l'inégalité $\chi(G) \leqslant \Delta(G) + 1$ est toujours vraie. Le célèbre théorème de Brooks caractérise les graphes G pour lesquels il y a égalité, c'est-à-dire $\chi(G) = \Delta(G) + 1$.

Pour un digraphe D et un sommet $v \in V(D)$ de D, on note par $d^+(v)$ et $d^-(v)$ respectivement le nombre d'arcs sortants et le nombre d'arcs entrants de v. On obtient alors trois définitions naturelles du degré maximum d'un digraphe D :

- $\Delta^+(D) = \max\{d^+(v) : v \in V(D)\}$,
- $\Delta^-(D) = \max\{d^-(v) : v \in V(D)\}$, et
- $\Delta_{\max}(D) = \max(\Delta^+(D), \Delta^-(D))$.

Mohar [45] a étendu le théorème de Brooks en montrant que l'inégalité

$$\vec{\chi}(D) \leqslant \Delta_{\max}(D) \tag{1}$$

est vraie pour tout digraphe connexe D, à l'exception des digraphes suivants : les cycles dirigés, les cycles impairs symétriques, et les digraphes complets. Comme montré par Aboulker et Aubian [1], ce théorème peut être obtenu en adaptant directement l'une des différentes preuves du théorème de Brooks. Cependant, il est possible d'obtenir des versions plus précises de ce théorème, et parfois plus fortes, notamment lorsque les digraphes considérés sont peu symétriques.

Dans un travail réalisé avec Gonçalves et Reinald [25], nous proposons par exemple une version plus générale de ce théorème, en introduisant la notion de *dégénérescence bivariable*. Notre résultat généralise à la fois des analogues du théorème de Brooks pour la liste-coloration et des résultats portant sur la partition d'un (di)graphe en sous-(di)graphes dégénérés. De plus, nous montrons qu'une partition en digraphes dégénérés peut être obtenue en temps linéaire en la taille du digraphe considéré. Ce résultat implique l'existence d'algorithmes linéaires pour l'ensemble des résultats que nous généralisons. C'est le cas, en particulier, de résultats obtenus par Borodin, Kostochka et Toft [17] puis par Bang-Jensen, Schweser et Stiebitz [10], pour lesquels l'existence d'un algorithme linéaire était ouverte. Plus récemment, dans un travail avec Stiebitz [50], nous montrons une version encore plus générale de ce théorème, qui implique en plus des analogues du théorème de Brooks pour la *DP-coloration*, une généralisation de la liste-coloration récemment introduite par Dvořák et Postle [20].

1. Renforcer le théorème de Brooks sur les graphes orientés.

seul auteur, publié dans Journal of Graph Theory *en 2023.*

Dans la première publication jointe à ce dossier [49], j'ai démontré que l'inégalité (1) pouvait être améliorée sur une large classe de digraphes. Un digraphe est *complet* s'il contient deux arcs (un dans chaque direction) entre chaque paire de sommets. Le digraphe complet à n sommets est noté $\overleftrightarrow{K_n}$. Pour un digraphe D, un ensemble de sommets induisant un digraphe complet est une *biclique*, et on note $\overleftrightarrow{\omega}(D)$ la taille d'une plus grande biclique de D. On note enfin

$$\Delta_{\min}(D) = \min(\Delta^+(D), \Delta^-(D)).$$

Remarquons que, pour tout digraphe D, $\Delta_{\min}(D) \leqslant \Delta_{\max}(D)$, et que $\Delta_{\max}(D)$ peut être arbitrairement plus grand que $\Delta_{\min}(D)$. Nous obtenons le résultat suivant.

Théorème 1 (Picasarri-Arrieta [49]). *Tout digraphe D tel que $\overleftrightarrow{\omega}(D) < \lceil(\Delta_{\min}(D)+1)/2\rceil$ satisfait :*

$$\vec{\chi}(D) \leqslant \Delta_{\min}(D).$$

La condition $\overleftrightarrow{\omega}(D) < \lceil(\Delta_{\min}(D)+1)/2\rceil$ du théorème est optimale car, comme nous le montrons, il existe des digraphes D pour lesquels $\overleftrightarrow{\omega}(D) = \lceil(\Delta_{\min}(D)+1)/2\rceil$ et $\vec{\chi}(D) = \Delta_{\min}(D)+1$. Plus encore, nous montrons qu'il est NP-difficile de décider, étant donné un digraphe D tel que $\overleftrightarrow{\omega}(D) = \lceil(\Delta_{\min}(D)+1)/2\rceil$, si $\vec{\chi}(D) \leqslant \Delta_{\min}(D)$.

Pour démontrer le Théorème 1, nous montrons que, pour tout digraphe D, il existe un digraphe auxiliaire H tel que

- $\vec{\chi}(H) \geqslant \vec{\chi}(D)$,
- $\overleftrightarrow{\omega}(H) \leqslant 2 \cdot \overleftrightarrow{\omega}(D)$ et
- $\Delta_{\max}(H) \leqslant \Delta_{\min}(D)$,

ce qui permet intuitivement de réduire le problème au cas plus symétrique où le degré entrant et sortant de chaque sommet est borné. Le résultat découle ensuite de l'inégalité (1) appliquée à H.

Un graphe orienté D est un digraphe tel qu'il existe au plus un arc entre chaque paire de sommets, c'est-à-dire un digraphe sans 2-cycle. Le résultat ci-dessus implique en particulier que tout graphe orienté dont le graphe sous-jacent est de degré maximum 5 est 2-dicolorable, ce qui répond à une question de Harutyunyan. Cette remarque, combinée avec un lemme de Lovász, implique le corollaire suivant. Étant donné un digraphe D, on note $\Delta(D)$ le degré maximum du graphe sous-jacent de D, et $\tilde{\Delta}(D) = \max\{\sqrt{d^+(v)\cdot d^-(v)} \mid v \in V(D)\}$ est le *degré géométrique maximum* de D.

Corollaire 2 (Picasarri-Arrieta, voir [37]). *Tout graphe orienté D satisfait*

$$\vec{\chi}(D) \leqslant \frac{1}{3}\Delta(D) + 2 \quad \textit{et} \quad \vec{\chi}(D) \leqslant \frac{1}{\sqrt{2}}\tilde{\Delta}(D) + 2.$$

Cela améliore deux bornes indépendantes, obtenues respectivement par Golowich [23] et Steiner [54], et est à ce jour la meilleure borne supérieure sur le nombre dichromatique des graphes orientés en fonction de ces deux paramètres. Le même argument permet d'obtenir les meilleures bornes supérieures connues sur $\vec{\chi}(D)$ en fonction de $\Delta_{\max}(D)$.

Le Théorème 1 permet également de déduire de nouvelles bornes sur la *recolorabilité* des graphes dirigés de degré borné, améliorant des bornes précédentes que j'ai obtenues avec Bousquet, Havet, Nisse et Reinald [18].

2. Un analogue de la conjecture de Reed pour les digraphes.

avec K. Kawarabayashi, publié dans les actes de la conférence SODA *en 2025.*

La seconde borne du Corollaire 2 ci-dessus se rapproche de la borne $\vec{\chi}(D) \leqslant \lceil\frac{1}{2}(\tilde{\Delta}(D)+2)\rceil$ conjecturée par Harutyunyan et Mohar [30] pour tout graphe orienté D. Dans ce travail, réalisé avec Kawarabayashi [38], nous proposons une conjecture plus générale, qui si elle est vraie implique également une célèbre conjecture de Reed [52].

Conjecture 3 (Kawarabayashi et Picasarri-Arrieta [38]). *Tout digraphe D satisfait*

$$\vec{\chi}(D) \leqslant \left\lceil \frac{1}{2}\left(\tilde{\Delta}(D) + 1 + \overleftrightarrow{\omega}(D)\right)\right\rceil.$$

Il est facile de remarquer que tout digraphe D satisfait $\vec{\chi}(D) \leqslant \tilde{\Delta}(D) + 1$, et que $\overleftrightarrow{\omega}(D)$ est une borne inférieure triviale de $\vec{\chi}(D)$.

La conjecture ci-dessus suggère donc que, à un arrondi supérieur près, le nombre dichromatique est toujours borné par la moyenne de ces deux bornes triviales. Comme résultat partiel, nous obtenons en effet que le nombre dichromatique peut être borné par une combinaison convexe (non triviale) de ces deux bornes.

Théorème 4 (Kawarabayashi et Picasarri-Arrieta [38])**.** *Il existe $\varepsilon > 0$ tel que tout digraphe D satisfait $\vec{\chi}(D) \leqslant \left\lceil (1-\varepsilon)(\tilde{\Delta}(D)+1) + \varepsilon\overleftrightarrow{\omega}(D) \right\rceil$.*

Lorsque restreint aux graphes orientés, le résultat ci-dessus implique un résultat obtenu par Harutyunyan et Mohar [30]. De l'autre côté du spectre, lorsque restreint aux digraphes symétriques, il implique le résultat analogue obtenu par Reed [52] pour les graphes non dirigés.

Notre preuve combine des arguments probabilistes et combinatoires, et est inspirée d'une nouvelle preuve du résultat de Reed obtenue par King et Reed [40]. L'étape cruciale consiste à généraliser un résultat de King [39] en montrant que tout digraphe D satisfaisant $\overleftrightarrow{\omega}(D) > \frac{2}{3}(\Delta_{\max}(D)+1)$ admet un ensemble de sommets acyclique X qui intersecte toutes les bicliques maximum de D. Dans le cas non dirigé, la preuve originale de King repose sur une propriété des ensembles dominants qui ne s'adapte pas directement au cas dirigé. En introduisant une nouvelle notion de domination dans les digraphes, plus asymétrique que la notion classique, nous parvenons à adapter la preuve de King. Il découle ensuite de ce résultat qu'un contre-exemple minimum D au Théorème 4 satisfait $\overleftrightarrow{\omega}(D) \leqslant \frac{2}{3}(\Delta_{\max}(D)+1)$. En utilisant un résultat classique d'Erdős, Rubin et Taylor sur la liste-coloration des graphes denses, nous montrons qu'un tel contre-exemple minimum D doit en plus être suffisamment épars.

Ceci nous permet de conclure en montrant, par le Lemme Local de Lovász, qu'une coloration pseudo-aléatoire de D est une dicoloration utilisant au plus $\left\lceil (1-\varepsilon)(\tilde{\Delta}(D)+1) + \varepsilon\overleftrightarrow{\omega}(D) \right\rceil$ couleurs avec probabilité positive.

3. Digraphes $(\Delta-1)$-dicolorables

avec A. Harutyunyan, K. Kawarabayashi et G. Puig i Surroca, soumis à SIDMA *en 2026.*

Lorsque $\Delta(G)$ est suffisamment grand, le théorème de Brooks a été amélioré par Reed [51], qui a démontré que $\chi(G) \geqslant \Delta(G)$ si et seulement si G contient un graphe complet de taille $\Delta(G)$, confirmant ainsi, asymptotiquement, une célèbre conjecture posée par Borodin et Kostochka [16].

Récemment, avec Harutyunyan, Kawarabayashi, et Puig i Surroca [29], nous étendons le résultat de Reed aux digraphes de la façon suivante. Pour tout entier $\Delta \geqslant 2$, soit H_Δ le digraphe obtenu à partir d'une copie K de $\overleftrightarrow{K}_{\Delta-2}$ en ajoutant un cycle dirigé C de longueur 3 disjoint, ainsi que tous les arcs, dans les deux directions, entre K et C.

Il est facile de vérifier que $\Delta^+(H_\Delta) \leqslant \Delta$, que $\overleftrightarrow{\omega}(H_\Delta) \leqslant \Delta - 1$, et que $\vec{\chi}(H_\Delta) \geqslant \Delta$. Le graphe H_Δ forme donc une obstruction à la $(\Delta-1)$-dicoloration qui n'existe pas dans le cas non dirigé. Nous conjecturons que tout digraphe D tel que $\vec{\chi}(D) \geqslant \Delta^+(D) \geqslant 9$ contient l'une des deux obstructions évidentes à la $(\Delta-1)$-dicoloration, ce qui généralise la conjecture de Borodin et Kostochka.

Conjecture 5 (Harutyunyan et al. [29])**.** *Si D est un digraphe tel que $\vec{\chi}(D) \geqslant \Delta^+(D) \geqslant 9$, alors l'une des affirmations suivantes est vraie :*

— $\overleftrightarrow{\omega}(D) \geqslant \Delta^+(D)$, *ou*

— *D contient $H_{\Delta^+(D)}$ en tant que sous-digraphe.*

Nous montrons que cette conjecture est vérifiée lorsque $\Delta^+(D)$ est suffisamment grand, ce qui généralise le résultat de Reed [51], à la valeur de Δ_0 près.

Théorème 6 (Harutyunyan et al. [29])**.** *Il existe $\Delta_0 \in \mathbb{N}$ tel que ce qui suit est correct. Si D est un digraphe tel que $\vec{\chi}(D) \geqslant \Delta^+(D) \geqslant \Delta_0$, alors l'une des affirmations suivantes est vraie :*

— *$\overleftrightarrow{\omega}(D) \geqslant \Delta^+(D)$, ou*

— *D contient $H_{\Delta^+(D)}$ en tant que sous-digraphe.*

Ce résultat est une première étape vers la résolution d'un problème que l'on a posé avec Kawarabayashi [38] (Problème 7 du projet de recherche ci-dessous).

La méthode utilisée par Reed dans le cas non dirigé ne peut pas s'appliquer directement pour obtenir le Théorème 6, car celle-ci repose sur la méthode probabiliste, et notamment sur le Lemme Local de Lovász. Ce dernier ne peut en effet pas être appliqué dans notre cas, car dans un digraphe de degré sortant borné, il peut exister des sommets de degré entrant arbitrairement grand, ce qui empêche de borner le degré de dépendance de *mauvais événements*.

Pour prouver ce théorème, nous procédons donc en deux étapes. D'abord, nous démontrons une version plus symétrique de celui-ci, où Δ^+ est remplacé par $\tilde{\Delta}$. Dans ce cas précis, nous parvenons à reprendre une partie des idées de Reed, et à conclure par la méthode probabiliste. L'analyse structurelle reste cependant plus délicate, car il peut exister des sommets de degré entrant ou sortant plus grand que $\tilde{\Delta}$ (cependant pas arbitrairement plus grand), et parce qu'il existe une nouvelle obstruction H_Δ à traiter.

Dans un second temps, nous dérivons le Théorème 6 à partir de sa version analogue avec $\tilde{\Delta}$ en utilisant des arguments de comptage sur un contre-exemple minimal bien choisi, ce qui permet de ne pas recourir à la méthode probabiliste ici.

Projet de recherche : partition de graphes dirigés

Comme indiqué dans le rapport sur les travaux effectués, les digraphes modélisent efficacement de nombreux problèmes du monde réel, tels que les réseaux de transport, les graphes de réseaux sociaux ou les dépendances dans les tâches. Leur capacité à capturer des relations asymétriques en fait un outil essentiel pour une variété croissante d'applications. Pourtant, la théorie des digraphes demeure largement moins développée que celle des graphes non dirigés. Cela s'explique en partie par la nature intrinsèquement plus complexe des digraphes : de nombreux problèmes classiques des graphes non dirigés apparaissent comme un cas particulier d'un problème plus général pour les digraphes, en général lorsque l'on se restreint aux digraphes symétriques.

Sur le plan algorithmique, il est ainsi courant que des problèmes polynomiaux sur les graphes non dirigés se transforment en problèmes NP-complets lorsqu'ils sont étendus aux digraphes. Une illustration classique de cette différence est le problème consistant à décider si un graphe G admet k chemins disjoints $P_1, \ldots, P_k$ de sorte que, pour tout $i \in [k]$, P_i relie deux terminaux s_i, t_i donnés. Un célèbre résultat de Robertson et Seymour affirme que ce problème est résoluble en temps $f(k) \cdot n^3$ (descendu à $g(k) \cdot n^2$ par Kawarabayashi, Kobayashi et Reed [36]). Le problème analogue pour les digraphes est largement plus difficile (sous l'hypothèse $P \neq NP$), puisque Fortune, Hopcroft et Wyllie [22] ont montré que ce problème est NP-complet pour tout $k \geqslant 2$ fixé.

Cette complexité se reflète également dans la démarche théorique : les outils développés pour les graphes ne se transposent pas toujours directement aux digraphes. Ces derniers nécessitent

souvent le développement de méthodes et concepts qui leur sont propres, comme l'illustre le problème suivant.

Conjecture 7 (Stiebitz [55]). *Il existe une fonction f telle que tout digraphe D de degré sortant minimum $f(s,t)$ admet une partition (V_s, V_t) de son ensemble de sommets tel que $D[V_s]$ et $D[V_t]$ ont degré sortant minimum au moins s et t respectivement.*

Dans le cas non dirigé, Stiebitz [56] a montré que $f(s,t) = s+t+1$, mais la simple existence de f peut être montrée de façon succincte à l'aide du Lemme Local de Lovász. Dans le cas dirigé, cette méthode ne fonctionne pas car le degré entrant d'un sommet peut être arbitrairement plus grand que son degré sortant.

Mon projet de recherche s'articule autour différents problèmes de partition de digraphes pour lesquels les méthodes usuelles de la théorie des graphes ne semblent pas s'adapter directement. L'objectif principal de ce projet est donc d'enrichir la théorie des digraphes en développant de nouveaux outils propres au cas dirigé. Le problème fondamental posé par Stiebitz s'inscrit exactement dans ce cadre.

Dans ce qui suit, je détaille plusieurs autres problèmes de partition de digraphes sur lesquels je compte travailler spécifiquement ces prochaines années. Par soucis de concision, je ne présente que des problèmes liés à la dicoloration, que j'ai déjà introduite ci-dessus. Ceux-ci s'inscrivent dans un projet de recherche plus large que le lecteur pourra retrouver ici : `projet de recherche long`.

1. Dicoloration de graphes orientés de degré maximum borné.

Comme indiqué dans le rapport sur les travaux effectués, mes recherches antérieures démontrent que de nombreux résultats classiques portant sur la coloration de graphes se généralisent à la dicoloration des digraphes. La dicoloration semble ainsi être la notion qui généralise correctement, aux digraphes, la notion de coloration propre. Il reste en revanche plusieurs grandes questions sur la dicoloration des digraphes qui résistent aux efforts de nombreux chercheurs. Je détaille dans cette section puis dans la section suivante deux questions fondamentales qui consistent à étudier comment une borne générale sur le nombre dichromatique des digraphes peut être améliorée lorsque restreint aux graphes orientés.

On peut aisément montrer, en utilisant un argument glouton, que tout graphe orienté D satisfait $\vec{\chi}(D) \leqslant \frac{1}{2}\Delta(D) + 1$. Cette borne supérieure n'est pas optimale lorsque $\Delta(D)$ devient suffisamment grand. Harutyunyan et Mohar [30] ont en effet démontré l'existence de constantes ε et C telles que tout graphe orienté D satisfait $\vec{\chi}(D) \leqslant (\frac{1}{2} - \varepsilon)\Delta(D) + C$. Comme expliqué précédemment, en combinant un théorème de Lovász avec l'un de mes résultats, on peut obtenir le même résultat pour des valeurs de ε significativement plus grandes. En effet, tout graphe orienté D satisfait $\vec{\chi}(D) \leqslant \frac{1}{3}\Delta(D)+2$. À nouveau, cette borne supérieure ne semble pas optimale. Erdős et Neumann-Lara [21] ont proposé la conjecture suivante en 1979.

Conjecture 8 (Erdős et Neumann-Lara [21]). *Tout graphe orienté D satisfait*

$$\vec{\chi}(D) = O\left(\frac{\Delta(D)}{\log \Delta(D)}\right).$$

Cette conjecture est un analogue dirigé du célèbre résultat de Johansson [34] affirmant que tout graphe G sans triangle satisfait $\chi(G) = O\left(\frac{\Delta(G)}{\log \Delta(G)}\right)$. Cependant, les différentes preuves du résultat obtenu par Johansson ne s'adaptent pas au cas dirigé. En effet, celles-ci reposent implicitement sur le fait que, dans le contexte de la coloration de graphes, les contraintes sont

locales : deux sommets ne peuvent pas recevoir la même couleur lorsqu'ils sont adjacents. En revanche, dans le cas dirigé, les contraintes peuvent être globales : les sommets d'un ensemble induisant un cycle dirigé, de taille arbitraire, ne peuvent pas tous recevoir la même couleur.

À plus court terme, afin de mieux comprendre ce problème, il conviendrait de répondre à une question plus simple que nous avons posée avec Kawarabayashi [38].

Problème 9 (Kawarabayashi et Picasarri-Arrieta [38])**.** *Montrer qu'il existe $\varepsilon > 0$ tel que tout graphe orienté D satisfait*

$$\vec{\chi}(D) \leqslant (1-\varepsilon)\Delta^+(D) + O(1).$$

Ce problème semble plus proche de ce qui est déjà connu, puisqu'il consiste seulement à gagner un facteur multiplicatif sur la borne triviale obtenue par un argument glouton. Comme expliqué précédemment, le résultat est en effet connu lorsque Δ^+ est remplacé par $\frac{1}{2}\Delta$ ou $\tilde{\Delta}$. Cependant, la quasi-totalité des démonstrations de résultats analogues utilisent le célèbre Lemme Local de Lovász, qui ne peut a priori pas être utilisé ici car, en général, $\Delta(D)$ n'est pas borné par une fonction de $\Delta^+(D)$ (il peut exister des sommets de degré entrant arbitrairement grand). Remarquons que cela fait écho au problème rencontré pour une résolution de la Conjecture 7.

Lorsque l'on considère un contre-exemple minimal au Problème 9, on peut montrer que si $\Delta(D)$ est arbitrairement grand comparé à $\Delta^+(D)$, c'est-à-dire si on ne peut pas conclure directement avec les méthodes probabilistes habituelles, alors il existe des cycles dirigés induits de taille arbitrairement grande. Le cœur de la difficulté de ce problème est donc similaire à celui de la Conjecture 8. Ainsi, résoudre ce problème apportera une meilleure compréhension des graphes orientés de degré maximum borné et nous rapprochera d'une résolution de la Conjecture 8.

Comme indiqué dans le descriptif des publications jointes, avec Harutyunyan, Kawarabayashi et Puig i Surroca nous avons caractérisé, lorsque $\Delta^+(D)$ est suffisamment grand, les digraphes D tels que $\vec{\chi}(D) \geqslant \Delta^+(D)$. En particulier, nous avons le résultat suivant.

Corollaire 10 (Harutyunyan et al. [29])**.** *Il existe $\Delta_0 \in \mathbb{N}$ tel que tout graphe orienté D satisfaisant $\Delta^+(D) \geqslant \Delta_0$ satisfait $\vec{\chi}(D) \leqslant \Delta^+(D) - 1$.*

Avec le résultat ci-dessus en main, une façon de résoudre le Problème 9 serait de répondre au problème suivant.

Problème 11. *Montrer l'existence de $d_0, \varepsilon > 0$ tels que l'affirmation suivante soit correcte pour tout entier $d \geqslant d_0$. Tout graphe orienté D admet une partition $(V_1, \ldots, V_s)$ de son ensemble de sommets tel que, pour tout i,*

$$\Delta^+(D[V_i]) \leqslant d,$$

où $s \leqslant \dfrac{\Delta^+(D)}{d-1+\varepsilon}$.

En effet, si l'énoncé ci-dessus est correct, en le combinant avec le Corollaire 10 nous obtenons que tout digraphe D tel que $\Delta^+(D)$ soit suffisamment grand satisfait

$$\vec{\chi}(D) \leqslant \left(1 - \frac{\varepsilon}{\Delta_1 - 1 + \varepsilon}\right)\Delta^+(D) + o(\Delta^+(D)),$$

où $\Delta_1 = \max(\Delta_0, d_0)$.

Dans le cas non dirigé, le Problème 11 se résout aisément, pour $\varepsilon = 2$ et $d_0 = 0$, en prenant une partition des sommets qui maximise le nombre d'arêtes entre des parties distinctes, comme l'a montré Lovász [43]. Dans le cas dirigé, Alon [2, 3] a montré que ce résultat général ne s'étend pas directement, et que l'énoncé du Problème 11 est correct pour $s \geqslant 2\frac{\Delta^+(D)}{d+1}$.

2. Coloration de digraphes planaires

Le théorème des quatre couleurs [4], obtenu par Appel et Haken en 1976, est sans doute le théorème le plus connu de la théorie des graphes. Il affirme que tout graphe planaire (c'est-à-dire, tout graphe représentable dans le plan de sorte à ce qu'aucune paire d'arêtes ne se croise) a nombre chromatique au plus 4. En 1959, Grötzsch [27] a démontré que, pour les graphes planaires sans triangle, cette borne pouvait être améliorée.

Théorème 12 (Grötzsch [27])**.** *Tout graphe planaire sans triangle G satisfait $\chi(G) \leqslant 3$.*

Le théorème des quatre couleurs implique directement que tout digraphe planaire a nombre dichromatique au plus 4. De façon analogue au théorème de Grötzsch, on peut se demander si cette borne est améliorable pour les graphes orientés planaires. Une conséquences directe de la faible densité des graphes planaires implique que ces digraphes sont 3-dicolorables. Neumann-Lara a proposé la conjecture suivante en 1982.

Conjecture 13 (Neumann-Lara [47])**.** *Tout graphe orienté planaire D satisfait $\vec{\chi}(D) \leqslant 2$.*

À nouveau, la résolution de cette conjecture semble nécessiter de nouvelles méthodes. En effet, la plupart des résultats portant sur la coloration de graphes planaires sont obtenus par la célèbre méthode du déchargement. Cette méthode a été adaptée par Harutyunyan et Mohar [31] pour démontrer la Conjecture 13 dans le cas des graphes orientés planaires sans cycle dirigé de longueur 3 et 4, mais ne semble pas s'adapter au cas général. Par la suite, Li et Mohar [42] ont démontré la conjecture pour les graphes orientés sans cycle dirigé de longueur 3 avec une approche complètement différente, reposant sur une décomposition obtenue par Thomassen [57]. À nouveau, cette méthode ne semble pas s'adapter au cas général.

Pour se rapprocher de la résolution de cette conjecture, il serait intéressant de montrer que la classe des digraphes planaires 3-dicolorables est en fait beaucoup plus large que celle des graphes orientés. Pour cela, j'aimerais résoudre le problème intermédiaire suivant. On dénote par $\mathcal{T}_3$ le digraphe obtenu à partir du digraphe complet à 3 sommets en retirant un arc.

Problème 14. *Montrer que tout digraphe planaire D sans $\mathcal{T}_3$ satisfait $\vec{\chi}(D) \leqslant 3$.*

L'objectif de la résolution de ce problème est multiple. En plus de montrer que la classe des digraphes 3-dicolorables est en effet plus large que celle des graphes orientés, un tel résultat formerait une généralisation élégante du théorème de Grötzsch. Ensuite, on peut remarquer que les différentes preuves du théorème de Grötzsch ne s'adaptent pas directement à ce problème. En effet, la preuve la plus courte du théorème de Grötzsch est une conséquence directe de la densité minimum des graphes 4-critiques sans triangle démontrée par Moore et Smith-Roberge [46]. Dans le cas dirigé, une telle preuve n'est pas attendue, car on sait construire des digraphes 4-critiques sans $\mathcal{T}_3$ dont la densité est significativement plus faible que celle des digraphes planaires sans $\mathcal{T}_3$. La quasi-totalité des autres preuves du théorème de Grötzsch repose sur le fait qu'il suffit de le montrer pour les graphes planaires sans triangle ni cycle de taille 4. À nouveau, on peut aisément vérifier que la réduction utilisée ne s'adapte pas dans le cas dirigé. Ainsi, la résolution de ce problème requiert le développement de nouvelles méthodes, qui formeront un outil supplémentaire pour la résolution de la Conjecture 13.

Toujours dans l'objectif de résoudre la conjecture de Neumann-Lara, il serait intéressant de mieux comprendre l'ensemble des 3-dicolorations d'un graphe orienté planaire, et plus spécifiquement comment l'une d'entre elles peut être obtenue à partir d'une autre. Une première direction consiste à comprendre quelles propriétés vérifie $R_3(D)$, le graphe non dirigé dont les sommets sont les 3-dicolorations de D, et dans lequel deux colorations sont adjacentes si elles

diffèrent sur exactement un sommet de D. Le premier problème ouvert sur $R_3(D)$ à résoudre est la conjecture suivante, que nous avons proposée avec Bousquet, Havet, Nisse et Reinald [18].

Conjecture 15 (Bousquet et al. [18])**.** *Si D est un graphe orienté planaire, alors $R_3(D)$ est connexe.*

Enfin, un outil classique en coloration de graphes est la notion de *changement de Kempe*, qui permet entre autre de démontrer, de façon succinte, la 5-colorabilité des graphes planaires. Il existe au moins trois façons distinctes d'étendre cette notion aux digraphes, et aucune d'entre elles n'a été étudiée jusqu'à présent. Afin d'obtenir d'autres méthodes pour la résolution de la conjecture de Neumann-Lara, il serait intéressant de comprendre quelles propriétés structurelles des graphes planaires orientés nous pouvons démontrer à l'aide de ces notions.

Projet d'intégration au sein du LIRMM

En cas de recrutement, j'intégrerai l'équipe *AlGCo* du département informatique du LIRMM. Mon projet de recherche s'inscrit pleinement dans les thématiques développées au sein de l'équipe. Les membres de l'équipe travaillent en effet en théorie des graphes ; je pourrai donc interagir scientifiquement avec chacun d'entre eux. Ci-dessous, je détaille quelques pistes concrètes de collaboration, en lien direct avec mon projet de recherche.

Tout d'abord, les travaux de Stéphane Bessy, membre de l'équipe AlGCo et co-encadrant de ma thèse, présentent une forte complémentarité avec les miens. Stéphane Bessy est en effet un expert de la théorie des graphes dirigés, domaine qui constitue le cœur de mon projet de recherche. Nos intérêts scientifiques communs, illustrés par nos nombreuses collaborations [7, 8, 12, 13, 14] , rendent donc particulièrement naturelle la poursuite de collaborations sur ces thématiques.

Ensuite, rejoindre l'équipe AlGCo serait particulièrement propice au développement de la seconde partie de mon projet de recherche, consacrée à la coloration des graphes orientés planaires. Je pourrais notamment interagir avec Petru Valicov, qui a récemment contribué à de nouvelles avancées sur la conjecture de Neumann-Lara (Conjecture 13), voir [19]. Il co-encadre par ailleurs actuellement, avec Alexandre Pinlou, la thèse de Simon Dreyer sur le sujet *Acyclic sets in (di)graphs*, directement lié à ces questions. Il a également participé au développement d'outils permettant d'obtenir, de façon assistée par ordinateur, des preuves en coloration de graphes planaires [41]. Avec Petru Valicov et Hoang La, nous avons déjà commencé à échanger sur la manière dont ces outils pourraient être utilisés pour aborder le Problème 14 de mon projet.

Ces questions pourraient également donner lieu à des interactions avec Daniel Gonçalves, qui a obtenu de nombreux résultats structurels sur les graphes planaires et leur partition (voir par exemple [24, 26]). Nous avons par ailleurs déjà obtenu, avec Amadeus Reinald, plusieurs résultats sur la coloration de graphes dirigés [25], en lien direct avec les problèmes considérés ici. Concernant la conjecture de Neumann-Lara, la seconde direction de recherche que je propose consiste à étudier la connexité et le diamètre des graphes de reconfiguration associés aux 3-dicolorations des graphes orientés planaires. Cet axe de recherche pourrait également être exploré avec Ignasi Sau, avec qui j'ai déjà collaboré sur des questions de recoloration dans les graphes dirigés (voir [48]).

De plus, mon projet de recherche pourrait également donner lieu à des interactions avec des membres d'autres équipes du LIRMM. Comme je l'explique dans le projet, la difficulté principale de certaines questions dans le cadre dirigé (notamment les Conjectures 7 et 8 ainsi que le Problème 9) tient au fait que les méthodes probabilistes usuelles échouent dans le cadre dirigé, car le Lemme Local de Lovász ne peut pas y être appliqué. Dans ce contexte, les travaux de Matthieu Rosenfeld, membre de l'équipe ESCAPE, sont particulièrement pertinents. Celui-ci a en effet proposé une méthode de comptage qui a permis d'obtenir de nouvelles démonstrations de plusieurs résultats, notamment en coloration de graphes, qui reposaient initialement sur le lemme local, conduisant souvent à des preuves plus courtes et à de meilleures bornes (voir par exemple [11, 35, 53]). Il est donc naturel de se demander si cette approche pourrait être adaptée au cadre dirigé afin d'aborder les problèmes considérés dans la première partie de mon projet.

Enfin, mon projet de recherche s'intègre naturellement dans le cadre du projet ANR : «*Graphes Ordonnés, Décompositions, Algorithmes et Structures* »(GODASse, ANR-24-CE48-4377) auquel l'équipe AlGCo participe et qui porte sur les graphes ordonnés. Les digraphes et les graphes ordonnés sont en effet étroitement liés : tout ordre $\prec$ sur les sommets d'un digraphe D définit un graphe ordonné $D^{\prec}$, dont les arêtes sont exactement les paires $\{u, v\}$ telles

que (u, v) est un arc de D et $v \prec u$. Le nombre dichromatique de D est par exemple égal au minimum, sur tout ordre $\prec$, du nombre chromatique de $D^{\prec}$.

Pour conclure, comme je l'ai déjà indiqué dans mon projet d'intégration au sein du département d'enseignement, j'ai pleinement conscience que la vie d'un laboratoire repose également sur un investissement collectif dans des responsabilités administratives et organisationnelles. Je souhaite ainsi m'impliquer dans les activités de lu LIRMM au-delà de mes travaux de recherche.

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Received: 10 February 2023 | Revised: 11 October 2023 | Accepted: 23 November 2023

DOI: 10.1002/jgt.23066

ARTICLE

WILEY

Strengthening the directed Brooks' theorem for oriented graphs and consequences on digraph redicolouring

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Funding information
National Research Agency (ANR), Grant/Award Numbers: ANR-19-CE48-0013, ANR-17-EURE-0004

Abstract

Let $D = (V, A)$ be a digraph. We define $\Delta_{\max}(D)$ as the maximum of $\{\max(d^+(v), d^-(v)) | v \in V\}$ and $\Delta_{\min}(D)$ as the maximum of $\{\min(d^+(v), d^-(v)) | v \in V\}$. It is known that the dichromatic number of D is at most $\Delta_{\min}(D) + 1$. In this work, we prove that every digraph D which has dichromatic number exactly $\Delta_{\min}(D) + 1$ must contain the directed join of $\overleftrightarrow{K_r}$ and $\overleftrightarrow{K_s}$ for some r, s such that $r + s = \Delta_{\min}(D) + 1$, except if $\Delta_{\min}(D) = 2$ in which case D must contain a digon. In particular, every oriented graph $\overrightarrow{G}$ with $\Delta_{\min}(\overrightarrow{G}) \geq 2$ has dichromatic number at most $\Delta_{\min}(\overrightarrow{G})$. Let $\overrightarrow{G}$ be an oriented graph of order n such that $\Delta_{\min}(\overrightarrow{G}) \leq 1$. Given two 2-dicolourings of $\overrightarrow{G}$, we show that we can transform one into the other in at most n steps, by recolouring exactly one vertex at each step while maintaining a dicolouring at any step. Furthermore, we prove that, for every oriented graph $\overrightarrow{G}$ on n vertices, the distance between two k-dicolourings is at most $2\Delta_{\min}(\overrightarrow{G})n$ when $k \geq \Delta_{\min}(\overrightarrow{G}) + 1$. We then extend a theorem of Feghali, Johnson and Paulusma to digraphs. We prove that, for every digraph D with $\Delta_{\max}(D) = \Delta \geq 3$ and every $k \geq \Delta + 1$, the k-dicolouring graph of D consists

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of isolated vertices and at most one further component that has diameter at most $c_\Delta n^2$, where $c_\Delta = O(\Delta^2)$ is a constant depending only on Δ.

KEYWORDS
dichromatic number, directed graphs, graph recolouring, oriented graphs, reconfiguration

1 | INTRODUCTION

1.1 | Graph (re)colouring

Given a graph $G = (V, E)$, a *k-colouring* of G is a function $c : V \to \{1, ..., k\}$ such that, for every edge $xy \in E$, we have $c(x) \neq c(y)$. So for every $i \in \{1, ..., k\}$, $c^{-1}(i)$ induces an independent set on G. The *chromatic number* of G, denoted by $\chi(G)$, is the smallest k such that G admits a k-colouring. The *maximum degree* of G, denoted by $\Delta(G)$, is the degree of the vertex with the greatest number of edges incident to it. A simple greedy procedure shows that, for any graph $G, \chi(G) \leq \Delta(G) + 1$. The celebrated theorem of Brooks [7] characterizes the graphs for which equality holds.

Theorem 1 (Brooks [7]). *A connected graph G satisfies $\chi(G) = \Delta(G) + 1$ if and only if G is an odd cycle or a complete graph.*

For any $k \geq \chi(G)$, the *k-colouring graph* of G, denoted by $\mathcal{C}_k(G)$, is the graph whose vertices are the k-colourings of G and in which two k-colourings are adjacent if they differ by the colour of exactly one vertex. A path (or a walk) between two given colourings α, β in $\mathcal{C}_k(G)$ corresponds to a *recolouring sequence*. The existence of such a recolouring sequence witnesses the fact that β can be obtained from α by recolouring exactly one vertex at each step while maintaining a colouring at any step. If $\mathcal{C}_k(G)$ is connected, we say that G is *k-mixing*. A k-colouring of G is *k-frozen* if it is an isolated vertex in $\mathcal{C}_k(G)$. The graph G is *k-freezable* if it admits a k-frozen colouring. In the last fifteen years, since the papers of Cereceda, van den Heuvel and Johnson [8, 9], graph recolouring has been studied by many researchers in graph theory. We refer the reader to the Ph.D. thesis of Bartier [2] for a complete overview on graph recolouring and to the surveys of van Heuvel [15] and Nishimura [17] for reconfiguration problems in general. Feghali, Johnson and Paulusma [11] proved the following analogue of Brooks' theorem for graphs recolouring.

Theorem 2 (Feghali et al. [11]). *Let $G = (V, E)$ be a connected graph with $\Delta(G) = \Delta \geq 3$, $k \geq \Delta + 1$, and α, β two k-colourings of G. Then at least one of the following holds:*

- α is k-frozen, or
- β is k-frozen, or
- there is a recolouring sequence of length at most $c_\Delta |V|^2$ between α and β, where $c_\Delta = O(\Delta)$ is a constant depending on Δ.

Considering graphs of bounded maximum degree, Theorem 2 has been very recently improved by Bousquet, Feuilloley, Heinrich and Rabie, who proved the following.

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Theorem 3 (Bousquet et al. [5]). *Let $G = (V, E)$ be a connected graph with $\Delta(G) = \Delta \geq 3$, $k \geq \Delta + 1$, and α, β two k-colourings of G. Then at least one of the following holds:*

- α is k-frozen, or
- β is k-frozen, or
- there is a recolouring sequence of length at most $O(\Delta^{c\Delta}|V|)$ between α and β, where c is a constant.

In this work, we are looking for extensions of Theorems 1 and 2 to the directed case.

1.2 | Digraph colouring

Let D be a digraph. A *digon* is a pair of arcs in opposite directions between the same vertices. A *simple arc* is an arc which is not in a digon. For any two vertices $x, y \in V(D)$, the digon $\{xy, yx\}$ is denoted by $[x, y]$. The *digon graph* of D is the undirected graph with vertex set $V(D)$ in which uv is an edge if and only if $[u, v]$ is a digon of D. An *oriented graph* is a digraph with no digon. The *bidirected graph* associated with a graph G, denoted by $\overleftrightarrow{G}$, is the digraph obtained from G, by replacing every edge with a digon. The *underlying graph* of D, denoted by $UG(D)$, is the undirected graph G with vertex set $V(D)$ in which uv is an edge if and only if uv or vu is an arc of D.

Let v be a vertex of a digraph D. The *out-degree* (respectively, *in-degree*) of v, denoted by $d^+(v)$ (respectively, $d^-(v)$), is the number of arcs leaving (respectively, entering) v. We define the *maximum degree* of v as $d_{\max}(v) = \max\{d^+(v), d^-(v)\}$, and the *minimum degree* of v as $d_{\min}(v) = \min\{d^+(v), d^-(v)\}$. We can then define the corresponding maximum degrees of D: $\Delta_{\max}(D) = \max_{v \in V(D)}(d_{\max}(v))$ and $\Delta_{\min}(D) = \max_{v \in V(D)}(d_{\min}(v))$. A digraph D is Δ-*diregular* if, for every vertex $v \in V(D)$, $d^-(v) = d^+(v) = \Delta$.

In 1982, Neumann-Lara [16] introduced the notions of dicolouring and dichromatic number, which generalize the ones of colouring and chromatic number. A k-*dicolouring* of D is a function $c : V(D) \to \{1, ..., k\}$ such that $c^{-1}(i)$ induces an acyclic subdigraph in D for each $i \in \{1, ..., k\}$. The *dichromatic number* of D, denoted by $\vec{\chi}(D)$, is the smallest k such that D admits a k-dicolouring. There is a one-to-one correspondence between the k-colourings of a graph G and the k-dicolourings of the associated bidirected graph $\overleftrightarrow{G}$, and in particular $\chi(G) = \vec{\chi}(\overleftrightarrow{G})$. Hence every result on graph colourings can be seen as a result on dicolourings of bidirected graphs, and it is natural to study whether the result can be extended to all digraphs.

The directed version of Brooks' theorem was first proved by Harutyunyan and Mohar in [13]. Aboulker and Aubian gave four new proofs of the following theorem in [1].

Theorem 4 (Directed Brooks' theorem). *Let D be a connected digraph. Then $\vec{\chi}(D) \leq \Delta_{\max}(D) + 1$ and equality holds if and only if one of the following occurs:*

- D is a directed cycle, or
- D is a bidirected odd cycle, or
- D is a bidirected complete graph (of order at least 4).

It is easy to prove, by a simple greedy procedure, that every digraph D can be dicoloured with $\Delta_{\min}(D) + 1$ colours. Hence, one can wonder if Brooks' theorem can be extended to

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digraphs using $\Delta_{\min}(D)$ instead of $\Delta_{\max}(D)$. Unfortunately, Aboulker and Aubian [1] proved that, given a digraph D, deciding whether D is $\Delta_{\min}(D)$-dicolourable is NP-complete. Thus, unless $P = \mathrm{NP}$, we cannot expect an easy characterization of digraphs satisfying $\vec{\chi}(D) = \Delta_{\min}(D) + 1$.

Let the *maximum geometric mean* of a digraph D be $\tilde{\Delta}(D) = \max\{\sqrt{d^+(v)d^-(v)} \mid v \in V(D)\}$. By definition we have $\Delta_{\min}(D) \leq \tilde{\Delta}(D) \leq \Delta_{\max}(D)$. Restricted to oriented graphs, Harutyunyan and Mohar [14] have strengthened Theorem 4 by proving the following.

Theorem 5 (Harutyunyan and Mohar [14]). *There is an absolute constant Δ_1 such that every oriented graph $\vec{G}$ with $\tilde{\Delta}(\vec{G}) \geq \Delta_1$ has $\vec{\chi}(\vec{G}) \leq (1 - e^{-13})\tilde{\Delta}(\vec{G})$.*

In Section 2, we prove another strengthening of Theorem 4 on a large class of digraphs which contains oriented graphs. The *directed join* of H_1 and H_2, denoted by $H_1 \Rightarrow H_2$, is the digraph obtained from disjoint copies of H_1 and H_2 by adding all arcs from the copy of H_1 to the copy of H_2 (H_1 or H_2 may be empty).

Theorem 6. *Let D be a digraph. If D is not $\Delta_{\min}(D)$-dicolourable, then one of the following holds:*

- $\Delta_{\min}(D) \leq 1$, or
- $\Delta_{\min}(D) = 2$ and D contains $\overleftrightarrow{K_2}$, or
- $\Delta_{\min}(D) \geq 3$ and D contains $\overleftrightarrow{K_r} \Rightarrow \overleftrightarrow{K_s}$, for some $r, s \geq 0$ such that $r + s = \Delta_{\min}(D) + 1$.

In particular, the following is a direct consequence of Theorem 6.

Corollary 7. *Let D be a digraph. If $\vec{\chi}(D) = \Delta_{\min}(D) + 1$, then D contains the complete bidirected graph on $\left\lceil \frac{\Delta_{\min}+1}{2} \right\rceil$ vertices as a subdigraph.*

Moreover, since an oriented graph does not contain any digon, Corollary 7 implies the following, which answers a question of Harutyunyan [12]:

Corollary 8. *Let $\vec{G}$ be an oriented graph. If $\Delta_{\min}(\vec{G}) \geq 2$, then $\vec{\chi}(\vec{G}) \leq \Delta_{\min}(\vec{G})$.*

We also show in Section 2 that Corollary 7 is best possible by showing the following.

Theorem 9. *For every fixed $k \geq 2$, k-Dicolourability remains NP-complete when restricted to digraphs D satisfying $\Delta_{\min}(D) = k$ and not containing the bidirected complete graph on $\left\lceil \frac{k+1}{2} \right\rceil + 1$ vertices.*

1.3 | Digraph redicolouring

For any $k \geq \vec{\chi}(D)$, the *k-dicolouring graph* of D, denoted by $\mathcal{D}_k(D)$, is the graph whose vertices are the k-dicolourings of D and in which two k-dicolourings are adjacent if they differ by the colour of exactly one vertex. Observe that $\mathcal{C}_k(G) = \mathcal{D}_k(\overleftrightarrow{G})$ for any bidirected graph $\overleftrightarrow{G}$. A

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redicolouring sequence between two dicolourings is a path between these dicolourings in $\mathcal{D}_k(D)$. The digraph D is *k-mixing* if $\mathcal{D}_k(D)$ is connected. A k-dicolouring of D is *k-frozen* if it is an isolated vertex in $\mathcal{D}_k(D)$. The digraph D is *k-freezable* if it admits a k-frozen dicolouring. A vertex v is *blocked* to its colour in a dicolouring α if, for every colour $c \neq \alpha(v)$, recolouring v to c in α creates a monochromatic directed cycle.

Digraph redicolouring was first introduced in [6], where the authors generalized different results on graph recolouring to digraphs, and proved some specific results on oriented graphs redicolouring. In particular, they studied the k-dicolouring graph of digraphs with bounded degeneracy or bounded maximum average degree, and they show that finding a redicolouring sequence between two given k-dicolourings of a digraph is PSPACE-complete. Dealing with the maximum degree of a digraph, they proved that, given an orientation of a subcubic graph $\vec{G}$ on n vertices, its 2-dicolouring graph $\mathcal{D}_2(\vec{G})$ is connected and has diameter at most $2n$ and they asked if this bound can be improved. We answer this question in Section 3 by proving the following theorem.

Theorem 10. *Let $\vec{G}$ be an oriented graph of order n such that $\Delta_{\min}(\vec{G}) \leq 1$. Then $\mathcal{D}_2(\vec{G})$ is connected and has diameter exactly n.*

In particular, if $\vec{G}$ is an orientation of a subcubic graph, then $\Delta_{\min}(\vec{G}) \leq 1$ (because $d^+(v) + d^-(v) \leq 3$ for every vertex v), and so $\mathcal{D}_2(\vec{G})$ has diameter exactly n. Furthermore, we prove the following as a consequence of Corollary 8 and Theorem 10.

Corollary 11. *Let $\vec{G}$ be oriented graph of order n with $\Delta_{\min}(\vec{G}) = \Delta \geq 1$, and let $k \geq \Delta + 1$. Then $\mathcal{D}_k(\vec{G})$ is connected and has diameter at most $2\Delta n$.*

Corollary 11 does not hold for digraphs in general: indeed, $\overleftrightarrow{P_n}$, the bidirected path on n vertices, satisfies $\Delta_{\min}(\overleftrightarrow{P_n}) = 2$ and $\mathcal{D}_3(\overleftrightarrow{P_n}) = \mathcal{C}_3(P_n)$ has diameter $\Omega(n^2)$, as proved in [4].

Finally in Section 4, we extend Theorem 2 to digraphs as follows.

Theorem 12. *Let $D = (V, A)$ be a connected digraph with $\Delta_{\max}(D) = \Delta \geq 3$, $k \geq \Delta + 1$, and α, β two k-dicolourings of D. Then at least one of the following holds:*

- α is k-frozen, or
- β is k-frozen, or
- there is a redicolouring sequence of length at most $c_\Delta |V|^2$ between α and β, where $c_\Delta = O(\Delta^2)$ is a constant depending only on Δ.

Furthermore, we prove that a digraph D is k-freezable only if D is bidirected and its underlying graph is k-freezable. Thus, an obstruction in Theorem 12 is exactly the bidirected graph of an obstruction in Theorem 2.

2 | PROOFS OF THEOREMS 6 AND 9

A digraph D is *k-dicritical* if $\vec{\chi}(D) = k$ and for every vertex $v \in V(D), \vec{\chi}(D - v) < k$. Observe that every digraph with dichromatic number at least k contains a k-dicritical subdigraph.

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Let $\mathcal{F}_2$ be $\{\overleftrightarrow{K_2}\}$, and for each $\Delta \geq 3$, we define $\mathcal{F}_\Delta = \{\overleftrightarrow{K_r} \Rightarrow \overleftrightarrow{K_s} | r, s \geq 0 \text{ and } r + s = \Delta + 1\}$. A digraph D is $\mathcal{F}_\Delta$-*free* if it does not contain F as a subdigraph, for any $F \in \mathcal{F}_\Delta$. Theorem 6 can then be reformulated as follows:

Theorem 6. *Let D be a digraph with $\Delta_{\min}(D) = \Delta \geq 2$. If D is $\mathcal{F}_\Delta$-free, then $\vec{\chi}(D) \leq \Delta$.*

Proof. Let D be a digraph such that $\Delta_{\min}(D) = \Delta \geq 2$ and $\vec{\chi}(D) = \Delta + 1$. We will show that D contains some $F \in \mathcal{F}_\Delta$ as a subdigraph.

Let (X, Y) be a partition of $V(D)$ such that for each $x \in X$, $d^+(x) \leq \Delta$, and for each $y \in Y$, $d^-(y) \leq \Delta$. We define the digraph $\tilde{D}$ as follows:

- $V(\tilde{D}) = V(D)$,
- $A(\tilde{D}) = A(D\langle X\rangle) \cup A(D\langle Y\rangle) \cup \{xy, yx | xy \in A(D), x \in X, y \in Y\}$.

Claim 6.1. $\vec{\chi}(\tilde{D}) \geq \Delta + 1$.

Proof of claim. Assume for a contradiction that there exists a Δ-dicolouring c of $\tilde{D}$. Then D, coloured with c, must contain a monochromatic directed cycle C. Now C is not contained in X nor Y, for otherwise C would be a monochromatic directed cycle of $D\langle X\rangle$ or $D\langle Y\rangle$ and so a monochromatic directed cycle of $\tilde{D}$. Thus C contains an arc xy from X to Y. But then, $[x, y]$ is a monochromatic digon in $\tilde{D}$, a contradiction. □

Since $\vec{\chi}(\tilde{D}) \geq \Delta + 1$, there is a $(\Delta + 1)$-dicritical subdigraph H of $\tilde{D}$. By dicriticality of H, for every vertex $v \in V(H)$, $d^+_H(v) \geq \Delta$ and $d^-_H(v) \geq \Delta$, for otherwise a Δ-dicolouring of $H - v$ could be extended to H by choosing for v a colour which is not appearing in its out-neighbourhood or in its in-neighbourhood. We define X_H as $X \cap V(H)$ and Y_H as $Y \cap V(H)$. Note that both $H\langle X_H\rangle$ and $H\langle Y_H\rangle$ are subdigraphs of D.

Claim 6.2. H is Δ-diregular.

Proof of claim. Let ℓ be the number of digons between X_H and Y_H in H. Observe that, by definition of X and H, for each vertex $x \in X_H$, $d^+_H(x) = \Delta$. Note also that, in H, ℓ is exactly the number of arcs leaving X_H and exactly the number of arcs entering X_H. We get

$$\begin{aligned}\Delta|X_H| &= \sum_{x \in X_H} d^+_H(x)\\ &= \ell + |A(H\langle X_H\rangle)|\\ &= \sum_{x \in X_H} d^-_H(x),\end{aligned}$$

which implies, since H is dicritical, $d^+_H(x) = d^-_H(x) = \Delta$ for every vertex $x \in X_H$. Using a symmetric argument, we prove that $\Delta|Y_H| = \sum_{y \in Y_H} d^+_H(y)$, implying $d^+_H(y) = d^-_H(y) = \Delta$ for every vertex $y \in Y_H$. □

Since H is Δ-diregular, then in particular $\Delta_{\max}(H) = \Delta$. Hence, because $\vec{\chi}(H) = \Delta + 1$, by Theorem 4, either $\Delta = 2$ and H is a bidirected odd cycle, or $\Delta \geq 3$ and H is the bidirected complete graph on $\Delta + 1$ vertices.

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- If $\Delta = 2$ and H is a bidirected odd cycle, then at least one digon of H belongs to $H\langle X_H\rangle$ or $H\langle Y_H\rangle$, for otherwise H would be bipartite (with bipartition (X_H, Y_H)). Since both $H\langle X_H\rangle$ and $H\langle Y_H\rangle$ are subdigraphs of D, this shows, as desired, that D contains a copy of $\overleftrightarrow{K_2}$.
- If $k \geq 3$ and H is the bidirected complete graph on $\Delta + 1$ vertices, let A_H be all the arcs from Y_H to X_H. Then $D\langle V(H)\rangle \setminus A_H$ is a subdigraph of D which belongs to $\mathcal{F}_\Delta$. □

Now we will justify that Corollary 7 is the best possible. To do so, we prove that given a digraph D which does not contain the bidirected complete graph on $\left\lceil \frac{\Delta_{\min}(D)+1}{2} \right\rceil + 1$ vertices, deciding if it is $\Delta_{\min}(D)$-dicolourable is NP-complete. We shall use a reduction from k-Dicolourability which is defined as follows:

k-Dicolourability
Input: A digraph D
Question: Is D k-dicolourable?

k-Dicolourability is NP-complete for every fixed $k \geq 2$ [3]. It remains NP-complete when we restrict to digraphs D with $\Delta_{\min}(D) = k$ [1].

Theorem 9. *For all $k \geq 2$, k-Dicolourability remains NP-complete when restricted to digraphs D satisfying $\Delta_{\min}(D) = k$ and not containing the bidirected complete graph on $\left\lceil \frac{k+1}{2} \right\rceil + 1$ vertices.*

Proof. Let $D = (V, A)$ be an instance of k-Dicolourability for some fixed $k \geq 2$. Then we build $D' = (V', A')$ as follows:

- For each vertex $x \in V$, we associate a copy of $S_x^- \Rightarrow S_x^+$ where S_x^- is the bidirected complete graph on $\left\lfloor \frac{k+1}{2} \right\rfloor$ vertices, and S_x^+ is the bidirected complete graph on $\left\lceil \frac{k+1}{2} \right\rceil$ vertices.
- For each arc $xy \in A$, we associate all possible arcs x^+y^- in A', such that $x^+ \in S_x^+$ and $y^- \in S_y^-$.

First observe that $\Delta_{\min}(D') = k$. Let v be a vertex of D', if v belongs to some S_x^+, then $d^-(v) = k$, otherwise it belongs to some S_x^- and then $d^+(v) = k$. Then observe that D' does not contain the bidirected complete graph on $\left\lceil \frac{k+1}{2} \right\rceil + 1$ vertices since every digon in D' is contained in some S_x^+ or S_x^-. Thus we only have to prove that $\vec{\chi}(D) \leq k$ if and only if $\vec{\chi}(D') \leq k$ to get the result.

- Let us first prove that $\vec{\chi}(D) \leq k$ implies $\vec{\chi}(D') \leq k$. Assume that $\vec{\chi}(D) \leq k$. Let $\phi : V \to \{1, ..., k\}$ be a k-dicolouring of D. Let ϕ' be the k-dicolouring of D' defined as

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follows: for each vertex $x \in V$, choose arbitrarily $x^- \in S_x^-, x^+ \in S_x^+$, and set $\phi'(x^-) = \phi'(x^+) = \phi(x)$. Then choose a distinct colour for every other vertex v in $S_x^- \cup S_x^+$, and set $\phi'(v)$ to this colour. We get that ϕ' must be a k-dicolouring of D': for each $x \in V$, every vertex but x^- in S_x^- must be a sink in its colour class, and every vertex but x^+ in S_x^+ must be a source in its colour class. Thus if D', coloured with ϕ', contains a monochromatic directed cycle C', then C' must be of the form $x_1^- x_1^+ x_2^- x_2^+ \cdots x_\ell^- x_\ell^+ x_1^-$. But then $C = x_1 x_2 \cdots x_\ell x_1$ is a monochromatic directed cycle in D coloured with ϕ: a contradiction.

- Reciprocally, let us prove that $\vec{\chi}(D') \leq k$ implies $\vec{\chi}(D) \leq k$. Assume that $\vec{\chi}(D') \leq k$. Let $\phi' : V' \to \{1, ..., k\}$ be a k-dicolouring of D'. Let ϕ be the k-dicolouring of D defined as follows. For each vertex $x \in V$, we know that $|S_x^+ \cup S_x^-| = k + 1$, thus there must be two vertices x^+ and x^- in $S_x^+ \cup S_x^-$ such that $\phi'(x^+) = \phi'(x^-)$. Moreover, since both S_x^+ and S_x^- are bidirected, one of these two vertices belongs to S_x^+ and the other one belongs to S_x^-. We assume without loss of generality $x^+ \in S_x^+$ and $x^- \in S_x^-$. Then we set $\phi(x) = \phi'(x^+)$. We get that ϕ must be a k-dicolouring of D. If D, coloured with ϕ, contains a monochromatic directed cycle $C = x_1 x_2 \cdots x_\ell x_1$, then $C' = x_1^- x_1^+ x_2^- x_2^+ \cdots x_\ell^- x_\ell^+ x_1^-$ is a monochromatic directed cycle in D' coloured with ϕ', a contradiction.

□

3 | PROOFS OF THEOREM 10 AND COROLLARY 11

In this section, we restrict to oriented graphs. We first prove Theorem 10, let us restate it.

Theorem 10. *Let $\vec{G}$ be an oriented graph of order n such that $\Delta_{\min}(\vec{G}) \leq 1$. Then $\mathcal{D}_2(\vec{G})$ is connected and has diameter exactly n.*

Observe that, if $\mathcal{D}_2(\vec{G})$ is connected, then its diameter must be at least n: for any 2-dicolouring α, we can define its mirror $\bar{\alpha}$ where, for every vertex $v \in V(\vec{G})$, $\alpha(v) \neq \bar{\alpha}(v)$; then every redicolouring sequence between α and $\bar{\alpha}$ has length at least n.

Lemma 14. *Let C be a directed cycle of length at least 3. Then $\mathcal{D}_2(C)$ is connected and has diameter exactly n.*

Proof. Let α and β be any two 2-dicolourings of C. Let $x = \mathrm{diff}(\alpha, \beta) = |\{v \in V(C) | \alpha(v) \neq \beta(v)\}|$. By induction on $x \geq 0$, let us show that there exists a path of length at most x from α to β in $\mathcal{D}_2(C)$. This clearly holds for $x = 0$ (i.e., $\alpha = \beta$). Assume $x > 0$ and the result holds for $x - 1$. Let $v \in V(C)$ be such that $\alpha(v) \neq \beta(v)$.

If v can be recoloured in $\beta(v)$, then we recolour it and reach a new 2-dicolouring α' such that $\mathrm{diff}(\alpha', \beta) = x - 1$ and the result holds by induction. Else if v cannot be recoloured, then recolouring v must create a monochromatic directed cycle, which must be C. Then there must be a vertex v', different from v, such that $\beta(v) = \alpha(v') \neq \beta(v')$, and

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v' can be recoloured. We recolour it and reach a new 2-dicolouring α' such that $\mathrm{diff}(\alpha', \beta) = x - 1$. The result then holds by induction. □

We are now ready to prove Theorem 10.

Proof of Theorem 10. Let α and β be any two 2-dicolourings of $\overrightarrow{G}$. We will show that there exists a redicolouring sequence of length at most n between α and β. We may assume that $\overrightarrow{G}$ is strongly connected, otherwise we consider each strongly connected component independently. This implies in particular that $\overrightarrow{G}$ does not contain any sink nor source. Let (X, Y) be a partition of $V(\overrightarrow{G})$ such that, for every $x \in X$, $d^+(x) = 1$, and for every $y \in Y$, $d^-(y) = 1$.

Assume first that $\overrightarrow{G}\langle X\rangle$ contains a directed cycle C. Since every vertex in X has exactly one out-neighbour, there is no arc leaving C. Thus, since $\overrightarrow{G}$ is strongly connected, $\overrightarrow{G}$ must be exactly C, and the result holds by Lemma 14. Using a symmetric argument, we get the result when $\overrightarrow{G}\langle Y\rangle$ contains a directed cycle.

Assume now that both $\overrightarrow{G}\langle X\rangle$ and $\overrightarrow{G}\langle Y\rangle$ are acyclic. Thus, since every vertex in X has exactly one out-neighbour, $\overrightarrow{G}\langle X\rangle$ is the union of disjoint and independent in-trees, that are oriented trees in which all arcs are directed towards the root. We denote by X_r the set of roots of these in-trees. Symmetrically, $\overrightarrow{G}\langle Y\rangle$ is the union of disjoint and independent out-trees (oriented trees in which all arcs are directed away from the root), and we denote by Y_r the set of roots of these out-trees. Set $X_\ell = X \setminus X_r$ and $Y_\ell = Y \setminus Y_r$. Observe that the arcs from X to Y form a perfect matching directed from X_r to Y_r. We denote by M_r this perfect matching. Observe also that there can be any arc from Y to X. Now we define X_r^1 and Y_r^1 two subsets of X_r and Y_r respectively, depending on the two 2-dicolourings α and β, as follows:

$$X_r^1 = \{x | xy \in M_r, \alpha(x) = \beta(y) \neq \alpha(y) = \beta(x)\},$$
$$Y_r^1 = \{y | xy \in M_r, \alpha(x) = \beta(y) \neq \alpha(y) = \beta(x)\}.$$

Set $X_r^2 = X_r \setminus X_r^1$ and $Y_r^2 = Y_r \setminus Y_r^1$. We denote by M_r^1 (respectively, M_r^2) the perfect matching from X_r^1 to Y_r^1 (respectively, from X_r^2 to Y_r^2). Figure 1 shows a partitioning of $V(\overrightarrow{G})$ into $X_r^1, X_r^2, X_\ell, Y_r^1, Y_r^2, Y_\ell$.

Claim 10.1. There exists a redicolouring sequence of length s_α from α to some 2-dicolouring α' and a redicolouring sequence of length s_β from β to some 2-dicolouring β' such that each of the following holds:

(i) for any arc $xy \in M_r$, $\alpha'(x) \neq \alpha'(y)$ and $\beta'(x) \neq \beta'(y)$,
(ii) for any arc $xy \in M_r^2$, $\alpha'(x) = \beta'(x)$ (and so $\alpha'(y) = \beta'(y)$ by (i)) and
(iii) $s_\alpha + s_\beta \leq |X_r^2| + |Y_r^2|$.

Proof of claim. We consider the arcs xy of M_r^2 one after another and do the following recolourings depending on the colours of x and y in both α and β to get α' and β'.

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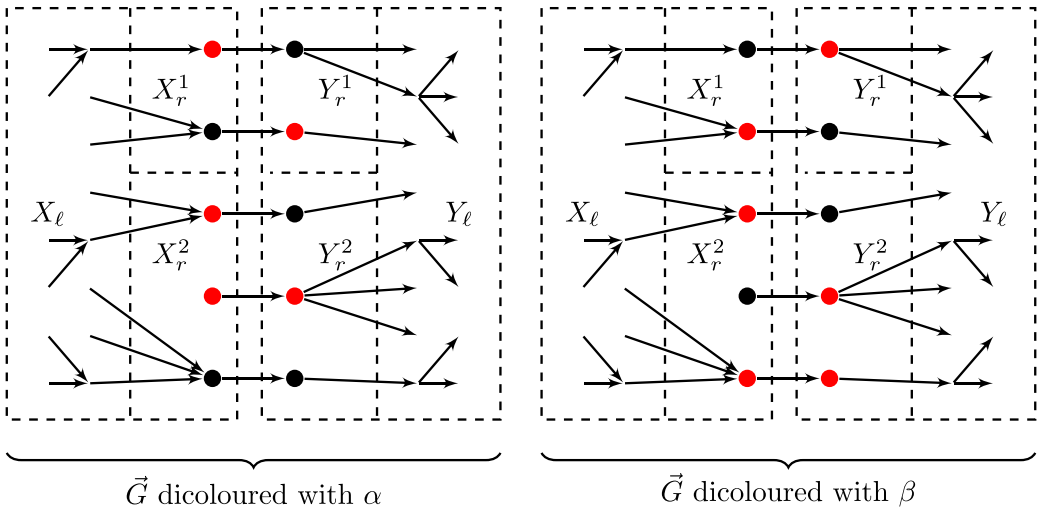

FIGURE 1 Partitioning of $V(\vec{G})$ into $X_r^1, X_r^2, X_\ell, Y_r^1, Y_r^2, Y_\ell$.

- If $\alpha(x) = \alpha(y) = \beta(x) = \beta(y)$, then we recolour x in both α and β.
- Else if $\alpha(x) = \alpha(y) \neq \beta(x) = \beta(y)$, then we recolour x in α and we recolour y in β.
- Else if $\alpha(x) = \beta(x) \neq \alpha(y) = \beta(y)$, then we do nothing.
- Else if $\alpha(x) \neq \alpha(y) = \beta(x) = \beta(y)$, then we recolour x in β.
- Finally if $\alpha(y) \neq \alpha(x) = \beta(x) = \beta(y)$, then we recolour y in β.

Each of these recolourings is valid because, when a vertex in X_r^2 (respectively, Y_r^2) is recoloured, it gets a colour different from its only out-neighbour (respectively, in-neighbour). Let α' and β' be the two resulting 2-dicolourings. By construction, α' and β' agree on $X_r^2 \cup Y_r^2$. For each arc $xy \in M_r^2$, either $\alpha(x) = \alpha'(x)$ or $\alpha(y) = \alpha'(y)$, and the same holds for β and β'. This implies that $s_\alpha + s_\beta \leq 2|M_r^2| = |X_r^2| + |Y_r^2|$. □

Claim 10.2. There exists a redicolouring sequence from α' to some 2-dicolouring $\tilde{\alpha}$ of length s'_α and a redicolouring sequence from β' to some 2-dicolouring $\tilde{\beta}$ of length s'_β such that each of the following holds:

(i) $\tilde{\alpha}$ and $\tilde{\beta}$ agree on $V(\vec{G}) \setminus (X_r^1 \cup Y_r^1)$,
(ii) α' and $\tilde{\alpha}$ agree on $X_r \cup Y_r$,
(iii) β' and $\tilde{\beta}$ agree on $X_r \cup Y_r$,
(iv) $X_\ell \cup Y_\ell$ is monochromatic in $\tilde{\alpha}$ (and in $\tilde{\beta}$ by (i)) and
(v) $s'_\alpha + s'_\beta \leq |X_\ell| + |Y_\ell|$.

Proof of claim. Observe that in both 2-dicolourings α' and β', we are free to recolour any vertex of $X_\ell \cup Y_\ell$ since there is no monochromatic arc from X to Y and both $\vec{G}\langle X\rangle$ and $\vec{G}\langle Y\rangle$ are acyclic. Let n_1 (respectively, n_2) be the number of vertices in $X_\ell \cup Y_\ell$ that are coloured 1 (respectively, 2) in both α' and β'. Without loss of generality, assume that $n_1 \leq n_2$. Then we set each vertex of $X_\ell \cup Y_\ell$ to colour 2 in both α' and β'. Let $\tilde{\alpha}$ and $\tilde{\beta}$ the resulting 2-dicolouring. Then $s'_\alpha + s'_\beta$ is exactly $|X_\ell| + |Y_\ell| + n_1 - n_2 \leq |X_\ell| + |Y_\ell|$. □

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Claim 10.3. There is a redicolouring sequence between $\tilde{\alpha}$ and $\tilde{\beta}$ of length $|X_r^1| + |Y_r^1|$.

Proof of claim. By construction of $\tilde{\alpha}$ and $\tilde{\beta}$, we only have to exchange the colours of x and y for each arc $xy \in M_r^1$. Without loss of generality, we may assume that the colour of all vertices in $X_\ell \cup Y_\ell$ by $\tilde{\alpha}$ and $\tilde{\beta}$ is 1.

We first prove that, by construction, we can recolour any vertex of $X_r^1 \cup Y_r^1$ from 1 to 2. Assume not, then there is such a vertex $x \in X_r^1 \cup Y_r^1$ such that recolouring x from 1 to 2 creates a monochromatic directed cycle C. Since both $\overrightarrow{G}\langle X\rangle$ and $\overrightarrow{G}\langle Y\rangle$ are acyclic, C must contain an arc of M_r. Since M_r does not contain any monochromatic arc in $\tilde{\alpha}$, then this arc must be incident to x. Now observe that colour 2, in $\tilde{\alpha}$, induces an independent set on both $\overrightarrow{G}\langle X\rangle$ and $\overrightarrow{G}\langle Y\rangle$. This implies that C must contain at least 2 arcs in M_r. This is a contradiction since recolouring x creates exactly one monochromatic arc in M_r.

Then, for each arc $xy \in M_r^1$, we can first recolour the vertex coloured 1 and then the vertex coloured 2. Note that we maintain the invariant that colour 2 induces an independent set on both $\overrightarrow{G}\langle X\rangle$ and $\overrightarrow{G}\langle Y\rangle$. We get a redicolouring sequence from $\tilde{\alpha}$ to $\tilde{\beta}$ in exactly $2|M_r^1| = |X_r^1| + |Y_r^1|$ steps. □

Combining the three claims, we finally proved that there exists a redicolouring sequence between α and β of length at most n. □

In the following, when α is a dicolouring of a digraph D, and H is a subdigraph of D, we denote by $\alpha_{|H}$ the restriction of α to H. We will now prove Corollary 11, let us restate it first.

Corollary 11. *Let $\overrightarrow{G}$ be an oriented graph of order n with $\Delta_{\min}(\overrightarrow{G}) = \Delta \geq 1$, and let $k \geq \Delta + 1$. Then $\mathcal{D}_k(\overrightarrow{G})$ is connected and has diameter at most $2\Delta n$.*

Proof. We will show the result by induction on Δ.

Assume first that $\Delta = 1$, let $k \geq 2$. Let α be any k-dicolouring of $\overrightarrow{G}$ and γ be any 2-dicolouring of $\overrightarrow{G}$. To ensure that $\mathcal{D}_k(\overrightarrow{G})$ is connected and has diameter at most $2n$, it is sufficient to prove that there is a redicolouring sequence between α and γ of length at most n. Let H be the digraph induced by the set of vertices coloured 1 or 2 in α, and let J be $V(\overrightarrow{G})\setminus V(H)$. By Theorem 10, since $\Delta_{\min}(H) \leq \Delta_{\min}(\overrightarrow{G}) \leq 1$, we know that there exists a redicolouring sequence, in H, from $\alpha_{|H}$ to $\gamma_{|H}$ of length at most $|V(H)|$. This redicolouring sequence extends in $\overrightarrow{G}$ because it only uses colours 1 and 2. Let α' be the obtained dicolouring of $\overrightarrow{G}$. Since $\alpha'(v) = \gamma(v)$ for every $v \in H$, we can recolour each vertex in J to its colour in γ. This shows that there is a redicolouring sequence between α and γ of length at most $|V(H)| + |J| = |V(\overrightarrow{G})|$. This ends the case $\Delta = 1$.

Assume now that $\Delta \geq 2$ and let $k \geq \Delta + 1$. Let α and β be two k-dicolourings of $\overrightarrow{G}$. By Corollary 8, we know that $\overrightarrow{\chi}(\overrightarrow{G}) \leq \Delta \leq k - 1$. We first show that there is a redicolouring sequence of length at most $2n$ from α to some $(k-1)$-dicolouring γ of $\overrightarrow{G}$. From α, whenever it is possible we recolour each vertex coloured $1, 2$ or k with a colour of

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$\{3, ..., k-1\}$ (when $k = 3$ we do nothing). Let $\tilde{\alpha}$ be the obtained dicolouring, and let M be the set of vertices coloured in $\{3, ..., k-1\}$ by $\tilde{\alpha}$ (when $k = 3$, M is empty). We get that $H = \overrightarrow{G} - M$ satisfies $\Delta_{\min}(H) \leq 2$, since every vertex in H has at least one in-neighbour and one out-neighbour coloured c for every $c \in \{3, ..., k-1\}$. By Corollary 8, there exists a 2-dicolouring $\gamma_{|H}$ of H. From $\tilde{\alpha}_{|H}$, whenever it is possible, we recolour a vertex coloured 1 or 2 to colour k. Let $\hat{\alpha}$ be the resulting dicolouring, and $\hat{H}$ be the subdigraph of H induced by the vertices coloured 1 or 2 in $\hat{\alpha}$. We get that $\Delta_{\min}(\hat{H}) \leq 1$ since every vertex in $\hat{H}$ has, in $\overrightarrow{G}$, at least one in-neighbour and one out-neighbour coloured c for every $c \in \{3, ..., k\}$. In at most $|V(\hat{H})|$ steps, using Theorem 10, we can recolour the vertices of $V(\hat{H})$ to their colour in $\gamma_{|H}$ (using only colours 1 and 2). Then we can recolour each vertex coloured k to its colour in $\gamma_{|H}$. This results in a redicolouring sequence of length at most $2n$ from α to some $(k-1)$-dicolouring γ of $\overrightarrow{G}$, since colour k is not used in the resulting dicolouring (recall that M is coloured with $\{3, ..., k-1\}$).

Now, from β, whenever it is possible we recolour each vertex to colour k. Let $\tilde{\beta}$ be the obtained k-dicolouring, and let N be the set of vertices coloured k in $\tilde{\beta}$. We get that $J = \overrightarrow{G} - N$ satisfies $\Delta_{\min}(J) \leq \Delta - 1$. Thus, by induction, there exists a redicolouring sequence from $\tilde{\beta}_{|J}$ to $\gamma_{|J}$, in at most $2(\Delta - 1)|V(J)|$ steps (using only colours $\{1, ..., k-1\}$). Since N is coloured k in $\tilde{\beta}$, this extends to a redicolouring sequence in $\overrightarrow{G}$. Now, since γ does not use colour k, we can recolour each vertex in N to its colour in γ. We finally get a redicolouring sequence from β to γ of length at most $2(\Delta - 1)n$. Concatenating the redicolouring sequence from α to γ and the one from γ to β, we get a redicolouring sequence from α to β in at most $2\Delta n$ steps. □

4 | PROOF OF THEOREM 12

Let us restate Theorem 12.

Theorem 12. *Let D be a connected digraph with $\Delta_{\max}(D) = \Delta \geq 3$, $k \geq \Delta + 1$, and α, β two k-dicolourings of D. Then at least one of the following holds:*

- α is k-frozen, or
- β is k-frozen, or
- there is a redicolouring sequence of length at most $c_\Delta |V|^2$ between α and β, where $c_\Delta = O(\Delta^2)$ is a constant depending only on Δ.

An *L-assignment* of a digraph D is a function which associates to every vertex a list of colours. An *L-dicolouring* of D is a dicolouring α where, for every vertex v of D, $\alpha(v) \in L(v)$. An *L-redicolouring sequence* is a redicolouring sequence $\gamma_1, ..., \gamma_r$, such that for every $i \in \{1, ..., r\}$, γ_i is an L-dicolouring of D.

Lemma 15. *Let $D = (V, A)$ be a digraph and L be a list assignment of D such that, for every vertex $v \in V$, $|L(v)| \geq d_{\max}(v) + 1$. Let α be an L-dicolouring of D. If $u \in V$ is blocked in α, then for each colour $c \in L(u)$ different from $\alpha(u)$, u has exactly one out-neighbour u_c^+ and one in-neighbour u_c^- coloured c. Moreover, if $u_c^+ \neq u_c^-$, there must be a*

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monochromatic directed path from u_c^+ to u_c^-. In particular, u is not incident to a monochromatic arc.

Proof. Since u is blocked to its colour in α, for each colour $c \in L(u)$ different from $\alpha(u)$, recolouring u to c must create a monochromatic directed cycle C. Let v be the out-neighbour of u in C and w be the in-neighbour of u in C. Then $\alpha(v) = \alpha(w) = c$, and there is a monochromatic directed path (in C) from v to w.

This implies that, for each colour $c \in L(u)$ different from $\alpha(u)$, u has at least one out-neighbour and at least one in-neighbour coloured c. Since $|L(u)| \geq d_{\max}(u) + 1$, then $|L(u)| = d_{\max}(u) + 1$, and u must have exactly one out-neighbour and exactly one in-neighbour coloured c. In particular, u cannot be incident to a monochromatic arc. □

Lemma 16. *Let $D = (V, A)$ be a digraph such that for every vertex $v \in V, N^+(v) \backslash N^-(v) \neq \emptyset$ and $N^-(v) \backslash N^+(v) \neq \emptyset$. Let L be a list assignment of D, such that for every vertex $v \in V$, $|L(v)| \geq d_{\max}(v) + 1$. Then for any pair of L-dicolourings α, β of D, there is an L-redicolouring sequence of length at most $(|V| + 3)|V|$.*

Proof. Let $x = \mathrm{diff}(\alpha, \beta) = |\{v \in V | \alpha(v) \neq \beta(v)\}|$. We will show by induction on x that there is an L-redicolouring sequence from α to β of length at most $(|V| + 3)x$. The result clearly holds for $x = 0$ (i.e., $\alpha = \beta$). Let $v \in V$ be such that $\alpha(v) \neq \beta(v)$. We denote $\alpha(v)$ by c and $\beta(v)$ by c'. If v can be recoloured to c', then we recolour it and we get the result by induction.

Assume now that v cannot be recoloured to c'. Whenever v is contained in a directed cycle C of length at least 3, such that every vertex of C but v is coloured c', we do the following: we choose w a vertex of C different from v, such that $\beta(w) \neq c'$. We know that such a w exists, for otherwise C would be a monochromatic directed cycle in β. Now, since w is incident to a monochromatic arc in C, and because $|L(w)| \geq d_{\max}(w) + 1$, by Lemma 15, we know that w can be recoloured to some colour different from c'. Thus we recolour w to this colour. Observe that it does not increase x.

After repeating this process, maybe v cannot be recoloured to c' because it is adjacent by a digon to some vertices coloured c'. We know that these vertices are not coloured c' in β. Thus, whenever such a vertex can be recoloured, we recolour it. After this, let η be the obtained dicolouring. If v can be recoloured to c' in η, we are done. Otherwise, there must be some vertices, blocked to colour c' in η, adjacent to v by a digon. Let S be the set of such vertices. Observe that, by Lemma 15, for every vertex $s \in S$, c belongs to $L(s)$, for otherwise s would not be blocked in η. We distinguish two cases, depending on the size of S.

- If $|S| \geq 2$, then by Lemma 15, v can be recoloured to a colour c'', different from both c and c', because v is adjacent by a digon with two neighbours coloured c'. Hence we can successively recolour v to c'', and every vertex of S to c. This does not create any monochromatic directed cycle because for each $s \in S$, since s is blocked in η, by Lemma 15 v must be the only neighbour of s coloured c in η. We can finally recolour v to c'.
- If $|S| = 1$, let w be the only vertex in S. If v can be recoloured to any colour (different from c' since w is coloured c'), then we first recolour v, allowing us to recolour w to c,

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because v is the single neighbour of w coloured c in η by Lemma 15. We finally can recolour v to c'. Assume then that v is blocked to colour c in η. Let us fix $w^+ \in N^+(w)\backslash N^-(w)$. Since w is blocked to c' in η, by Lemma 15, there exists exactly one vertex $w^- \in N^-(w)\backslash N^+(w)$ such that $\eta(w^+) = \eta(w^-) = c''$ and there must be a monochromatic directed path from w^+ to w^-. Since v is blocked to colour c in η, either $vw^- \notin A$ or $w^+v \notin A$, otherwise, by Lemma 15, there must be a monochromatic directed path from w^- to w^+, which is blocking v to its colour. But since there is also a monochromatic directed path from w^+ to w^- (blocking w) there would be a monochromatic directed cycle, a contradiction (see Figure 2). We distinguish the two possible cases:

- if $vw^- \notin A$, then we start by recolouring w^- with a colour that does not appear in its in-neighbourhood. This is possible because w^- has a monochromatic entering arc, and because $|L(w^-)| \geq d_{\max}(w^-) + 1$. We first recolour w with c'', since c'' does not appear in its in-neighbourhood anymore (w^- was the only one by Lemma 15). Next we recolour v with c': this is possible because v does not have any out-neighbour coloured c' since w was the only one by Lemma 15 and w^- is not an out-neighbour of v. We can finally recolour w to colour c and w^- to c''. After all these operations, we exchanged the colours of v and w.
- if $w^+v \notin A$, then we use a symmetric argument.

Observe that we found an L-redicolouring sequence from α to an α', in at most $|V| + 3$ steps, such that $\mathrm{diff}(\alpha', \beta) < \mathrm{diff}(\alpha, \beta)$. Thus by induction, we get an L-redicolouring sequence of length at most $(|V| + 3)x$ between α and β. □

We are now able to prove Theorem 12. The idea of the proof is to divide the digraph D into two parts. One of them is bidirected and we will use Theorem 2 as a black box on it. In the other part, we know that each vertex is incident to at least two simple arcs, one leaving and one entering, and we will use Lemma 16 on it.

Proof of Theorem 12. Let $D = (V, A)$ be a connected digraph with $\Delta_{\max}(D) = \Delta$, $k \geq \Delta + 1$. Let α and β be two k-dicolourings of D. Assume that neither α nor β is k-frozen.

We first make a simple observation. For any simple arc $xy \in A$, we may assume that $N^+(y)\backslash N^-(y) \neq \emptyset$ and $N^-(x)\backslash N^+(x) \neq \emptyset$. If this is not the case, then every directed cycle containing xy must contain a digon, implying that the k-dicolouring graph of D is

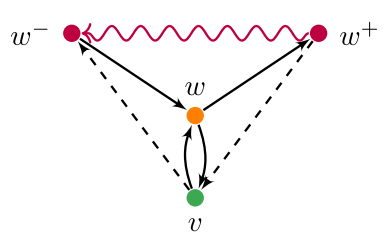

FIGURE 2 Vertices v, w, w^+ and w^-.

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also the k-dicolouring graph of $D\setminus\{xy\}$. Then we may look for a redicolouring sequence in $D\setminus\{xy\}$.

Let $X = \{v \in V | N^+(v) = N^-(v)\}$ and $Y = V\setminus X$. Observe that $D\langle X\rangle$ is bidirected, and thus the dicolourings of $D\langle X\rangle$ are exactly the colourings of $UG(D\langle X\rangle)$. We first show that $\alpha_{|D\langle X\rangle}$ and $\beta_{|D\langle X\rangle}$ are not frozen k-colourings of $D\langle X\rangle$. If Y is empty, then $D\langle X\rangle = D$ and $\alpha_{|D\langle X\rangle}$ and $\beta_{|D\langle X\rangle}$ are not k-frozen by assumption. Otherwise, since D is connected, there exists $x \in X$ such that, in $D\langle X\rangle$, $d^+(x) = d^-(x) \leq \Delta - 1$, implying that x is not blocked in any dicolouring of $D\langle X\rangle$. Thus, by Theorem 2, there is a redicolouring sequence $\gamma'_1, ..., \gamma'_r$ in $D\langle X\rangle$ from $\alpha_{|D\langle X\rangle}$ to $\beta_{|D\langle X\rangle}$, where $r \leq c_\Delta |X|^2$, and $c_\Delta = O(\Delta)$ is a constant depending on Δ.

We will show that, for each $i \in \{1, ..., r-1\}$, if γ_i is a k-dicolouring of D which agrees with γ'_i on X, then there exist a k-dicolouring γ_{i+1} of D that agrees with γ'_{i+1} on X and a redicolouring sequence from γ_i to γ_{i+1} of length at most $\Delta + 2$.

Observe that α agrees with γ'_1 on X. Now assume that there is such a γ_i, which agrees with γ'_i on X, and let $v_i \in X$ be the vertex for which $\gamma'_i(v_i) \neq \gamma'_{i+1}(v_i)$. We denote by c (respectively, c') the colour of v_i in γ'_i (respectively, γ'_{i+1}). If recolouring v_i to c' in γ_i is valid then we have the desired γ_{i+1}. Otherwise, we know that v_i is adjacent with a digon (since v_i is only adjacent to digons) to some vertices (at most Δ) coloured c' in Y. Whenever such a vertex can be recoloured to a colour different from c', we recolour it. Let η_i be the reached k-dicolouring after these operations. If v_i can be recoloured to c' in η_i we are done. If not, then the neighbours of v_i coloured c' in Y are blocked to colour c' in η_i. We denote by S the set of these neighbours. We distinguish two cases:

- If $|S| \geq 2$, then by Lemma 15, v_i can be recoloured to a colour c'', different from both c and c', because v_i has two neighbours with the same colour. Then we successively recolour v_i to c'', and every vertex of S to c. This does not create any monochromatic directed cycle because, by Lemma 15, for each $s \in S$, v_i is the only neighbour of s coloured c in η_i. We can finally recolour v_i to c' to reach the desired γ_{i+1}.
- If $|S| = 1$, let y be the only vertex in S. Since y belongs to Y and is blocked to its colour in η_i, by Lemma 15, we know that y has an out-neighbour $y^+ \in N^+(y)\setminus N^-(y)$ and an in-neighbour $y^- \in N^-(y)\setminus N^+(y)$ such that there is a monochromatic directed path from y^+ to y^-. Observe that both y^+ and y^- are recolourable in η_i by Lemma 15, because there are incidents to a monochromatic arc.
  - If v_i is not adjacent to y^+, then we recolour y^+ to any possible colour, and we recolour y to $\eta_i(y^+)$. We can finally recolour v_i to c' to reach the desired γ_{i+1}.
  - If v_i is not adjacent to y^-, then we recolour y^- to any possible colour, and we recolour y to $\eta_i(y^-)$. We can finally recolour v_i to c' to reach the desired γ_{i+1}.
  - Finally if v_i is adjacent to both y^+ and y^-, since $\eta_i(y^+) = \eta_i(y^-)$, then v_i can be recoloured to a colour c'' different from c and c'. This allows us to recolour y to c, and we finally can recolour v_i to c' to reach the desired γ_{i+1}.

We have shown that there is a redicolouring sequence of length at most $(\Delta + 2)c_\Delta n^2$ from α to some α' that agrees with β on X. Now we define the list assignment: for each $y \in Y$,

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$$L(y) = \{1, ..., k\} \setminus \{\beta(x) | x \in N(y) \cap X\}.$$

Observe that, for every $y \in Y$,

$$|L(y)| \geq k - |N^+(y) \cap X| \geq \Delta + 1 - \left(\Delta - d_Y^+(y)\right) \geq d_Y^+(y) + 1.$$

Symmetrically, we get $|L(y)| \geq d_Y^-(y) + 1$. This implies, in $D\langle Y\rangle$, $|L(y)| \geq d_{\max}(y) + 1$. Note also that both $\alpha'_{|D\langle Y\rangle}$ and $\beta_{|D\langle Y\rangle}$ are L-dicolourings of $D\langle Y\rangle$. Note finally that, for each $y \in Y$, $N^+(y) \setminus N^-(y) \neq \emptyset$ and $N^+(y) \setminus N^-(y) \neq \emptyset$ by choice of X and Y and by the initial observation. By Lemma 16, there is an L-redicolouring sequence in $D\langle Y\rangle$ between $\alpha'_{|D\langle Y\rangle}$ and $\beta_{|D\langle Y\rangle}$, with length at most $(|Y| + 3)|Y|$. By choice of L, this extends directly to a redicolouring sequence from α' to β on D of the same length.

The concatenation of the redicolouring sequence from α to α' and the one from α' to β leads to a redicolouring sequence from α to β of length at most $c'_\Delta |V|^2$, where $c'_\Delta = O(\Delta^2)$ is a constant depending on Δ. □

Remark 17. If α is a k-frozen dicolouring of a digraph D, with $k \geq \Delta_{\max}(D) + 1$, then D must be bidirected. If D is not bidirected, then we choose v a vertex incident to a simple arc. If v cannot be recoloured in α, by Lemma 15, since v is incident to a simple arc, there exists a colour c for which v has an out-neighbour w and an in-neighbour u both coloured c, such that $u \neq w$ and there is a monochromatic directed path from w to u. But then, every vertex on this path is incident to a monochromatic arc, and it can be recoloured by Lemma 15. Thus, α is not k-frozen. This shows that an obstruction of Theorem 12 is exactly the bidirected graph of an obstruction of Theorem 2.

5 | FURTHER RESEARCH

In this paper, we established some analogues of Brooks' theorem for the dichromatic number of oriented graphs and for digraph redicolouring. Many open questions arise, we detail a few of them.

Restricted to oriented graphs, Erdős and Neumann-Lara (see [10]) conjectured that the directed Brooks' theorem can be improved to the following.

Conjecture 18 (Erdős and Neumann-Lara). *Every oriented graph* $\vec{G}$ *has* $\vec{\chi}(\vec{G}) = O\left(\frac{\Delta_{\max}}{\log(\Delta_{\max})}\right)$.

Concerning digraph redicolouring, we believe that Corollary 11 and Theorem 12 can be improved. We pose the following conjecture.

Conjecture 19. *There is an absolute constant* c *such that for every integer* k *and every oriented graph* $\vec{G}$ *on* n *vertices, such that* $k \geq \Delta_{\min}(\vec{G}) + 1$, *the diameter of* $\mathcal{D}_k(\vec{G})$ *is bounded by* cn.

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Conjecture 20. *There is an absolute constant d such that for every integer k and every digraph D on n vertices, with $k \geq \Delta_{\max}(D) + 1 \geq 4$, the diameter of $\mathcal{D}_k(D)$ is bounded by dn^2.*

It would also be nice to extend Theorem 3 to directed graphs, that is to show that the diameter of $\mathcal{D}_k(D)$ is bounded by $f(\Delta_{\max}(D))n$ for some computable function f whenever $k \geq \Delta_{\max}(D) + 1 \geq 4$. To prove it, it would be sufficient to show the analogue of Lemma 16 with an L-redicolouring sequence of length at most $f(\Delta_{\max}(D))|V|$, and then follow the proof of Theorem 12 (using Theorem 3 instead of Theorem 2).

Given an orientation $\vec{G}$ of a planar graph, a celebrated conjecture from Neumann-Lara [16] states that the dichromatic number of $\vec{G}$ is at most 2. It is known that it must be 4-mixing because planar graphs are 5-degenerate [6]. It is also known that there exist 2-freezable orientations of planar graphs [6]. Thus the following problem, stated in [6], remains open:

Question 21. Is every oriented planar graph 3-mixing?

ACKNOWLEDGEMENTS

I am grateful to Frédéric Havet and Nicolas Nisse for stimulating discussions. Research supported by research grant DIGRAPHS and by the French government, through the EUR DS4H Investments in the Future project managed by the National Research Agency (ANR) with the reference numbers ANR-19-CE48-0013, ANR-17-EURE-0004.

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How to cite this article: L. Picasarri-Arrieta, *Strengthening the directed Brooks' theorem for oriented graphs and consequences on digraph redicolouring*, J. Graph Theory. 2024;**106**:5–22. https://doi.org/10.1002/jgt.23066

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An analogue of Reed's conjecture for digraphs

Ken-ichi Kawarabayashi* Lucas Picasarri-Arrieta†

Abstract

Reed in 1998 conjectured that every graph G satisfies $\chi(G) \leqslant \lceil \frac{\Delta(G)+1+\omega(G)}{2} \rceil$. As a partial result, he proved the existence of $\varepsilon > 0$ for which every graph G satisfies $\chi(G) \leqslant \lceil (1-\varepsilon)(\Delta(G)+1) + \varepsilon\omega(G) \rceil$. We propose an analogue conjecture for digraphs. Given a digraph D, we denote by $\vec{\chi}(D)$ the dichromatic number of D, which is the minimum number of colours needed to partition D into acyclic induced subdigraphs. We let $\overleftrightarrow{\omega}(D)$ denote the size of a largest biclique (a set of vertices inducing a complete digraph) of D and $\tilde{\Delta}(D) = \max_{v \in V(D)} \sqrt{d^+(v) \cdot d^-(v)}$. We conjecture that every digraph D satisfies $\vec{\chi}(D) \leqslant \lceil \frac{\tilde{\Delta}(D)+1+\overleftrightarrow{\omega}(D)}{2} \rceil$, which if true implies Reed's conjecture. As a partial result, we prove the existence of $\varepsilon > 0$ for which every digraph D satisfies $\vec{\chi}(D) \leqslant \lceil (1-\varepsilon)(\tilde{\Delta}(D)+1) + \varepsilon\overleftrightarrow{\omega}(D) \rceil$. This implies both Reed's result and an independent result of Harutyunyan and Mohar for oriented graphs.

To obtain this upper bound on $\vec{\chi}$, we prove that every digraph D with $\overleftrightarrow{\omega}(D) > \frac{2}{3}(\Delta_{\max}(D)+1)$ admits an acyclic set of vertices intersecting each biclique of D, where $\Delta_{\max}(D) = \max_{v \in V(D)} \max(d^+(v), d^-(v))$, which generalises a result of King.

1 Introduction

In 1998, Reed [31] posed the following celebrated conjecture, which draws a connection between the maximum degree $\Delta(G)$, the clique number $\omega(G)$, and the chromatic number $\chi(G)$ of a graph G.

Conjecture 1.1. (Reed [31]) *Every graph G satisfies* $\chi(G) \leqslant \left\lceil \frac{\Delta(G)+1+\omega(G)}{2} \right\rceil$.

As a partial result towards this conjecture, Reed proved the following.

Theorem 1.1. (Reed [31]) *There exists $\varepsilon > 0$ such that every graph G satisfies*

$$\chi(G) \leqslant \lceil (1-\varepsilon)(\Delta(G)+1) + \varepsilon\omega(G) \rceil.$$

Observe that Conjecture 1.1 is exactly Theorem 1.1 for $\varepsilon = \frac{1}{2}$. Restricted to graphs with sufficiently large maximum degree, Hurley *et al.* [21] proved that Theorem 1.1 holds for $\varepsilon = 0.119$, improving on earlier results obtained respectively by Bonamy *et al.* [9] and by Delcourt and Postle [11].

The purpose of this work is to propose analogues of Conjecture 1.1 and Theorem 1.1 for digraphs. We thus need three digraph parameters that extend Δ, ω, and χ respectively. By extension, we mean that, if D is a symmetric digraph, these digraph parameters on D must coincide with Δ, ω, and χ on the underlying graph of D. One can easily check that this is actually the case for $\tilde{\Delta}$, $\overleftrightarrow{\omega}$, and $\vec{\chi}$, defined as follows.

Let D be a digraph. We let $\tilde{\Delta}(D)$ denote the maximum geometric mean of the in- and out-degrees of the vertices of D, that is $\tilde{\Delta}(D) = \max\{\sqrt{d^+(v) \cdot d^-(v)} \mid v \in V(D)\}$. A *biclique* of D is a set of vertices inducing a complete digraph. The *biclique number* $\overleftrightarrow{\omega}(D)$ of D is the size of a largest biclique of D. The notions of dicolouring and dichromatic number were introduced in the late 1970s by Erdős and Neumann-Lara [28, 12]. A *k-dicolouring* of D is a function $c : V(D) \to [k]$ such that $c^{-1}(i)$ induces an acyclic subdigraph on D for each $i \in [k]$. The *dichromatic number* of D, denoted by $\vec{\chi}(D)$, is the smallest $k \in \mathbb{N}$ such that D admits a k-dicolouring. These two concepts were rediscovered by Mohar in the 2000s [26, 8] and received a lot of attention since then. It turned out that many classical results on graph colouring can be extended to digraphs with these notions, which shows that

*National Institute of Informatics, The University of Tokyo, Japan. Research supported by JSPS KAKENHI JP20A402 and 22H05001 and by JST ASPIRE JPMJAP2302.

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dicolouring is somehow the right extension of proper colouring to digraphs. For instance, people generalised the Strong Perfect Graph Theorem [6], Gyárfás-Sumner's Conjecture [4, 3, 10], or Brooks-type results [18, 1, 2, 16]. Harutyunyan and Mohar [19] posed the following analogue of Conjecture 1.1 for oriented graphs, that are digraphs with biclique number at most 1.

Conjecture 1.2. (Harutyunyan and Mohar [19]) *Let D be an oriented graph, then* $\vec{\chi}(D) \leqslant \left\lceil \frac{\tilde{\Delta}(D)}{2} \right\rceil + 1$.

As a partial result towards their conjecture, they proved the following for $\varepsilon = e^{-13}$. It was later strengthened by Golowich [15], who proved it for $\varepsilon = 0.183$.

Theorem 1.2. (Harutyunyan and Mohar [19]) *There exists $\varepsilon > 0$ such that every oriented graph D with $\tilde{\Delta}(D)$ large enough satisfies $\vec{\chi}(D) \leqslant (1-\varepsilon)\tilde{\Delta}(D)$.*

We actually believe that the analogue of Conjecture 1.1 holds not only for oriented graphs but for all digraphs. We thus propose the following conjecture, which if true implies both Conjectures 1.1 and 1.2.

Conjecture 1.3. *Every digraph D satisfies* $\vec{\chi}(D) \leqslant \left\lceil \frac{\tilde{\Delta}(D)+1+\overleftrightarrow{\omega}(D)}{2} \right\rceil$.

The main result of this work is the following, which implies both Theorem 1.1 (when restricted to symmetric digraphs) and Theorem 1.2 (when restricted to oriented graphs), and provide some support to Conjecture 1.3.

Theorem 1.3. *There exists $\varepsilon > 0$ such that every digraph D satisfies* $\vec{\chi}(D) \leqslant \lceil (1-\varepsilon)(\tilde{\Delta}(D)+1) + \varepsilon\overleftrightarrow{\omega}(D) \rceil$.

We made no attempt to maximize the value of ε in our proof. In particular, applying Theorem 1.3 to obtain Theorems 1.1 and 1.2 yields values of ε that are much smaller than the known ones.

Our proof of Theorem 1.3 is inspired by a short proof of Theorem 1.1 due to King and Reed [24]. Along the way, we generalise to digraphs a collection of intermediate results that we find of independent interest. We refer the reader to the recent book of Stiebitz, Schweser, and Toft [32, Chapter 6] for a complete proof of Theorem 1.1 which includes proofs of all intermediate results (on undirected graphs) mentioned later on.

The paper is organised as follows. In Section 2, we give notation on digraphs and preliminary results used in the paper. In Section 3, using the probabilistic method, we prove the following upper bound on the dichromatic number of sparse digraphs, which generalises to digraphs a classical result of Molloy and Reed [27]. Given a digraph D with maximum degree $\Delta_{\max}(D) = \Delta$ and a vertex v of D, we denote by $m^+(v)$ and $m^-(v)$ the number of arcs of $D[N^+(v)]$ and $D[N^-(v)]$ respectively. We say that D is *B-sparse* if, for every vertex $v \in V(D)$, we have $\min(m^+(v), m^-(v)) \leqslant \Delta(\Delta-1) - B$.

Theorem 1.4. *There exists an integer $\Delta_0 \geqslant 1$ such that the following holds. For every digraph D with maximum degree $\Delta_{\max}(D) = \Delta > \Delta_0$, if D is B-sparse with $B > \Delta \log^3 \Delta$, then* $\vec{\chi}(D) \leqslant (\Delta+1) - \frac{B}{4e^7\Delta}$.

In Section 4, we give a sufficient condition for a digraph to admit an acyclic set of vertices intersecting each of its bicliques, by proving the following generalisation of a result of King [23].

Theorem 1.5. *Let D be a digraph with $\overleftrightarrow{\omega}(D) > \frac{2}{3}(\Delta_{\max}(D)+1)$. Then D has an acyclic set of vertices I such that $\overleftrightarrow{\omega}(D-I) = \overleftrightarrow{\omega}(D) - 1$.*

This result implies that, in a minimum counterexample of Theorem 1.3, there must be a significant gap between the biclique number and the maximum degree. In Section 5, we deal with vertices having a dense neighbourhood in such counterexamples. We extend a result of King and Reed [24] by proving the following.

Theorem 1.6. *Let a and ε be constants satisfying $0 < \varepsilon \leqslant \frac{1}{6} - 4\sqrt{a}$. There exists $\Delta(a) \geqslant 1$ such that the following holds. Let D be a digraph with maximum degree $\Delta_{\max}(D) = \Delta \geqslant \Delta(a)$ satisfying $\overleftrightarrow{\omega}(D) \leqslant \frac{2}{3}(\Delta+1)$ and let $v \in V(D)$ be such that $\max(m^-(v), m^+(v)) > (1-a)\Delta(\Delta-1)$. Then*

$$\vec{\chi}(D) \leqslant \max\left(\vec{\chi}(D-v), (1-\varepsilon)(\Delta+1)\right).$$

We put the pieces together and prove Theorem 1.3 in Section 6. In particular, we use a trick introduced by Harutyunyan and Mohar [19] to obtain the result with $\tilde{\Delta}$ instead of $\Delta_{\max}$. We finally propose some further research directions in Section 7. In particular, we discuss possible analogues of Conjecture 1.3 with $\tilde{\Delta}$ being replaced by smaller parameters.

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2 Preliminaries

2.1 General notation on digraphs Our notation on digraphs follows [7]. A *digon* is a pair of arcs in opposite directions between the same vertices. A *simple arc* is an arc that is not in a digon. A digraph is *symmetric* if it contains no simple arc. An *oriented graph* is a digraph with no digon. Given an undirected graph G, the digraph $\overleftrightarrow{G}$ is the symmetric digraph obtained from G by replacing every edge with a digon. For an integer n, $\overleftrightarrow{K_n}$ is called the *complete digraph* on n vertices.

Let D be a digraph. The *underlying graph* of D, denoted by $\mathrm{UG}(D)$, is the undirected graph with vertex-set $V(D)$ in which uv is an edge if and only if uv or vu is an arc of D. The *symmetric part* of D, denoted by $S(D)$, is the undirected graph with vertex-set $V(D)$ in which uv is an edge if and only if $\{uv, vu\}$ is a digon of D. Given X, Y two disjoint set of vertices of D, we denote by $A(X, Y)$ the set of arcs $uv \in A(D)$ such that $u \in X$ and $v \in Y$.

Let v be a vertex of D. The *out-degree* (resp. *in-degree*) of v, denoted by $d^+(v)$ (resp. $d^-(v)$), is the number of arcs leaving (resp. entering) v. We define the *maximum degree* of v as $d_{\max}(v) = \max\{d^+(v), d^-(v)\}$, and the *minimum degree* of v as $d_{\min}(v) = \min\{d^+(v), d^-(v)\}$. We can then define the corresponding maximum degrees of D:

$$\Delta_{\max}(D) = \max\{d_{\max}(v) \mid v \in V(D)\} \qquad \text{and} \qquad \Delta_{\min}(D) = \max\{d_{\min}(v) \mid v \in V(D)\}.$$

The *maximum geometric mean* of D is defined as $\tilde{\Delta}(D) = \max_{v \in V(D)} \sqrt{d^-(v) \cdot d^+(v)}$. By definition, observe that we have $\Delta_{\min}(D) \leqslant \tilde{\Delta}(D) \leqslant \Delta_{\max}(D)$. We say that D is Δ-*diregular* if, for every vertex $v \in V(D)$, $d^-(v) = d^+(v) = \Delta$. Note that D being diregular implies $\Delta_{\min}(D) = \Delta_{\max}(D)$ but the converse is not true.

A *list assignment* L of D is a function that associates a list of colours to every vertex v of D. It is a k-*list assignment* if $|L(v)| \geqslant k$ holds for every vertex v of D. Let L be a list assignment. An L-*dicolouring* of D is a dicolouring α of D such that, for every vertex v of D, $\alpha(v) \in L(v)$. We say that D is L-*dicolourable* if D admits an L-dicolouring. Given an integer k, we say that D is k-*dichoosable* if, for every k-list assignment L of D, D is L-dicolourable.

2.2 Probabilistic tools Our proof of Theorem 1.3 contains some probabilistic arguments. We briefly recall here two classical results used later. The reader unfamiliar with the probabilistic method is referred to [5] and [27]. We first need the symmetric version of the Lovász Local Lemma, due to Erdős and Lovász [13].

Lemma 2.1. (Lovász Local Lemma) *Let $A_1, A_2, \ldots, A_n$ be events in an arbitrary probability space. Suppose that each event A_i is mutually independent of a set of all the other events but at most d, and that $\mathbf{Pr}(A_i) \leqslant p$ for all $1 \leqslant i \leqslant n$. If*

$$ep(d+1) \leqslant 1,$$

then $\mathbf{Pr}\left(\bigwedge_{i=1}^{n} \overline{A_i}\right) > 0$.

The second result we need is the celebrated concentration bound due to Talagrand [33]. We use it in the following form (see [22, Theorem 6.3]).

Lemma 2.2. (Talagrand's Inequality) *Let X be a non-negative random variable, not identically 0, which is determined by n independent trials and satisfying the following for some $c, r > 0$:*

1. *Changing the outcome of any one trial can affect X by at most c, and*
2. *for any s, if $X \geqslant s$ then there is a set of at most rs trials whose outcomes certify that $X \geqslant s$.*

Then

$$\mathbf{Pr}\left(|X - \mathbf{E}(X)| > t\right) \leqslant 4 \exp\left(-\frac{t^2}{8rc^2(4\mathbf{E}(X) + t)}\right)$$

for any real number t with $t > 96c\sqrt{r\mathbf{E}(X)} + 128rc^2$.

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3 Colouring sparse digraphs

Our first goal is to prove Theorem 1.4, which is the directed analogue of the following result due to Molloy and Reed [27, Theorem 10.5] on the chromatic number of sparse graphs. For a vertex v of a graph G, we denote by $m(v)$ the number of edges of $G[N(v)]$. We say that G is *B-sparse* if $m(v) \leqslant \binom{\Delta(G)}{2} - B$ for every $v \in V(G)$.

THEOREM 3.1. (MOLLOY AND REED [27]) *Let Δ be an integer large enough and G be a graph with maximum degree Δ. If G is B-sparse with $B > \Delta \log^3 \Delta$, then*

$$\chi(G) \leqslant (\Delta + 1) - \frac{B}{e^6 \Delta}.$$

We first recall Theorem 1.4 here for the reader's convenience.

THEOREM 3.2. *There exists an integer $\Delta_0 \geqslant 1$ such that the following holds. For every digraph D with maximum degree $\Delta_{\max}(D) = \Delta > \Delta_0$, if D is B-sparse with $B > \Delta \log^3 \Delta$, then*

$$\vec{\chi}(D) \leqslant (\Delta + 1) - \frac{B}{4e^7 \Delta}.$$

We will actually use Theorem 3.2 in the following form in Section 6.

COROLLARY 3.1. *There exists $\Delta_1 \geqslant 2$ such that the following holds. Let D be a digraph with maximum degree $\Delta_{\max}(D) = \Delta \geqslant \Delta_1$ and $a \in \mathbb{R}$ be such that $a > \frac{\log^3 \Delta}{\Delta - 1}$, and for every vertex $v \in V(D)$, $\min(m^+(v), m^-(v)) \leqslant (1-a)\Delta(\Delta - 1)$. Then*

$$\vec{\chi}(D) \leqslant \left(1 - \frac{a}{5e^7}\right)(\Delta + 1).$$

Proof. By assumption D is B_a-sparse for $B_a = a\Delta(\Delta - 1) > \Delta \log^3 \Delta$. Hence, by taking $\Delta_1 \geqslant \Delta_0$, we obtain by Theorem 3.2 that

$$\vec{\chi}(D) \leqslant \left(1 - \frac{a}{4e^7}\right)(\Delta + 1) + \frac{a}{2e^7} \leqslant \left(1 - \frac{a}{5e^7}\right)(\Delta + 1),$$

where in the second inequality we used $\Delta_1 \geqslant 9$. ∎

The remainder of this section is devoted to the proof of Theorem 3.2. We first briefly justify that we only have to prove it for Δ-diregular digraphs.

LEMMA 3.1. *Let D be a B-sparse digraph with maximum degree $\Delta_{\max}(D) = \Delta$. There exists a B-sparse Δ-diregular digraph H such that $\vec{\chi}(D) \leqslant \vec{\chi}(H)$.*

Proof. We repeat the following process until D is Δ-diregular. Add to D a disjoint copy of $\overleftarrow{D}$, the digraph obtained from D by reversing every arc. Then for every vertex $v \in V(D)$ such that $d^+_D(v) < \Delta$, we add the arc vv', where v' is the copy of v in $\overleftarrow{D}$. Similarly, for every vertex v such that $d^-_D(v) < \Delta$, we add the arc $v'v$. Let D' be the obtained digraph. Note that $\vec{\chi}(D) \leqslant \vec{\chi}(D')$ trivially holds as D is a subdigraph of D'. Observe also that, for every vertex $v \in V(D)$, we have $m^+_D(v) = m^+_{D'}(v) = m^-_{D'}(v')$ and $m^-_D(v) = m^-_{D'}(v) = m^+_{D'}(v')$. Hence, as D is B-sparse, so is D'. Observe finally that, for every vertex $v \in V(D)$, $d^+_{D'}(v) = d^-_{D'}(v') = \min(d^+_D(v) + 1, \Delta)$ and $d^-_{D'}(v) = d^+_{D'}(v') = \min(d^-_D(v) + 1, \Delta)$. Hence, repeating this process at most Δ times, we eventually find the desired digraph H. ∎

A *partial (k, ℓ)-dicolouring* α of a digraph D is a k-dicolouring of an induced subdigraph of D such that, for every vertex v, either ℓ colours appear twice in $N^-(v)$ or ℓ colours appear twice in $N^+(v)$. We formally have

$$\begin{aligned} & \left|\{c \in [k] \mid \exists u, w \in N^-(v), u \neq w, \alpha(u) = \alpha(w) = c\}\right| \geqslant \ell \\ \text{or} \quad & \left|\{c \in [k] \mid \exists u, w \in N^+(v), u \neq w, \alpha(u) = \alpha(w) = c\}\right| \geqslant \ell. \end{aligned}$$

LEMMA 3.2. *Let D be a Δ-diregular digraph and k, ℓ be two integers such that $k \leqslant \Delta + 1 - \ell$. If D admits a partial (k, ℓ)-dicolouring, then $\vec{\chi}(D) \leqslant \Delta + 1 - \ell$.*

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Proof. Assume that D admits a partial (k,ℓ)-dicolouring. We let α be a partial $(\Delta+1-\ell,\ell)$-dicolouring of D for which the number of uncoloured vertices is minimum, the existence of which is guaranteed as $k \leqslant \Delta+1-\ell$. We claim that α is a dicolouring of D, hence implying $\vec{\chi}(D) \leqslant \Delta+1-\ell$. Assume this is not the case, so there is a vertex v which is not coloured by α. By definition of a partial $(\Delta+1-\ell,\ell)$-dicolouring, we have $|\alpha(N^-(v))| \leqslant \Delta-\ell$ or $|\alpha(N^+(v))| \leqslant \Delta-\ell$. Hence we can choose for v a colour of $[\Delta+1-\ell]$ that is not appearing either in its in- or in its out-neighbourhood. Colouring v with this specific colour extends α into a partial $(\Delta+1-\ell,\ell)$-dicolouring of D with fewer uncoloured vertices, a contradiction to the choice of α. ∎

We are now ready to prove Theorem 3.2.

Proof. We do not give the explicit value of Δ_0, we simply assume in what follows that it is arbitrarily large. We assume that D is Δ-diregular, for otherwise we may prove the result for the B-sparse Δ-diregular digraph H obtained by Lemma 3.1, and conclude that $\vec{\chi}(D) \leqslant \vec{\chi}(H) \leqslant (\Delta+1) - \frac{B}{4e^7\Delta}$.

We use the probabilistic method. Let us fix $k = \lfloor \frac{\Delta}{2} \rfloor$. We randomly colour every vertex of D with a colour of $[k]$. Then we simultaneously uncolour every vertex v having at least one in-neighbour and at least one out-neighbour sharing its colour. Our goal is to show that with non-zero probability, the obtained colouring α is a partial (k,ℓ)-dicolouring of D, where $\ell = \frac{B}{4e^7\Delta}$. Observe that we have $k \leqslant \Delta+1-\ell$ as $B \leqslant \Delta(\Delta-1)$. Hence, if α is indeed a partial (k,ℓ)-dicolouring of D, the result follows from Lemma 3.2. Note that α is necessarily a dicolouring of an induced subdigraph of D, for otherwise D, partially coloured with α, contains a monochromatic directed cycle, the vertices of which should have been uncoloured in the process. It remains to show that, for every vertex of D, ℓ colours are repeated either in its in-neighbourhood or in its out-neighbourhood.

For each vertex v, we let N_v be $N^+(v)$ if $m^+(v) \leqslant m^-(v)$ and $N^-(v)$ otherwise. We also let B_v be the number of missing arcs in $D[N_v]$, that is $B_v = \Delta(\Delta-1) - \min(m^-(v), m^+(v))$. Note that $B_v \geqslant B > \Delta \log^3 \Delta$. Let X_v be the random variable corresponding to the number of colours that are assigned to at least two vertices in N_v and are retained by all these vertices. We also define Y_v as the number of colours assigned to at least two vertices in N_v that are not linked by a digon, but not necessarily retained. Finally, we define Z_v as the number of colours assigned to at least two vertices in N_v that are not linked by a digon and one of these vertices does not retain its colour. We thus have $X_v = Y_v - Z_v$.

We let A_v be the event that $X_v < \ell$. Our goal is to show that none of the events A_v occurs with non-zero probability. To do so, we will prove that, for each vertex v, the probability that A_v occurs is small enough and then conclude by the Lovász Local Lemma (Lemma 2.1).

Claim 3.1. *For every vertex v, we have $\mathbf{E}(X_v) \geqslant \frac{B_v}{2e^7\Delta}$*

Proof of claim. For every pair of vertices $u, w \in N_v$ and every colour $c \in [k]$, we let $A^c_{u,w}$ be the event that u, w are exactly the vertices coloured c in N_v and they both retain their colour. We let $X^c_{u,w}$ be the binary random variable equals to 1 if $A^c_{u,w}$ occurs and 0 otherwise. We clearly have $X_v \geqslant \sum_{u,w,c} X^c_{u,w}$.

Assume u and w are not linked by a digon. If uw is a simple arc of D, then $A^c_{u,w}$ occurs if and only if both u and w are coloured c and no vertex in $N = (N_v \cup N^-(u) \cup N^+(w)) - \{u,w\}$ receives colour c. If the arc uw is missing, then $A^c_{u,w}$ occurs in particular if both u and w are coloured c and no vertex in $N = (N_v \cup N^+(u) \cup N^-(w)) - \{u,w\}$ receives colour c. In both cases, as $k = \lfloor \frac{\Delta}{2} \rfloor$, we have $|N| \leqslant 3\Delta \leqslant 7k$, which implies

$$\mathbf{Pr}(A^c_{u,w}) \geqslant \left(\frac{1}{k}\right)^2 \left(1 - \frac{1}{k}\right)^{7k}.$$

By definition of B_v we have at least $\frac{B_v}{2}$ distinct pair of vertices that are not linked by a digon in $N^+(v)$. We also have k choices for colour c. By linearity of the expectation we thus obtain

$$\mathbf{E}(X_v) \geqslant k\frac{B_v}{2}\left(\frac{1}{k}\right)^2\left(1-\frac{1}{k}\right)^{7k} \geqslant \frac{B_v}{\Delta}\left(1-\frac{1}{k}\right)^{7k} \geqslant \frac{B_v}{2e^7\Delta},$$

where in the last inequality we used that $k = \lfloor \frac{\Delta}{2} \rfloor$ is large enough and $\lim_{k\to+\infty} \left(1-\frac{1}{k}\right)^{7k} = \frac{1}{e^7} > \frac{1}{2e^7}$. ◊

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Claim 3.2. *For every vertex v, we have* $\mathbf{E}(X_v) \leqslant \mathbf{E}(Y_v) \leqslant \frac{3B_v}{\Delta}$

Proof of claim. Since $X_v = Y_v - Z_v$ it is clear that $\mathbf{E}(X_v) \leqslant \mathbf{E}(Y_v)$. Given two vertices $u, w \in N_v$ that are not linked by a digon and a colour $c \in [k]$, we let $Y^c_{u,w}$ be the binary random variable equals to 1 if both u and w are coloured c and 0 otherwise. We have $Y_v \leqslant \sum_{u,w,c} Y^c_{u,w}$ and $\mathbf{Pr}(Y^c_{u,w} = 1) = \frac{1}{k^2}$. Since we have at most B_v choices for the pair u, w and exactly k for colour c, by linearity of the expectation we obtain

$$\mathbf{E}(X_v) \leqslant \mathbf{E}(Y_v) \leqslant kB_v \left(\frac{1}{k}\right)^2 \leqslant \frac{3B_v}{\Delta},$$

where in the last inequality we used $k = \left\lfloor \frac{\Delta}{2} \right\rfloor \geqslant \frac{\Delta}{3}$. ◊

Claim 3.3. *For every vertex v,* $\mathbf{Pr}\left(|X_v - \mathbf{E}(X_v)| > \log \Delta \sqrt{\mathbf{E}(X_v)}\right) < \frac{1}{\Delta^5}$.

Proof of claim. Let us set $t = \frac{1}{2} \log \Delta \sqrt{\mathbf{E}(X_v)}$. Recall that $X_v = Y_v - Z_v$, so by linearity of the expectation we have $\mathbf{E}(X_v) = \mathbf{E}(Y_v) - \mathbf{E}(Z_v)$, and $|X_v - \mathbf{E}(X_v)| \leqslant |Y_v - \mathbf{E}(Y_v)| + |Z_v - \mathbf{E}(Z_v)|$. In particular, this implies

$$\mathbf{Pr}(|X_v - \mathbf{E}(X_v)| > 2t) \leqslant \mathbf{Pr}(|Y_v - \mathbf{E}(Y_v)| > t) + \mathbf{Pr}(|Z_v - \mathbf{E}(Z_v)| > t). \tag{3.1}$$

We will use Talagrand's inequality (Lemma 2.2) to bound both $\mathbf{Pr}(|Y_v - \mathbf{E}(Y_v)| > t)$ and $\mathbf{Pr}(|Z_v - \mathbf{E}(Z_v)| > t)$. Observe that changing the colour of exactly one vertex affects Y_v by at most 1. Also if $Y_v \geqslant s$ then there exists a set of $2s$ vertices whose colours certify $Y_v \geqslant s$: for each of the s colours we may choose two vertices not linked by a digon sharing this colour. Combining Claims 3.1 and 3.2 we obtain $\mathbf{E}(Y_v) \leqslant 6e^7 \mathbf{E}(X_v)$. Hence, Δ being is sufficiently large, we have $t > 96\sqrt{2\mathbf{E}(Y_v)} + 256$, which allows us to apply Talagrand's inequality to Y_v and obtain

$$\mathbf{Pr}(|Y_v - \mathbf{E}(Y_v)| > t) \leqslant 4 \exp\left(-\frac{t^2}{64\mathbf{E}(Y_v) + 16t}\right). \tag{3.2}$$

We now consider Z_v. Changing the colour of exactly one vertex affects Z_v by at most 2. Also if $Z_v \geqslant s$ then there exists a set of $4s$ vertices whose colours certify $Z_v \geqslant s$: for each of the s colours we may choose four vertices a, b, b^-, b^+ sharing this colour, $a \neq b$, such that a and b are not linked by a digon in D, and $b^- b, bb^+ \in A(D)$ (a, b^-, b^+ are not necessarily distinct). Observe that $Z_v \leqslant Y_v$, so in particular $\mathbf{E}(Z_v) \leqslant \mathbf{E}(Y_v) \leqslant 6e^7 \mathbf{E}(X_v)$ by Claims 3.1 and 3.2. Hence, Δ being sufficiently large, we have $t > 192\sqrt{4\mathbf{E}(Z_v)} + 2048$, which allows us to apply Talagrand's inequality to Z_v and obtain

$$\mathbf{Pr}(|Z_v - \mathbf{E}(Z_v)| > t) \leqslant 4 \exp\left(-\frac{t^2}{512\mathbf{E}(Z_v) + 128t}\right). \tag{3.3}$$

Combining (3.1), (3.2), and (3.3) together with $\mathbf{E}(Z_v) \leqslant \mathbf{E}(Y_v) \leqslant 6e^7 \mathbf{E}(X_v)$ we obtain

$$\begin{aligned}\mathbf{Pr}(|X_v - \mathbf{E}(X_v)| > 2t) &\leqslant 8 \exp\left(-\frac{t^2}{3072e^7 \mathbf{E}(X_v) + 128t}\right) = 8 \exp\left(-\frac{\log^2 \Delta}{12288e^7 + 256 \frac{\log \Delta}{\sqrt{\mathbf{E}(X_v)}}}\right) \\ &\leqslant 8 \exp\left(-\frac{\log^2 \Delta}{12288e^7 + 256\sqrt{2e^7} \frac{1}{\sqrt{\log \Delta}}}\right),\end{aligned}$$

where in the last inequality we used $\mathbf{E}(X) \geqslant \frac{B_v}{2e^7 \Delta}$ and $B_v \geqslant \Delta \log^3 \Delta$. As Δ is sufficiently large, we obtain

$$\mathbf{Pr}\left(|X_v - \mathbf{E}(X_v)| > \log \Delta \sqrt{\mathbf{E}(X_v)}\right) < \frac{1}{\Delta^5},$$

which concludes the proof of the claim. ◊

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By Claim 3.1 together with the fact that $B_v \geqslant B \geqslant \Delta \log^3 \Delta$, we have

$$\mathbf{E}(X_v) - \log \Delta \sqrt{\mathbf{E}(X_v)} \geqslant \frac{B_v}{2e^7\Delta}\left(1 - \log\Delta\sqrt{\frac{2e^7\Delta}{B_v}}\right) \geqslant \frac{B}{2e^7\Delta}\left(1 - \sqrt{\frac{2e^7}{\log\Delta}}\right).$$

By choice of ℓ, we thus have $\mathbf{E}(X_v) - \log\Delta\sqrt{\mathbf{E}(X_v)} \geqslant 2\ell\left(1 - \sqrt{\frac{2e^7}{\log\Delta}}\right)$, which is larger than ℓ for Δ large enough. Hence we obtain

$$\mathbf{Pr}(A_v) = \mathbf{Pr}\left(X_v < \ell\right) \leqslant \mathbf{Pr}\left(X_v < \mathbf{E}(X) - \log\Delta\sqrt{\mathbf{E}(X)}\right) \leqslant \mathbf{Pr}\left(|X_v - \mathbf{E}(X)| > \log\Delta\sqrt{\mathbf{E}(X)}\right).$$

Therefore, by Claim 3.3, we have $\mathbf{Pr}(A_v) < \frac{1}{\Delta^5}$. Observe finally that the value of X_v depends only on the colour of vertices at distance at most 2 of v in $\mathrm{UG}(D)$. Hence, each event A_v is mutually independent of all other events A_u except at most $(2\Delta)^4 = d$ of them. For every vertex v, we have $e\mathbf{Pr}(A_v)(d+1) \leqslant e\frac{16\Delta^4+1}{\Delta^5}$, which is smaller than 1 if Δ is large enough. Hence by the Lovász Local Lemma (Lemma 2.1) none of the events A_v occurs with non-zero probability, which concludes the proof. ∎

4 Hitting every maximum biclique with an acyclic set of vertices

Let G be a graph with maximum degree $\Delta(G) \leqslant 3k-1$. Haxell [20] proved that, if the vertices of G can be partitionned into r cliques $C_1, \ldots, C_r$ of size $\geqslant 2k$, then G contains an independent set S of size r. In particular, S intersects each clique C_i. Using this result, Rabern [30] proved that every graph G satisfying $\omega(G) \geqslant \frac{3}{4}(\Delta(G)+1)$ contains an independent set intersecting every maximum clique of G. This was later improved to $\omega(G) > \frac{2}{3}(\Delta(G)+1)$ by King [23], which is best possible.

The goal of this section is to prove Theorem 1.5, which generalises King's result. We first prove the following result, which generalises to digraphs a more precise version of Haxell's result obtained by King [23, Theorem 5].

Theorem 4.1. *Let k be a positive integer and D be a digraph with vertices partitioned into bicliques $C_1, \ldots, C_r$. If for every $i \in [r]$ and every $v \in C_i$, v has at most k out-neighbours outside C_i and at most $|C_i| - k$ in-neighbours outside C_i, then D contains an acyclic set of size r.*

Given a digraph D whose vertices are partitionned into independent sets $V_1, \ldots, V_r$, an *acyclic system of representative* (with respect to $V_1, \ldots, V_r$), or *ASR* for conciseness, is an acyclic set of vertices X such that, for every $i \in [r]$, $|V_i \cap X| = 1$. If we relax this condition to $|V_i \cap X| \leqslant 1$, then X is a *partial ASR*. Theorem 4.1 can be restated in terms of ASR as follows.

Theorem 4.2. *Let k be a positive integer and D be a digraph with vertices partitioned into independent sets $V_1, \ldots, V_r$. If every vertex $v \in V_i$ satisfies $d^+(v) \leqslant k$ and $d^-(v) \leqslant |V_i| - k$, then D admits an ASR.*

Proof. For any subset of indices $I \subseteq [r]$ we denote by V_I the set of vertices $\bigcup_{i \in I} V_i$. Given a set of vertices $X \subseteq V(D)$, we denote by $\iota(X)$ the set of indices $\{i \in [r] \mid V_i \cap X \neq \varnothing\}$. For a single vertex v, we denote by $\iota(v)$ the integer i such that $v \in V_i$.

We proceed by induction on $r \geqslant 1$. The result being trivial when $r = 1$, we assume $r \geqslant 2$. Let us fix any vertex $x_1 \in V_r$, we will show as a stronger result that there exists an ASR containing x_1. Assume for a contradiction that no such ASR exists.

Let $I \subseteq [r-1]$ and $X, Y \subseteq (V_I \cup \{x_1\})$. We say that (I, X, Y) is a *good triplet* if each of the following properties holds:

(1) X and Y are two disjoint acyclic sets with $x_1 \in X$;

(2) Y is an ASR of $D[V_I]$ (with respect to $(V_i)_{i \in I}$);

(3) every vertex in Y has exactly one in-neighbour in X and every vertex in X has at least one out-neighbour in Y, in particular $|X| \leqslant |Y|$; and

(4) every vertex $v \in V_I \cup \{x_i\}$ has an in-neighbour in X or an out-neighbour in Y.

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We first justify that there exists no good triplet in D.

Claim 4.1. *D admits no good triplet (I, X, Y).*

Proof of claim. Assume for a contradiction that (I, X, Y) is a good triplet of D. The assumption on the degrees in D implies in particular $d^+(x) \leqslant k$ for every vertex $x \in X$ and $d^-(y) \leqslant |V_{\iota(y)}| - k$ for every vertex $y \in Y$. Observe that $|X| \leqslant |Y| = |I|$ by (3) and (2). We thus have

$$\sum_{x \in X} d^+(x) + \sum_{y \in Y} d^-(y) \leqslant k|I| + \sum_{i \in I}(|V_i| - k) = |V_I|.$$

On the other hand, we know by (4) that $(V_I \cup \{x_1\}) \subseteq (N^+(X) \cup N^-(Y))$. This implies

$$\sum_{x \in X} d^+(x) + \sum_{y \in Y} d^-(y) \geqslant |N^+(X) \cup N^-(Y)| \geqslant |V_I| + 1,$$

yielding the contradiction. ◊

Our goal is now to show that D admits an infinite sequence $(Y_i)_{i \geqslant 1}$ of partial ASR, such that each Y_i is a proper subset of Y_{i+1}. This will yield the contradiction since such a sequence has length at most r.

Let R_1 be an ASR of $D[V_{[r-1]}]$ (with respect to $V_1, \ldots, V_{r-1}$) for which the size of $Y_1 = Y_1' = N^+(x_1) \cap R_1$ is minimum. The existence of R_1 is guaranteed by induction on r. Observe that $Y_1 \neq \varnothing$ for otherwise $R_1 \cup \{x_1\}$ is an ASR of D containing x_1. We let X_1 be the singleton $\{x_1\}$. We will recursively build $(R_{i+1}, X_{i+1}, Y_{i+1}, Y_{i+1}')$ from (R_i, X_i, Y_i, Y_i') in such a way that, for every $i \geqslant 1$, each of the following properties holds.

(a) $X_i = \{x_1, \ldots, x_i\}$ is acyclic. If $i > 1$ then $x_i \in V_{\iota(Y_{i-1})}$.

(b) R_i is an ASR of $D[V_{[r-1]}]$ such that for every $1 \leqslant j \leqslant i-1$, $R_i \cap N^+(x_j) = Y_j'$. Subject to this property, $Y_i' = N^+(x_i) \cap R_i$ has minimum size.

(c) $Y_i = \bigsqcup_{1 \leqslant j \leqslant i} Y_j'$ and $Y_i' \neq \varnothing$.

Observe that (R_1, X_1, Y_1, Y_1') satisfies (a), (b), and (c). Also, the existence of such an infinite sequence $(R_i, X_i, Y_i, Y_i')_{i \geqslant 1}$ implies the desired infinite sequence $(Y_i)_{i \geqslant 1}$ of partial ASR, yielding the contradiction.

Hence we assume that for some $i \geqslant 1$, there exists $(R_i, X_i, Y_i, Y_i')_{1 \leqslant j \leqslant i}$ such that, for each $j \in [i]$, (a), (b), and (c) hold. We construct $(R_{i+1}, X_{i+1}, Y_{i+1}, Y_{i+1}')$ as follows.

- We let x_{i+1} be a vertex in $V_{\iota(Y_i)} - (X_i \cup Y_i)$ such that

  $$N^-(x_{i+1}) \cap X_i = \varnothing \qquad \text{and} \qquad N^+(x_{i+1}) \cap Y_i = \varnothing. \tag{4.4}$$

  If such a vertex x_{i+1} exists then $X_{i+1} = \{x_1, \ldots, x_{i+1}\}$ is acyclic as x_{i+1} is a source in $D[X_{i+1}]$ and $X_i = \{x_1, \ldots, x_i\}$ is acyclic by recurrence, so (a) holds.

  Let us justify the existence of x_{i+1}. If no such vertex exists, then X_i and Y_i are two disjoint acyclic sets such that $x_1 \in X_i$, with Y_i being an ASR of $V_{\iota(Y_i)}$. Combining conditions (b) and (c) we obtain that $(N^+(x_1), \ldots, N^+(x_i))$ partitions Y_i into non-empty parts, so in particular every vertex in Y_i has exactly one in-neighbour in X_i and every vertex in X_i has at least one out-neighbour in Y_i. Hence, if no vertex in $V_{\iota(Y_i)} - (X_i \cup Y_i)$ satisfies (4.4), we deduce that $(\iota(Y_i), X_i, Y_i)$ is a good triplet, a contradiction to Claim 4.1.

- We let R_{i+1} be an ASR of $D[V_{[r-1]}]$ such that for every $1 \leqslant j \leqslant i$, $R_{i+1} \cap N^+(x_j) = Y_j'$, and subject to this property, $Y_{i+1}' = N^+(x_{i+1}) \cap R_{i+1}$ has minimum size. The existence of R_{i+1} is guaranteed as R_i is such an ASR. Hence (b) is maintained.

- It remains to show that (c) is maintained, that is $Y_{i+1}' \neq \varnothing$ and $Y_{i+1}' \cap Y_i = \varnothing$. First we have $Y_{i+1}' \cap Y_i = \varnothing$ by (4.4). Now assume for a contradiction that $Y_{i+1}' = \varnothing$, which together with (4.4) implies that x_{i+1} has no out-neighbour in R_{i+1}.

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By recurrence we know that $(Y'_1, \dots, Y'_i)$ partitions Y_i. As x_{i+1} belongs to $V_{\iota(Y_i)} - (X_i \cup Y_i)$ there exists a unique $j \leqslant i$ such that $\iota(x_{i+1}) \in \iota(Y'_j)$. As Y'_j is a partial ASR, there exists a unique $y_j \in Y'_j \cap V_{\iota(x_{i+1})}$. Consider the set $\tilde{R}_j$ defined as

$$\tilde{R}_j = R_{i+1} - \{y_j\} \cup \{x_{i+1}\}.$$

Observe that $\tilde{R}_j$ is necessarily acyclic as x_{i+1} has no out-neighbour in R_{i+1} and R_{i+1} is an ASR of $D[V_{[r-1]}]$. Also, since $y_j \in V_{\iota(x_{i+1})}$, we obtain that $\tilde{R}_j$ is an ASR of $D[V_{[r-1]}]$.

Since x_{i+1} has no in-neighbour in X_i by (4.4), for every $1 \leqslant \ell < j$, we have $\tilde{R}_j \cap N^+(x_\ell) = R_{i+1} \cap N^+(x_\ell) = Y'_\ell$. We finally have $\tilde{R}_j \cap N^+(x_j) = (R_{i+1} \cap N^+(x_j)) - \{y_j\} = Y'_j - \{y_j\}$, a contradiction to the choice of R_j.

Hence (a), (b), and (c) are maintained, and the result follows. ∎

With Theorem 4.1 in hand, we are now going to prove Theorem 1.5. We need one additional intermediate result due to Hajnal. Let G be a (di)graph. We denote by $\mathcal{C}(G)$ the set of all maximum (bi)cliques of G. Given a collection $\mathcal{C} \subseteq \mathcal{C}(G)$ of maximum (bi)cliques of G, we denote by $\mathrm{IG}(\mathcal{C})$ the intersection graph of $\mathcal{C}$, *i.e.* the undirected graph with vertex-set $\mathcal{C}$ which contains an edge between C_1 and C_2 whenever $C_1 \cap C_2 \neq \varnothing$.

Lemma 4.1. (Hajnal [17]) *Let G be a graph and $\mathcal{C}$ be a collection of maximum cliques of G, then*

$$\left|\bigcap \mathcal{C}\right| + \left|\bigcup \mathcal{C}\right| \geqslant 2\omega(G).$$

We are now ready to prove Theorem 1.5, that we first restate here for the reader's convenience.

Theorem 4.3. *Let D be a digraph with $\overleftrightarrow{\omega}(D) > \frac{2}{3}(\Delta_{\max}(D) + 1)$. Then D has an acyclic set of vertices I such that $\overleftrightarrow{\omega}(D - I) = \overleftrightarrow{\omega}(D) - 1$.*

Proof. Let $\mathcal{H}$ be $\mathrm{IG}(\mathcal{C}(D))$. We denote by $\mathcal{H}_1, \dots, \mathcal{H}_r$ the connected components of $\mathcal{H}$. For each $i \in [r]$, let $X_i = \bigcap_{C \in V(\mathcal{H}_i)}$ and $Y_i = \bigcup_{C \in V(\mathcal{H}_i)}$.

We first show that, for every $i \in [r]$, $X_i \neq \varnothing$. If this is not the case, then there exist $C_1, C_2, C_3 \in \mathcal{C}(D)$ such that C_2 intersects both C_1 and C_3, and $C_1 \cap C_3 = \varnothing$. Since C_1 and C_3 are disjoint, we can assume without loss of generality that $|C_1 \cap C_2| \leqslant \frac{1}{2}\overleftrightarrow{\omega}(D)$. Therefore $|C_1 \cup C_2| \geqslant \frac{3}{2}\overleftrightarrow{\omega}(D) > \Delta + 1$, which implies that vertices in $C_1 \cap C_2$ have out-degree larger than Δ, a contradiction.

Hence $X_i = \bigcap_{C \in V(\mathcal{H}_i)} C \neq \varnothing$ holds for every $i \in [r]$. Let D' be the subdigraph of D induced by $\bigcup_{i \in [r]} X_i$. Let us show that we can apply Theorem 4.1 to D', the vertices of D' being partitioned into bicliques $X_1, \dots, X_r$, with $k = \left\lfloor \frac{1}{3}(\Delta + 1) \right\rfloor$. Let i be any index of $[r]$ and v_i be any vertex in X_i. Observe that v_i has at least $\overleftrightarrow{\omega}(D) - 1$ out-neighbours in $Y_i = \bigcup_{C \in V(\mathcal{H}_i)} C$, so the out-degree of v_i outside X_i in D' is at most $\Delta - \overleftrightarrow{\omega}(D) + 1 \leqslant \frac{1}{3}(\Delta + 1)$. Since it is an integer, the out-degree of v_i outside X_i in D' is thus at most k. On the other hand, v_i has at most $\Delta - |Y_i| + 1$ in-neighbours outside X_i in D'. By Lemma 4.1 applied to $S(D)$ we have $|Y_i| \geqslant 2\overleftrightarrow{\omega}(D) - |X_i| > \frac{4}{3}(\Delta + 1) - |X_i|$. Hence v_i has at most $|X_i| - \frac{1}{3}(\Delta + 1) \leqslant |X_i| - k$ in-neighbours outside X_i in D'.

Therefore, we can apply Theorem 4.1 to D', and conclude that D' admits an acyclic set I such that $\overleftrightarrow{\omega}(D' - I) = \overleftrightarrow{\omega}(D) - 1$. By construction of D', I is also an acyclic set of D such that $\overleftrightarrow{\omega}(D - I) = \overleftrightarrow{\omega}(D) - 1$, a contradiction. ∎

5 Handling vertices with a dense neighbourhood

In the short proof of Theorem 1.1 due to King and Reed [24], Theorem 3.1 handles the case of graphs in which every vertex has a sparse neighbourhood. The following result handles the remaining case, that is graphs admitting a vertex with a dense neighbourhood.

Theorem 5.1. (King and Reed [24]) *Let a and ε be constants satisfying $0 < \varepsilon < \frac{1}{6} - 2\sqrt{a}$. Let G be a graph with maximum degree Δ satisfying $\omega(G) \leqslant \frac{2}{3}(\Delta + 1)$ and let v be such that $m(v) > (1 - a)\binom{n}{2}$. Then*

$$\chi(G) \leqslant \max\left(\chi(G - v), (1 - \varepsilon)(\Delta + 1)\right).$$

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The goal of this section is to prove Theorem 1.6, which is the immediate analogue of Theorem 5.1 for digraphs. We first recall it here for convenience.

THEOREM 5.2. *Let a and ε be constants satisfying $0 < \varepsilon \leqslant \frac{1}{6} - 4\sqrt{a}$. There exists $\Delta(a) \geqslant 1$ such that the following holds. Let D be a digraph with maximum degree $\Delta_{\max}(D) = \Delta \geqslant \Delta(a)$ satisfying $\overleftrightarrow{\omega}(D) \leqslant \frac{2}{3}(\Delta+1)$ and let $v \in V(D)$ be such that $\max(m^-(v), m^+(v)) > (1-a)\Delta(\Delta-1)$. Then*

$$\vec{\chi}(D) \leqslant \max\left(\vec{\chi}(D-v), (1-\varepsilon)(\Delta+1)\right).$$

We will use the following classical result due to Erdős, Rubin, and Taylor [14].

LEMMA 5.1. (ERDŐS *et al.* [14]) *Let G be a graph obtained from K_n by removing a matching of size p. Then G is $(n-p)$-choosable.*

The following is an immediate consequence of Lemma 5.1. A *matching* in a digraph is a set of pairwise disjoint arcs.

LEMMA 5.2. *Let D be a digraph obtained from $\overleftrightarrow{K_n}$ by removing a matching of size p. Then D is $(n-p)$-dichoosable.*

Proof. Let L be any $(n-p)$-list assignment of D. By Lemma 5.1, there exists an L-colouring α of $S(D)$, because $S(D)$ is obtained from K_n by removing a matching of size p. As D is obtained from $\overleftrightarrow{S(D)}$ by adding an acyclic set of arcs, α must be an L-dicolouring of D. ∎

We are now ready to prove Theorem 5.2.

Proof. We take $\Delta(a) = \max\left(\frac{1-a}{\sqrt{a}-a}, \frac{1-a}{a}\right)$. By directional duality we assume without loss of generality that $m^+(v) \geqslant (1-a)\Delta(\Delta-1)$ and let $N = N^+(v) \cup \{v\}$ and $\bar{N} = V(D) - N$. We assume that $|N| = \Delta+1$ for otherwise we add new vertices dominated by v. We partition N into three parts as follows:

$$N_1 = \left\{u \in N \mid |N^+(u) \cap \bar{N}| \geqslant \frac{\Delta}{2}\right\},$$
$$N_2 = \left\{u \in (N - N_1) \mid |N^+(u) \cap (\bar{N} \cup N_1)| \geqslant 2\sqrt{a}\Delta\right\}, \text{and}$$
$$N_3 = N - (N_1 \cup N_2).$$

Observe first that

$$\Delta^2 \geqslant \sum_{u \in N - \{v\}} d^+(u) = m^+(v) + |A(N, \bar{N})|.$$

Since $m^+(v) > (1-a)\Delta(\Delta-1)$, we obtain

$$|A(N, \bar{N})| < \Delta^2 - (1-a)\Delta(\Delta-1) = a\Delta^2 + (1-a)\Delta.$$

By definition of N_1, this implies

$$\frac{\Delta}{2}|N_1| < a\Delta^2 + (1-a)\Delta.$$

Using the fact that $\Delta \geqslant \Delta(a) \geqslant \frac{1-a}{\sqrt{a}-a}$, we thus have $|N_1| < 2\sqrt{a}\Delta$. By definition, $v \notin N_1$ as $N^+(v) \subseteq N$. We also have $v \notin N_2$ as $N^+(v) \cap (\bar{N} \cup N_1) = N_1$, and $|N_1| < 2\sqrt{a}\Delta$. We will now bound the size of N_2. Recall that every vertex $u \in N_1$ satisfies $2|N^+(u) \cap \bar{N}| \geqslant \Delta \geqslant d^-(u)$. We thus have the following.

$$\begin{aligned}
|A(N_2 \cup N_3, \bar{N} \cup N_1)| &= |A(N, \bar{N})| + |A(N_2 \cup N_3, N_1)| - |A(N_1, \bar{N})| \\
&\leqslant |A(N, \bar{N})| + \sum_{u \in N_1} d^-(u) - \sum_{u \in N_1} |N^+(u) \cap \bar{N}| \\
&\leqslant |A(N, \bar{N})| + \sum_{u \in N_1} |N^+(u) \cap \bar{N}| \\
&= |A(N, \bar{N})| + |A(N_1, \bar{N})| \\
&\leqslant 2|A(N, \bar{N})| \\
&< 2a\Delta^2 + 2(1-a)\Delta.
\end{aligned}$$

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By definition of N_2, this implies

$$2\sqrt{a}\Delta|N_2| < 2a\Delta^2 + 2(1-a)\Delta,$$

which together with $\Delta \geqslant \Delta(a) \geqslant \frac{1-a}{a}$ implies $|N_2| < 2\sqrt{a}\Delta$. Let $k = \max(\vec{\chi}(D-v), \lfloor(1-\varepsilon)(\Delta+1)\rfloor)$. Since $v \in N_3$, $D - N_3$ is k-dicolourable, as it is a subdigraph of $D - v$. Let α be such a k-dicolouring and L be the list assignment of $D[N_3]$ defined as follows

$$L(u) = [k] - \alpha(N^+(u) - N_3).$$

We are now going to prove that $D[N_3]$ is L-dicolourable, implying that D has dichromatic number at most k (it is straightforward to check that any L-dicolouring of $D[N_3]$ extends α into a dicolouring of D). First note that, by construction, every vertex $u \in N_3$ satisfies $|N^+(u) - N_3| < 2\sqrt{a}\Delta + |N_2| < 4\sqrt{a}\Delta$. Hence every vertex $u \in N_3$ satisfies

$$|L(u)| > k - 4\sqrt{a}(\Delta+1) \geqslant (1-\varepsilon)(\Delta+1) - 1 - 4\sqrt{a}(\Delta+1) \geqslant \frac{5}{6}(\Delta+1) - 1,$$

As $|L(u)|$ is an integer, we thus have $|L(u)| \geqslant \lfloor\frac{5}{6}(\Delta+1)\rfloor$. Let M be a maximum matching in $\bar{D}[N_3]$ (the complementary of $D[N_3]$). Let X be the vertices of N_3 that are not covered M. By definition, X is a biclique of D so $|X| \leqslant \frac{2}{3}(\Delta+1)$. Note that $|N_3| = 2|M| + |X|$. If $|M| \leqslant \frac{1}{6}(\Delta+1)$ then $|M| + |X| \leqslant \frac{5}{6}(\Delta+1)$. If $|M| \geqslant \frac{1}{6}(\Delta+1)$ then $|M| + |X| = |N_3| - |M| \leqslant \frac{5}{6}(\Delta+1)$.

In both cases, we obtain $|M|+|X| \leqslant \frac{5}{6}(\Delta+1)$. Since $D[N_3]$ is a subdigraph of a digraph obtained from $\overleftrightarrow{K}_{|N_3|}$ by removing a matching of size $|M|$, $D[N_3]$ is $(|M|+|X|)$-dichoosable by Lemma 5.2. Therefore $D[N_3]$ must be $\lfloor\frac{5}{6}(\Delta+1)\rfloor$-dichoosable, so in particular it is L-dicolourable, implying the result. ∎

6 Proof of the main result

This section is devoted to the proof of our main result, namely Theorem 1.3. We need the following intermediate result proved by the second author [29]. Recall that an oriented graph is a digraph without any digon.

THEOREM 6.1. (PICASARRI-ARRIETA [29]) *Let D be an oriented graph, then $\vec{\chi}(D) \leqslant \max(2, \Delta_{\min}(D))$.*

We are now going to prove Theorem 1.3. Let us first restate it.

THEOREM 6.2. *There exists $\varepsilon > 0$ such that every digraph D satisfies*

$$\vec{\chi}(D) \leqslant \lceil(1-\varepsilon)(\tilde{\Delta}(D)+1) + \varepsilon\overleftrightarrow{\omega}(D)\rceil.$$

Proof. Let Δ_1 be as in Corollary 3.1 and, for a real number $a > 0$, let $\Delta(a)$ be as in Theorem 5.2. We define a, ε_0, Δ_2, ε, and γ_ε as follows:

- $a = \frac{1}{600}$,
- $\varepsilon_0 = \frac{1}{6} - 4\sqrt{a} > 0$,
- $\Delta_2 = \min\left\{\Delta \in \mathbb{N} \mid a > \frac{\log^3\Delta}{\Delta-1}\right\}$,
- $\varepsilon = \min\left(\frac{1}{\Delta_1}, \frac{1}{\Delta_2}, \frac{1}{\Delta(a)}, \frac{1}{3}\varepsilon_0, \frac{a}{16e^7}\right)$, and
- $\gamma_\varepsilon = \frac{\varepsilon}{1-\varepsilon} \leqslant 2\varepsilon$.

We claim that the statement holds for this specific value of ε. Assume this is not the case, and let D be a counterexample with minimum order. For the sake of conciseness let us denote $\tilde{\Delta}(D)$ by $\tilde{\Delta}$, $\Delta_{\max}(D)$ by $\Delta_{\max}$, $\overleftrightarrow{\omega}(D)$ by $\overleftrightarrow{\omega}$, and let $k = \lceil(1-\varepsilon)(\tilde{\Delta}+1) + \varepsilon\overleftrightarrow{\omega}\rceil$. As D is a minimum counter example, it satisfies $\vec{\chi}(D) > k$ and for every vertex $v \in V$ we have $\vec{\chi}(D-v) \leqslant k$. This directly implies that every vertex v of D satisfies $\min(d^-(v), d^+(v)) \geqslant k$, for otherwise a k-dicolouring of D can be obtained from any k-dicolouring of $D - v$ by choosing for v a colour that is not appearing either in its in-neighbourhood or in its out-neighbourhood. In particular, this implies that $\tilde{\Delta}$ must be close to $\Delta_{\max}$, as we justify in the following claim.

CLAIM 6.1. $\Delta_{\max} \leqslant (1+\gamma_\varepsilon)\tilde{\Delta}$.

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Proof of claim. Assume this is not the case, so there exists a vertex v such that $\max(d^+(v), d^-(v)) > (1+\gamma_\varepsilon)\tilde{\Delta}$. By directional duality, we assume without loss of generality that $d^+(v) \geqslant d^-(v)$. Therefore, we have $\tilde{\Delta}^2 \geqslant d^+(v) \cdot d^-(v) > (1+\gamma_\varepsilon)\tilde{\Delta} \cdot d^-(v)$, which implies $d^-(v) < \frac{1}{1+\gamma_\varepsilon}\tilde{\Delta} = (1-\varepsilon)\tilde{\Delta} \leqslant k$, a contradiction. ◊

With Theorem 1.5 in hands, we can also justify that there exists a significant gap between $\overleftrightarrow{\omega}$ and $\Delta_{\max}$.

Claim 6.2. $\overleftrightarrow{\omega} \leqslant \frac{2}{3}(\Delta_{\max}+1)$.

Proof of claim. Assume for a contradiction that $\overleftrightarrow{\omega} > \frac{2}{3}(\Delta_{\max}+1)$. By Theorem 1.5, there exists an acyclic set of vertices $I \subseteq V(D)$ which intersects every maximum biclique of D. Among all such sets I, we choose one with maximum size. In particular, this implies that every vertex in $V(D) - I$ has at least one in-neighbour and one out-neighbour in I. Let D' be $D - I$. As I is acyclic and intersects every maximum clique of D, we have $\overleftrightarrow{\omega}(D') = \overleftrightarrow{\omega} - 1$. The following shows that we also have $\tilde{\Delta}(D') \leqslant \tilde{\Delta} - 1$.

$$\begin{aligned}
\tilde{\Delta}(D') &= \max\left\{\sqrt{d^+_{D'}(v) \cdot d^-_{D'}(v)} \mid v \in V(D')\right\} \\
&\leqslant \max\left\{\sqrt{(d^+_D(v)-1)(d^-_D(v)-1)} \mid v \in V(D')\right\} \\
&\leqslant \max\left\{\sqrt{d^+_D(v)d^-_D(v)} - 1 \mid v \in V(D')\right\} \\
&\leqslant \tilde{\Delta} - 1,
\end{aligned}$$

where in the first inequality we used the maximality of I and in the second inequality we used that $\sqrt{(p-1)(q-1)} \leqslant \sqrt{pq}-1$ holds for every pair of real numbers $p, q \geqslant 1$. We briefly justify it here for completeness.

$$\begin{aligned}
& 0 \leqslant (\sqrt{p}-\sqrt{q})^2 \\
\Rightarrow \quad & -p-q \leqslant -2\sqrt{pq} \\
\Rightarrow \quad & pq-p-q+1 \leqslant pq - 2\sqrt{pq}+1 \\
\Rightarrow \quad & (p-1)(q-1) \leqslant (\sqrt{pq}-1)^2 \\
\Rightarrow \quad & \sqrt{(p-1)(q-1)} \leqslant \sqrt{pq}-1
\end{aligned}$$

Since I is acyclic, we have $\vec{\chi}(D) \leqslant \vec{\chi}(D') + 1$, as any dicolouring of D' can be extended to D by colouring I with an additional colour. Together with the minimality of D we obtain

$$\begin{aligned}
\vec{\chi}(D) &\leqslant \left\lceil (1-\varepsilon)(\tilde{\Delta}(D')+1) + \varepsilon\overleftrightarrow{\omega}(D') \right\rceil + 1 \\
&\leqslant \left\lceil (1-\varepsilon)\tilde{\Delta} + \varepsilon(\overleftrightarrow{\omega}-1) \right\rceil + 1 \\
&= \left\lceil (1-\varepsilon)(\tilde{\Delta}+1) - (1-\varepsilon) + \varepsilon\overleftrightarrow{\omega} - \varepsilon + 1 \right\rceil = k,
\end{aligned}$$

a contradiction. ◊

We finally claim that $\tilde{\Delta}$ is large enough.

Claim 6.3. $\Delta_{\max} \geqslant \tilde{\Delta} \geqslant \max(\Delta_1, \Delta_2, \Delta(a))$.

Proof of claim. It is clear from the definitions that $\Delta_{\max} \geqslant \tilde{\Delta}$. Assume for a contradiction that $\tilde{\Delta} < \max(\Delta_1, \Delta_2, \Delta(a))$. Since $\varepsilon \leqslant \min\left(\frac{1}{\Delta_1}, \frac{1}{\Delta_2}, \frac{1}{\Delta(a)}\right)$ we have $\varepsilon < \frac{1}{\tilde{\Delta}}$, which implies:

$$k = \left\lceil \tilde{\Delta} + 1 + \varepsilon(\overleftrightarrow{\omega} - \tilde{\Delta} - 1) \right\rceil \geqslant \left\lceil \tilde{\Delta} + \frac{1}{\tilde{\Delta}}(\overleftrightarrow{\omega} - 1) \right\rceil.$$

Using the facts that $\vec{\chi}(D) > k$ and that $\vec{\chi}(D) \leqslant \left\lfloor \tilde{\Delta} + 1 \right\rfloor$, we deduce

$$\left\lfloor \tilde{\Delta} + 1 \right\rfloor > \left\lceil \tilde{\Delta} + \frac{1}{\tilde{\Delta}}(\overleftrightarrow{\omega} - 1) \right\rceil.$$

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This is a contradiction unless $\overleftrightarrow{\omega} = 1$ and $\tilde{\Delta}$ is an integer. Henceforth we assume that $\overleftrightarrow{\omega} = 1$, *i.e.* D is an oriented graph, and $\tilde{\Delta} \in \mathbb{N}$. In particular, we have $k \geqslant \tilde{\Delta}$. As D is an oriented graph, by Theorem 6.1 we must have $\Delta_{\min}(D) \leqslant 1$, for otherwise $\vec{\chi}(D) \leqslant \Delta_{\min}(D) \leqslant \tilde{\Delta} \leqslant k$, a contradiction. Hence we have $\tilde{\Delta}(D) \leqslant 1$, for otherwise $\vec{\chi}(D) \leqslant 2 \leqslant \tilde{\Delta} \leqslant k$, a contradiction. If $\tilde{\Delta} = 1$ then $k = \lceil 2 - \varepsilon \rceil = 2$. If $\tilde{\Delta} = 0$ then $k = 1$. Both cases yield a contradiction, which concludes the proof of the claim. ◊

We now distinguish two cases, depending on the sparseness of D.

Case 1: *There exists a vertex v such that* $\max(m^-(v), m^+(v)) \geqslant (1-a)\Delta(\Delta-1)$.

By Claim 6.2 we have $\overleftrightarrow{\omega} \leqslant \frac{2}{3}(\Delta_{\max}+1)$. As $\varepsilon_0 = \frac{1}{6} - 4\sqrt{a}$, by Theorem 5.2 we have

$$\vec{\chi}(D) \leqslant \max(\vec{\chi}(D-v), (1-\varepsilon_0)(\Delta_{\max}+1)).$$

Recall that $\vec{\chi}(D-v) \leqslant k$ as D is a minimum counterexample. Since $\vec{\chi}(D) > k$, we thus necessarily have

$$\vec{\chi}(D) \leqslant (1-\varepsilon_0)(\Delta_{\max}+1) \leqslant (1-\varepsilon_0)(1+\gamma_\varepsilon)(\tilde{\Delta}+1) \leqslant (1-\varepsilon)(\tilde{\Delta}+1),$$

where in the second inequality we used Claim 6.1, and in the last inequality we used $\varepsilon \leqslant \varepsilon_0 - \gamma_\varepsilon$, which holds because $\gamma_\varepsilon \leqslant 2\varepsilon$ and $\varepsilon \leqslant \frac{1}{3}\varepsilon_0$. This yields the contradiction as $(1-\varepsilon)(\tilde{\Delta}+1) \leqslant k$.

Case 2: *For every vertex $v \in V(D)$ we have* $\max(m^-(v), m^+(v)) \leqslant (1-a)\Delta(\Delta-1)$.

A fortiori every vertex satisfies $\min(m^-(v), m^+(v)) \leqslant (1-a)\Delta(\Delta-1)$. By Claim 6.3 we know that $\Delta_{\max} \geqslant \max(\Delta_1, \Delta_2)$. Hence, by definition of Δ_2, we have $a > \frac{\log^3 \Delta}{\Delta - 1}$. We can thus apply Corollary 3.1, which together with Claim 6.1 gives

$$\vec{\chi}(D) \leqslant \left(1 - \frac{a}{5e^7}\right)(\Delta_{\max}+1) \leqslant \left(1 - \frac{a}{5e^7}\right)(1+\gamma_\varepsilon)(\tilde{\Delta}+1) \leqslant (1-\varepsilon)(\tilde{\Delta}+1).$$

where in the last inequality we used $\varepsilon \leqslant \frac{a}{5e^7} - \gamma_\varepsilon$, which holds because $\gamma_\varepsilon \leqslant 2\varepsilon$ and $\varepsilon \leqslant \frac{a}{15e^7}$. This contradicts $\vec{\chi}(D) > k$.

Since one of the two cases above occurs, and because each of them yields a contradiction, the result follows. ∎

7 Further research: analogues for $\Delta_{\min}$

In this work we proved an analogue of Reed's result for digraphs by taking the maximum degree of a digraph to be $\tilde{\Delta}$. A natural question then arises: can we strengthen this result by replacing $\tilde{\Delta}$ with $\Delta_{\min}$? The answer to this question turns out to be negative. Moreover, for every fixed real numbers $\varepsilon, \varepsilon'$ with $0 < \varepsilon' < 2\varepsilon$, there exists a digraph D with

$$\vec{\chi}(D) > \lceil (1-\varepsilon)(\Delta_{\min}(D)+1) + \varepsilon' \overleftrightarrow{\omega}(D) \rceil.$$

This is because the k-Dicolourability problem is NP-complete for every fixed $k \geqslant 2$, even when restricted to digraphs D satisfying $\Delta_{\min}(D) = k$ and $\overleftrightarrow{\omega}(D) = \lceil \frac{k+1}{2} \rceil$, as shown in [29]. In particular, the reduction shows that there exist such digraphs satisfying $\vec{\chi}(D) = k+1$. Hence, if we choose k to be at least $\frac{2\varepsilon'-2\varepsilon}{2\varepsilon-\varepsilon'}$, we deduce the existence of digraphs D with $\Delta_{\min}(D) = k$, $\overleftrightarrow{\omega}(D) = \lceil \frac{k+1}{2} \rceil$, and $\vec{\chi}(D) = k+1$, while $\lceil (1-\varepsilon)(k+1) + \varepsilon' \lceil \frac{k+1}{2} \rceil \rceil \leqslant k$.

However, we believe that an analogue of our result holds with $\Delta_{\min}$ instead of $\tilde{\Delta}$ if we also adapt the notion of clique number. Given a digraph D, a *directed clique* of D is a set of vertices X that can be partitioned into (X_1, X_2) such that both $D[X_1]$ and $D[X_2]$ are complete digraphs, and D contains every possible arc uv with $u \in X_1$ and $v \in X_2$. The *directed clique number* of D, denoted by $\vec{\omega}(D)$, is the size of a largest directed clique of D. We propose the following conjecture.

Conjecture 7.1. *There exists $\varepsilon > 0$ such that every digraph D satisfies*

$$\vec{\chi}(D) \leqslant \lceil (1-\varepsilon)\Delta_{\min}(D) + \varepsilon \vec{\omega}(D) \rceil.$$

In particular, the conjecture above implies the following slightly weaker one, which if true shows that the condition $\varepsilon' < 2\varepsilon$ mentioned above is indeed necessary.

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Conjecture 7.2. *There exists $\varepsilon > 0$ such that every digraph D satisfies*

$$\vec{\chi}(D) \leqslant \left\lceil (1-\varepsilon)\Delta_{\min}(D) + 2\varepsilon\overleftrightarrow{\omega}(D) \right\rceil.$$

We are inclined to believe that these two conjectures are true even for $\varepsilon = \frac{1}{2}$. One direction for proving Conjecture 7.1 would be to prove an analogue of Theorem 1.3 using the maximum out-degree Δ^+ instead of $\tilde{\Delta}$, that is to prove the existence of $\varepsilon > 0$ such that every digraph D satisfies $\vec{\chi}(D) \leqslant \lceil (1-\varepsilon)(\Delta^+(D)+1) + \varepsilon\overleftrightarrow{\omega}(D) \rceil$, where $\Delta^+(D) = \max\{d^+(v) \mid v \in V(D)\}$. It is of interest that already the case of oriented graphs is open.

Problem 7.1. *Show the existence of $\varepsilon > 0$ such that every oriented graph D with sufficiently large maximum out-degree $\Delta^+(D)$ satisfies $\vec{\chi}(D) \leqslant (1-\varepsilon)\Delta^+(D)$.*

The hardness of this problem mainly comes from the fact that the maximum in-degree of a digraph can be arbitrarily larger than its maximum out-degree. Hence, the proof of Theorem 3.2 cannot be directly adapted as we cannot bound the degree of dependence of bad events when applying the Lovász Local Lemma.

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$(\Delta - 1)$-dicolouring of digraphs

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arXiv:2507.10266v1 [math.CO] 14 Jul 2025

Abstract

In 1977, Borodin and Kostochka conjectured that every graph with maximum degree $\Delta \geqslant 9$ is $(\Delta - 1)$-colourable, unless it contains a clique of size Δ. In 1999, Reed confirmed the conjecture when $\Delta \geqslant 10^{14}$. We propose different generalisations of this conjecture for digraphs, and prove the analogue of Reed's result for each of them.

The *dichromatic number* of a digraph is the minimum number of colours needed to partition it into acyclic induced subdigraphs. A *biclique* is a set of vertices that are pairwise linked with two arcs in opposite directions. For a digraph D, we let $\tilde{\Delta}(D) = \max_{v \in V(D)} \sqrt{d^+(v) \cdot d^-(v)}$.

Let $\Delta \geqslant 9$ be any integer and D be any digraph with $\tilde{\Delta}(D) \leqslant \Delta$. We conjecture that D has dichromatic number at most $\Delta - 1$, unless either (i) D contains a biclique of size Δ, or (ii) D contains a biclique K of size $\Delta - 2$, a directed 3-cycle $\vec{C_3}$ disjoint from K, and all possible arcs in both directions between $\vec{C_3}$ and K. If true, this implies the conjecture of Borodin and Kostochka. We prove our conjecture when Δ is large enough, thereby generalising the result of Reed.

We also conjecture that every digraph D with maximum out-degree $\Delta^+ \geqslant 9$ has dichromatic number at most $\Delta^+ - 1$, unless D contains one of the two obstructions described above. Again, we show the conjecture when Δ^+ is large enough, yielding a second generalisation of Reed's result. We finally give a sufficient condition for a digraph D to have dichromatic number at most $\Delta_{\min}(D) - 1$, assuming that $\Delta_{\min}(D) = \max_{v \in V(D)} \min(d^+(v), d^-(v))$ is large enough. In particular, this holds when D has no biclique of size $\lceil \frac{1}{2}(\Delta_{\min}(D) - 1) \rceil$, and when the underlying graph of D has no clique of size $\Delta_{\min}(D)$, thus yielding a third independent generalisation of Reed's result. We further give a hardness result witnessing that our sufficient condition is best possible.

To obtain these new upper bounds on the dichromatic number, we prove a dense decomposition lemma for digraphs having large maximum degree, which generalises to the directed setting the so-called dense decomposition of graphs due to Molloy and Reed. We believe this may be of independent interest, especially as a tool in various applications.

1 Introduction

The relationships between the chromatic number $\chi(G)$, the clique number $\omega(G)$, and the maximum degree $\Delta(G)$ of a graph G have attracted a lot of attention during the last decades. Perhaps the first non-trivial result here is Brooks' Theorem [14], which states that, for a connected graph G, $\chi(G)$ is at most $\Delta(G)$, unless G is an odd cycle or a complete graph. Following this, in 1977,

[*]Research supported by ANR-21-CE48-0012.
[†]Research supported by JSPS KAKENHI JP20A402 and 22H05001, and by JST ASPIRE JPMJAP2302.

Borodin and Kostochka [13] made the following well-known conjecture, which, if true, would give a nice extension of Brooks' theorem for graphs with maximum degree at least 9.

Conjecture 1 (Borodin–Kostochka [13]). *Every graph G with maximum degree $\Delta(G) \geqslant 9$ and clique number $\omega(G) \leqslant \Delta(G) - 1$ satisfies $\chi(G) \leqslant \Delta(G) - 1$.*

The graph obtained from a 5-cycle by blowing up each vertex into a triangle witnesses that the lower bound on the maximum degree is tight. Twenty years later, Reed published two fundamental papers [38, 39] related to Borodin–Kostochka's conjecture. In the first one, Reed posed the following celebrated conjecture.

Conjecture 2 (Reed [38]). *Every graph G satisfies $\chi(G) \leqslant \lceil \frac{1}{2}\big(\Delta(G) + 1 + \omega(G)\big) \rceil$.*

If true, it implies in particular that every graph G with chromatic number at least $\Delta(G)$ contains a clique of size $\Delta(G) - 2$, thereby approaching Conjecture 1. As a partial result, Reed [38] shows that $\chi(G)$ is indeed bounded by a convex combination of $\Delta(G) + 1$ and $\omega(G)$ (up to ceiling).

Theorem 3 ([38]). *There exists $\varepsilon > 0$ such that every graph G satisfies*

$$\chi(G) \leqslant \lceil (1 - \varepsilon)(\Delta(G) + 1) + \varepsilon\omega(G) \rceil .$$

Theorem 3 implies the weaker version of Borodin–Kostochka's Conjecture stating that every graph G with chromatic number at least $\Delta(G)$ contains a clique of size $\Delta(G) - C$, for some absolute constant C. One year later, Reed [39] strengthened this by proving that Conjecture 1 holds for large values of Δ.

Theorem 4 ([39]). *There exists $\Delta_4 \in \mathbb{N}$ such that the following holds. Every graph G with maximum degree $\Delta(G) \geqslant \Delta_4$ and clique number $\omega(G) \leqslant \Delta(G) - 1$ satisfies $\chi(G) \leqslant \Delta(G) - 1$.*

$\overleftrightarrow{\omega}$, $\tilde{\Delta}$, and $\vec{\chi}$

The purpose of this work is to provide extensions of Conjecture 1 and Theorem 4 to the digraph setting. We thus introduce three digraph parameters extending Δ, ω, and χ. By an extension, we mean that if D is **symmetric**, that is, D is obtained from a graph G by replacing each edge by two opposite arcs, then these digraph parameters on D coincide with Δ, ω, and χ on G. It is straightforward to verify that this holds for $\tilde{\Delta}$, $\overleftrightarrow{\omega}$, and $\vec{\chi}$, defined as follows.

Let D be a digraph. We let $\tilde{\Delta}(D)$ denote the maximum geometric mean of the in- and out-degrees of the vertices of D, that is $\tilde{\Delta}(D) = \max\{\sqrt{d^-(v) \cdot d^+(v)} : v \in V(D)\}$. A **biclique** of D is a set of vertices inducing a complete digraph, which is a digraph containing all possible arcs (*i.e.*, between every pair of vertices there are two oppositely oriented arcs). The **biclique number** $\overleftrightarrow{\omega}(D)$ of D is the size of the largest biclique of D.

A **k-dicolouring** of D is a colouring of its vertex-set $\varphi\colon V(D) \to [k]$ such that each colour class $\varphi^{-1}(i)$ induces an acyclic subdigraph on D. The **dichromatic number** of D, denoted by $\vec{\chi}(D)$, is the smallest $k \in \mathbb{N}$ for which D admits a k-dicolouring. These colouring notions for digraphs were introduced in the late 1970s by Erdős and Neumann-Lara [35, 17], rediscovered by Mohar in the 2000s [29, 12], and have received a lot of attention since then. It has emerged that many classical results on graph colouring can be extended to digraphs under these definitions, reinforcing the idea that dicolouring constitutes an appropriate generalisation of proper colouring to digraphs. For instance, the Strong Perfect Graph Theorem [8], Gyárfás–Sumner's conjecture [5, 3, 16], Hadwiger's Conjecture [9, 28], and Kostochka–Yancey's type of bounds [27, 4, 6, 24, 37] have been generalised to digraphs via the notion of dicolouring.

Similar to the undirected case, relationships linking $\vec{\chi}$, $\tilde{\Delta}$, and $\overleftrightarrow{\omega}$ have recently gained interest. This line of research began with a result of Mohar [30], which states that $\vec{\chi}(D) \leqslant \lceil \tilde{\Delta}(D) \rceil$ for every connected digraph D which is not a directed cycle, a symmetric odd cycle, nor a complete digraph, thereby generalising Brooks' Theorem (see also [22, 2, 21] for stronger versions of this result). Recently, Kawarabayashi and Picasarri-Arrieta [26] posed that Reed's conjecture generalises to digraphs with these parameters.

Conjecture 5 ([26])**.** *Every digraph D satisfies* $\vec{\chi}(D) \leqslant \left\lceil \frac{1}{2}\left(\tilde{\Delta}(D) + 1 + \overleftrightarrow{\omega}(D)\right) \right\rceil$.

If true, this would imply both Reed's conjecture in the case of symmetric digraphs and an independent conjecture posed by Harutyunyan and Mohar [23] when applied to digraphs with biclique number 1. As an intermediate result, they generalised Theorem 3 to digraphs (up to the value of ε).

Theorem 6 ([26])**.** *There exists $\varepsilon > 0$ such that every digraph D satisfies*

$$\vec{\chi}(D) \leqslant \lceil (1-\varepsilon)(\tilde{\Delta}(D) + 1) + \varepsilon \overleftrightarrow{\omega}(D) \rceil.$$

One might expect that, like Reed's conjecture, Borodin–Kostochka's conjecture directly extends to digraphs with these parameters; that is, for every integer $\Delta \geqslant 9$, digraphs D with maximum degree $\tilde{\Delta}(D) \leqslant \Delta$ and biclique number $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$ have dichromatic number at most $\Delta - 1$. It turns out that this is not the case: for every integer $k \geqslant 1$, let $H_k = \vec{C}_3 \boxplus \overleftrightarrow{K}_{k-2}$ be the digraph made of a copy C of $\vec{C}_3$ (the directed cycle of length 3), a disjoint copy K of $\overleftrightarrow{K}_{k-2}$ (the complete digraph of order $k-2$), and all possible arcs in both directions between K and C. Then H_k satisfies $\tilde{\Delta}(H_k) = k$, $\overleftrightarrow{\omega}(H_k) = k - 1$, and $\vec{\chi}(D) = k$. We believe that this is the only obstruction arising from the directed setting and conjecture the following.

Conjecture 7. *Let $\Delta \geqslant 9$ be an integer and D a digraph satisfying $\tilde{\Delta}(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$. Then $\vec{\chi}(D) \leqslant \Delta - 1$ unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

If true, Conjecture 7 would imply Conjecture 1, since graphs in Conjecture 1 can be replaced by corresponding symmetric digraphs in Conjecture 7. In particular, condition $\Delta \geqslant 9$ cannot be relaxed, as it is already best possible is in the undirected case. Similarly to the undirected case, Theorem 6 implies the weaker version stating that digraphs D with $\vec{\chi}(D) \geqslant \tilde{\Delta}(D)$ have biclique number at least $\tilde{\Delta}(D) - C$ for some absolute constant C. Our first contribution is that Conjecture 7 holds for digraphs with large maximum degree, thereby generalising Theorem 4 to digraphs (up to the value of Δ_8).

Theorem 8. *There exists $\Delta_8 \in \mathbb{N}$ such that the following holds. Let $\Delta \geqslant \Delta_8$ be an integer and D a digraph with $\tilde{\Delta}(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$. Then $\vec{\chi}(D) \leqslant \Delta - 1$ unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Our proof of Theorem 8 combines structural and probabilistic arguments. In particular, we prove the following dense decomposition lemma, which generalises to the directed setting the so-called dense decomposition of graphs appearing in Molloy and Reed's series of papers [38, 32, 31, 39, 19, 34] (see also [33, Chapter 15]). We believe it is of independent interest, and that it can be used to extend to the directed setting many other results obtained by Molloy and Reed, as well as have applications in the algorithmic theory of digraphs.

Before we state it, we first need a few additional definitions. Let D be a digraph. We denote by $\Delta_{\max}(D)$ the maximum, overall vertices v of D, of $\max(d^+(v), d^-(v))$. Suppose that $\Delta_{\max}(D) = \Delta$, and let $0 \leqslant d \leqslant \Delta - 1$ be any real number. We say that a vertex v is **d-sparse** if the digraph induced by its out-neighbourhood contains at most $\Delta(\Delta - 1) - d\Delta$ arcs. A vertex

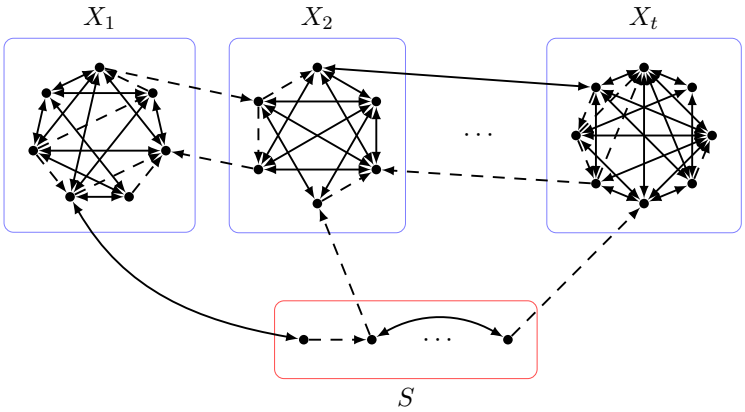

Figure 1: An illustration of the decomposition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ provided by Lemma 9. Each set X_i has size close to Δ, and each vertex in X_i has a large fraction of its out-neighbours inside X_i. Intuitively, each X_i is thus close to being a biclique on Δ vertices. Vertices in S are d-sparse.

that is not d-sparse is **d-dense**. Given a subset of vertices X, we denote by $\partial^+(X)$ (respectively $\partial^-(X)$) the set of arcs of D going out from X (respectively entering X). In what follows, by $\omega(1) \leqslant d \leqslant o(\Delta)$ we mean that d is any fixed sublinear function of Δ going to infinity.

Lemma 9 (Dense Decomposition Lemma)**.** *For every $0 < \varepsilon < \frac{1}{2}$ and $\omega(1) \leqslant d \leqslant o(\Delta_{\max})$, there exists Δ_9 such that the following holds. Every digraph D with $\Delta_{\max}(D) = \Delta_{\max} \geqslant \Delta_9$ admits a partition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ of its vertex-set such that:*

(i) *for every $i \in [t]$, $\Delta_{\max} - \frac{3}{\varepsilon}d < |X_i| < \Delta_{\max} + 1 + 4d$;*

(ii) *for every $i \in [t]$, $|\partial^+(X_i)| \leqslant \frac{8}{\varepsilon} \cdot d \cdot \Delta_{\max}$ and $|\partial^-(X_i)| \leqslant \frac{8}{\varepsilon} \cdot d \cdot \Delta_{\max}$;*

(iii) *for every $i \in [t]$ and every $u \in V(D)$, $u \in X_i$ if and only if $|N^+(u) \cap X_i| \geqslant (1-\varepsilon)\Delta_{\max}$; and*

(iv) *vertices in S are d-sparse.*

See Figure 1 for an illustration of a dense decomposition.

We now outline the proof of Theorem 8. Consider a minimal counterexample D along with one of its dense decompositions $(X_1, \ldots, X_t, S)$ generated by Lemma 9, where ε is sufficiently small and d is approximately $\log^3 \Delta$. In a minimal counterexample, it is easy to show that $\tilde{\Delta}$ and $\Delta_{\max}$ are close, so for clarity, let us assume here that $\Delta = \Delta_{\max}(D) = \tilde{\Delta}(D)$.

Consider the random colouring process which consists of (i) assigning each vertex of D a colour uniformly at random from $\{1, \ldots, \Delta - 1\}$, and (ii) simultaneously uncolouring all vertices belonging to a monochromatic directed cycle. The core idea is to show that the resulting partial dicolouring can be extended to D with positive probability. This method is now standard in the field, and has been used for instance in [23, 26]. For a d-sparse vertex v, applying this process, it is highly likely that many colours are repeated in its out-neighbourhood, thereby allowing us to extend the obtained dicolouring to v. This argument is central to the proof of [26, Theorem 1.7], which establishes that a sparse digraph H can be dicoloured using only a fraction of $\tilde{\Delta}(H)$ colours.

Thus, the main challenge in proving Theorem 8 lies in handling d-dense vertices. Furthermore, we know that all such vertices belong to some X_i, and that each X_i can be coloured almost

independently from the rest of the digraph. This is mainly due to property (iii) of Lemma 9 and the fact that ε is sufficiently small. Building on this observation, we progressively constrain the possible structure of each X_i until reaching a point where it becomes likely that the random colouring process described above yields a partial dicolouring that can be extended to X_i. At that stage, using the Lovász Local Lemma, we conclude that the obtained partial dicolouring can be extended simultaneously to all sparse vertices and all X_is with positive probability, leading to the desired contradiction.

Analogues for other definitions of Δ

A natural question is whether $\tilde{\Delta}$ could be replaced by other, potentially more natural, definitions of the maximum degree of a digraph in Conjecture 7 and Theorem 8. Observe that, by definition, all digraphs D satisfy $\Delta_{\max}(D) \geqslant \tilde{\Delta}(D)$. Hence, replacing $\tilde{\Delta}$ with $\Delta_{\max}$ yields weaker statements. A natural candidate is then the **maximum out-degree** Δ^+ of a digraph, defined as

$$\Delta^+(D) = \max\{d^+(v) : v \in V(D)\}.$$

Indeed, Δ^+ and $\tilde{\Delta}$ are independent parameters, in the sense that one cannot be bounded by any function of the other. Furthermore, when only the out-degree of a digraph is bounded, probabilistic methods are unlikely to work. Intuitively, if the maximum in-degree of a digraph is arbitrarily large compared to its maximum out-degree, one cannot expect to apply Lovász Local Lemma, as it seems unlikely to bound the degree of "bad events". That is why finding upper bounds on the dichromatic number of a digraph D involving $\Delta^+(D)$ is usually significantly harder than ones involving $\tilde{\Delta}(D)$. The most emblematic open problem of this kind is probably the following one, proposed by Kawarabayashi and Picasarri-Arrieta [26], whose version with $\tilde{\Delta}$ has been proved first by Harutyunyan and Mohar [23] (see also [20]), and appears to be a particular case of Theorem 6.

Problem 10 ([26])**.** Show the existence of $\varepsilon > 0$ such that every digraph D with $\overleftrightarrow{\omega}(D) = 1$ satisfies $\vec{\chi}(D) \leqslant (1-\varepsilon)\Delta^+(D) + O(1)$.

Nevertheless, we believe that Conjecture 7 remains valid when $\tilde{\Delta}$ is replaced by Δ^+. Furthermore, we are able to prove the analogue of Theorem 8 to support it.

Conjecture 11. *Every digraph D with maximum out-degree $\Delta^+(D) = \Delta \geqslant 9$ and biclique number $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$ satisfies $\vec{\chi}(D) \leqslant \Delta - 1$, unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Theorem 12. *There exists $\Delta_{12} \in \mathbb{N}$ such that the following holds. Let $\Delta \geqslant \Delta_{12}$ be an integer and D a digraph with $\Delta^+(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$. Then $\vec{\chi}(D) \leqslant \Delta - 1$ unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Theorem 12 yields a second independent generalisation of Theorem 4. The proof combines structural and discharging arguments on a well-chosen counterexample, which in particular does not satisfy the requirements of Theorem 8.

Finally, leveraging a technique introduced first in [36] and formally established in [25], we derive from Theorem 12 the following two sufficient conditions for a digraph D to satisfy $\vec{\chi}(D) \leqslant \Delta_{\min}(D) - 1$, where

$$\Delta_{\min}(D) = \max\{\min(d^-(v), d^+(v)) : v \in V(D)\}.$$

Note that, for digraphs satisfying one of these conditions, this is a strengthening of Theorems 8 and 12, as $\Delta_{\min}$ is always bounded above by both $\tilde{\Delta}$ and Δ^+.

Corollary 13. *For every integer $\Delta \geqslant \Delta_{12}$ and digraph D with $\Delta_{\min}(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) < \frac{1}{2}(\Delta - 1)$, $\vec{\chi}(D) \leqslant \Delta - 1$.*

Corollary 14. *The following holds for every integer $\Delta \geqslant \Delta_{12}$. Let D be a digraph with underlying graph G. If $\Delta_{\min}(D) \leqslant \Delta$ and $\omega(G) \leqslant \Delta - 1$, then $\vec{\chi}(D) \leqslant \Delta - 1$.*

Note that Corollary 14 gives a third independent generalisation of Theorem 4. We are actually able to prove a slightly more general result on digraphs D with $\vec{\chi}(D) \geqslant \Delta_{\min}(D)$, the interested reader is referred to Section 6 (see Corollary 30) for the exact statement. We finally justify that the condition $\overleftrightarrow{\omega}(D) < \frac{1}{2}(\Delta - 1)$ of Corollary 13 is tight. In fact, when restricted to digraphs D with $\overleftrightarrow{\omega}(D) = \lceil \frac{1}{2}(\Delta_{\min}(D) - 1) \rceil$, we prove that determining whether $\vec{\chi}(D) \leqslant \Delta_{\min}(D) - 1$ is even NP-complete.

Proposition 15. *For every integer $k \geqslant 2$, deciding $\vec{\chi}(D) \leqslant k$ remains NP-complete when restricted to digraphs D with $\Delta_{\min}(D) = k + 1$ and $\overleftrightarrow{\omega}(D) = \lceil \frac{k}{2} \rceil$.*

Organisation of the paper. All notation, definitions, and probabilistic tools used in the paper are provided in Section 2. Section 3 is dedicated to the proof of Lemma 9. Section 4 is devoted to the proof of Theorem 8. It is mainly divided into two parts: in Subsection 4.1, we use Lemma 9 to show a collection of structural properties on a minimum counterexample of Theorem 8. In Subsection 4.2, we use the probabilistic method to show that a digraph D satisfying these specific properties can actually be coloured with $\Delta - 1$ colours at most, thus showing the result. We prove Theorem 12 in Section 5. We finally derive Corollaries 13 and 14, and prove Proposition 15 in Section 6.

2 Preliminaries

2.1 General notation on digraphs

We only consider digraphs without loops and parallel arcs. Our notation follows [11]. A **digon** is a pair of arcs in opposite directions between the same vertices u, v, and is denoted $\langle u, v \rangle$. A **simple arc** is an arc that is not in a digon. A digraph is **symmetric** if it contains no simple arc. Given an undirected graph G, the digraph $\overleftrightarrow{G}$ is the symmetric digraph obtained from G by replacing every edge with a digon. For an integer n, $\overleftrightarrow{K}_n$ is called the **complete digraph** on n vertices, and a **biclique** of size k is thus a set of vertices inducing a copy of $\overleftrightarrow{K}_k$.

Let D be a digraph. The **underlying graph** of D, denoted by $\mathrm{UG}(D)$, is the undirected graph with vertex-set $V(D)$ in which uv is an edge if and only if uv or vu is an arc of D. We say that D is an **orientation** of G if D does not contain any digon and G is the underlying graph of D. A **tournament** is an orientation of a complete graph.

The **complement** of D, denoted by $\overline{D}$, is the digraph with vertex-set $V(D)$ and arc set $(V(D) \times V(D)) \setminus A(D) \setminus \{(v, v) : v \in V(D)\}$. A **matching** of D is a set of pairwise disjoint arcs of D. The maximum size of a matching of D is denoted $\nu(D)$.

Let xy be an arc of D. We call y an **out-neighbour** of x, and x and **in-neighbour** of y. If further xy is simple, we call y a **simple out-neighbour** of x, and x a **simple in-neighbour** of y. Finally, a **neighbour** of x is either an in- or out-neighbour of x. Let v be any vertex of D. The **out-neighbourhood** $N^+(v)$ of v is the set of out-neighbours of v, and its **in-neighbourhood** $N^-(v)$ is the set of its in-neighbours. We denote by $N^+[v]$ and $N^-[v]$ the **closed out-neighbourhood** $N^+(v) \cup \{v\}$ and the **closed in-neighbourhood** $N^-(v) \cup \{v\}$ of v respectively. Furthermore, we denote by $N(v)$ the union of $N^+(v)$ and $N^-(v)$, and by $N^{\pm}(v)$ their intersection. The **out-** and **in-degrees** of v, denoted by $d^+(v)$ and $d^-(v)$, are respectively

the number of out- and in-neighbours of v. From this, we can derive the four distinct definitions of the maximum degree of D used throughout the paper:

- $\Delta_{\max}(D) = \max\Big\{\max(d^-(v), d^+(v)) : v \in V(D)\Big\}$,
- $\tilde{\Delta}(D) = \max\Big\{\sqrt{d^-(v)\cdot d^+(v)} : v \in V(D)\Big\}$,
- $\Delta^+(D) = \max\Big\{d^+(v) : v \in V(D)\Big\}$, and
- $\Delta_{\min}(D) = \max\Big\{\min(d^-(v), d^+(v)) : v \in V(D)\Big\}$.

Given two subsets of vertices $X, Y \subseteq V(D)$, we say that X **dominates** Y if D contains every possible arc uv with $u \in X$ and $v \in Y$. For an integer k, D is called **k-critical** if $\vec{\chi}(D) = k$ and $\vec{\chi}(H) \leqslant k-1$ holds for all proper subdigraphs H of D. Given two digraphs H_1 and H_2, the digraph $H_1 \boxplus H_2$ is the one obtained from disjoint copies of H_1 and H_2 by adding all possible digons between them.

2.2 Probabilistic tools

Our proof of Theorem 8 contains some probabilistic arguments. We briefly recall here three classical results used later. The reader unfamiliar with the probabilistic method is referred to [7, 33]. We first need the symmetric version of the Lovász Local Lemma, due to Erdős and Lovász [18].

Lemma 16 (Lovász Local Lemma)**.** *Let $A_1, A_2, \ldots, A_n$ be events in an arbitrary probability space. Suppose that each event A_i is mutually independent of a set of all the other events but at most d, and that $\mathbb{P}(A_i) \leqslant p$ for all $1 \leqslant i \leqslant n$. If*

$$ep(d+1) \leqslant 1,$$

then $\mathbb{P}\left(\bigcap_{i=1}^n \overline{A_i}\right) > 0$.

The second result we need is the celebrated concentration bound due to Talagrand [40]. We use it in the following form, which is similar to that of [34].

Lemma 17 (Talagrand's Inequality)**.** *Let X be a random variable valued in $\mathbb{N}$, determined by n independent trials and satisfying the following for some integers $c, r \geqslant 1$:*

1. *changing the outcome of any one trial can affect X by at most c, and*
2. *for every $s \in \mathbb{N}$, if $X \geqslant s$ then there is a set of at most rs trials whose outcomes certify that $X \geqslant s$.*

Then

$$\mathbb{P}\left(|X - \mathbb{E}(X)| > t\right) \leqslant 4\exp\left(\frac{-t^2}{32c^2r(\mathbb{E}(X)+t)}\right)$$

for any real number $t > 126c\sqrt{r\mathbb{E}(X)} + 344c^2r$.

The difference between the version above and that of [34] is purely technical. In [34], the second condition is asked not only for integers s but for all real numbers $s > 0$. To prove that an integer-valued random variable X satisfies such a condition, one needs, in particular, to justify that X is always positive (as s might be smaller than $\frac{1}{r}$), and to use some rounding arguments. For the sake of clarity in the proofs of our main theorems, we decided to include these technical details directly in the proof of Lemma 17, which we give in Appendix A.

The last probabilistic tool we will use is the following consequence [33] of the celebrated concentration bound due to Azuma [10].

Lemma 18 (Azuma's Inequality). *Let X be a random variable determined by n trials $T_1, \ldots, T_n$, such that, for each $1 \leqslant i \leqslant n$ and any two possible sequences of outcomes $t_1, \ldots, t_i$ and $t_1, \ldots, t_{i-1}, t'_i$ differing only in the last one,*

$$\left| \mathbb{E}(X \mid T_1 = t_1, \ldots, T_i = t_i) - \mathbb{E}(X \mid T_1 = t_1, \ldots, T_{i-1} = t_{i-1}, T_i = t'_i) \right| \leqslant c_i.$$

Then, for any $\lambda \geqslant 0$,

$$\mathbb{P}(|X - \mathbb{E}(X)| > \lambda) \leqslant 2 \exp\left(\frac{-\lambda^2}{2 \sum_{i=1}^{n} c_i^2} \right).$$

2.3 The Directed Brooks' Theorem

In order to simplify the proof of Theorem 8, we will make use of the directed Brooks' Theorem, due to Mohar [30] (see also [1]).

Theorem 19 (Directed Brooks' Theorem). *Let D be a digraph with $\vec{\chi}(D) = \Delta_{\max}(D) + 1$, then one of the following holds for some connected component H of D:*

- *$\Delta_{\max}(D) = 1$ and H is a directed cycle,*
- *$\Delta_{\max}(D) = 2$ and H is a symmetric odd cycle, or*
- *H is a complete digraph on $\Delta_{\max}(D) + 1$ vertices.*

3 Directed dense decompositions

This section is devoted to the proof of Lemma 9. Let us first recall it for convenience.

Lemma 9 (Dense Decomposition Lemma). *For every $0 < \varepsilon < \frac{1}{2}$ and $\omega(1) \leqslant d \leqslant o(\Delta_{\max})$, there exists Δ_9 such that the following holds. Every digraph D with $\Delta_{\max}(D) = \Delta_{\max} \geqslant \Delta_9$ admits a partition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ of its vertex-set such that:*

(i) *for every $i \in [t]$, $\Delta_{\max} - \frac{3}{\varepsilon} d < |X_i| < \Delta_{\max} + 1 + 4d$;*

(ii) *for every $i \in [t]$, $|\partial^+(X_i)| \leqslant \frac{8}{\varepsilon} \cdot d \cdot \Delta_{\max}$ and $|\partial^-(X_i)| \leqslant \frac{8}{\varepsilon} \cdot d \cdot \Delta_{\max}$;*

(iii) *for every $i \in [t]$ and every $u \in V(D)$, $u \in X_i$ if and only if $|N^+(u) \cap X_i| \geqslant (1 - \varepsilon)\Delta_{\max}$; and*

(iv) *vertices in S are d-sparse.*

Proof. Let us fix $0 < \varepsilon < \frac{1}{2}$ and $\omega(1) \leqslant d \leqslant o(\Delta_{\max})$. We do not explicitly give the value of Δ_9; we simply assume that it is large enough so that all inequalities hold in what follows. For the sake of conciseness, let us denote $\Delta_{\max}$ by Δ along this proof. For every d-dense vertex v, we let X_v be the set of vertices obtained via the following process:

1. at the beginning, $X_v = N^+[v]$;
2. as long as there exists $u \in X_v$ with $|N^+(u) \cap X_v| < (1 - \varepsilon)\Delta$, remove u from X_v;
3. as long as there exists $u \in V(D) - X_v$ with $|N^+(u) \cap X_v| \geqslant (1 - \varepsilon)\Delta$, add u to X_v.

Claim 9.1. *Every vertex $u \in V(D)$ belongs to X_v if and only if $|N^+(u) \cap X_v| \geqslant (1 - \varepsilon)\Delta$.*

Proof of claim. This follows directly from the construction of X_v. ◊

Claim 9.2. *Every d-dense vertex v satisfies $d^+(v) \geqslant \Delta - d$.*

Proof of claim. If $d^+(v) < \Delta - d$, then the number of arcs in $D[N^+(v)]$ is less than

$$(\Delta - d)(\Delta - d - 1) < \Delta(\Delta - 1) - d\Delta,$$

and v is not a d-dense vertex. $\Diamond$

Claim 9.3. *For every d-dense vertex v, the number of vertices removed from X_v at step 2 is less than $\frac{2d}{\varepsilon}$. In particular, $|N^+(v) \setminus X_v| < \frac{2d}{\varepsilon}$.*

Proof of claim. Let $c_\varepsilon = \frac{2}{\varepsilon}$. Assume for a contradiction that at least $c_\varepsilon \cdot d$ vertices are removed at step 2 of the process described above. Let $\mathcal{U}$ be the $\lceil c_\varepsilon \cdot d \rceil$ first deleted vertices, and let $\mathcal{V}$ be $N^+(v) \setminus \mathcal{U}$. By Claim 9.2, $|\mathcal{V}| \geqslant \Delta - (c_\varepsilon + 1)d - 1$. By construction of $\mathcal{U}$, vertices in $\mathcal{U}$ have less than $(1 - \varepsilon)\Delta$ out-neighbours in $\mathcal{V}$. Therefore, for each vertex $u \in \mathcal{U}$, we have

$$|\mathcal{V} \setminus N^+(u)| > \varepsilon\Delta - (c_\varepsilon + 1)d - 1. \tag{1}$$

Observe that, in $D[N^+(v)]$, the number of non-arcs is at least $\sum_{u \in \mathcal{U}} |\mathcal{V} \setminus N^+(u)|$. By (1), the number of arcs in $D[N^+(v)]$ is thus less than

$$\Delta(\Delta - 1) - c_\varepsilon d \cdot (\varepsilon\Delta - (c_\varepsilon + 1)d - 1) < \Delta(\Delta - 1) - d\Delta.$$

This is a contradiction to v being d-dense. $\Diamond$

Claim 9.4. *For every d-dense vertex v, $|\partial^-(N^+[v])| \leqslant (d+2)\Delta$ and $|\partial^+(N^+[v])| \leqslant (d+2)\Delta$.*

Proof of claim. The vertex v being d-dense, there are at least $\Delta^2 - (d+1)\Delta$ arcs in $D[N^+[v]]$. Observe that $\sum_{u \in N^+[v]} d^-(u)$ is exactly $|\partial^-(N^+[v])| + |A(D[N^+[v]])|$. We thus have

$$\Delta(\Delta + 1) \geqslant \sum_{u \in N^+[v]} d^-(u) \geqslant |\partial^-(N^+[v])| + \Delta^2 - (d+1)\Delta,$$

which implies $|\partial^-(N^+[v])| \leqslant (d+2)\Delta$. Symmetrically, we have

$$\Delta(\Delta + 1) \geqslant \sum_{u \in N^+[v]} d^+(u) \geqslant |\partial^+(N^+[v])| + \Delta^2 - (d+1)\Delta,$$

which implies $|\partial^+(N^+[v])| \leqslant (d+2)\Delta$. $\Diamond$

Claim 9.5. *For every d-dense vertex v, the number of vertices added to X_v at step 3 is less than $4d$. In particular, $|X_v \setminus N^+[v]| < 4d$.*

Proof of claim. Assume for a contradiction that at least $4d$ vertices are added at step 3 of the process described above. Note that the ith added vertex has at least $(1 - \varepsilon)\Delta + 1 - i$ out-neighbours in $N^+[v]$. This implies

$$|\partial^-(N^+[v])| \geqslant \sum_{i=1}^{\lceil 4d \rceil} \Big((1 - \varepsilon)\Delta + 1 - i\Big) > (d+2)\Delta,$$

a contradiction to Claim 9.4. $\Diamond$

Claim 9.6. *For every d-dense vertex v, $|\partial^-(X_v)| < \dfrac{8}{\varepsilon} d\Delta$ and $|\partial^+(X_v)| < \dfrac{8}{\varepsilon} d\Delta$.*

Proof of claim. After step 1 of the process, $|\partial^-(X_v)| \leqslant (d+2)\Delta$ and $|\partial^+(X_v)| \leqslant (d+2)\Delta$ by Claim 9.4. During step 2, each time we remove a vertex u from X_v, both $|\partial^-(X_v)|$ and $|\partial^+(X_v)|$ increase by at most Δ. Similarly, every time we add a vertex during step 3, both increase by at most Δ. Since less than $\frac{2d}{\varepsilon}$ vertices are removed and less than $4d$ are added by Claims 9.3 and 9.5, the claim follows. ◊

It remains to show the existence of a set of d-dense vertices $\mathcal{W}$ such that $(X_v)_{v\in\mathcal{W}}$ are pairwise disjoint, and such that every d-dense vertex belongs to $\bigcup_{v\in\mathcal{W}} X_v$. The result then follows from Claims 9.1, 9.2, 9.3, 9.5, and 9.6. First observe that every d-dense vertex v indeed belongs to X_v by Claims 9.1, 9.2, and 9.3. The following thus implies that such a set $\mathcal{W}$ can be constructed greedily by adding to $\mathcal{W}$ a d-dense vertex $u \notin \bigcup_{v\in\mathcal{V}} X_v$.

Claim 9.7. *Let u and v be two d-dense vertices. If $X_v \cap X_u \neq \varnothing$, then $v \in X_u$ and $u \in X_v$.*

Proof of claim. By symmetry, we only have to show that $u \in X_v$. For this, it is sufficient to show that $|N^+(u) \cap X_v| \geqslant (1-\varepsilon)\Delta$, the claim then follows from Claim 9.1. Let x be any vertex in $X_u \cap X_v$. By Claim 9.1 applied to both X_u and X_v, we have:

$$\Delta \geqslant d^+(x) \geqslant 2(1-\varepsilon)\Delta - |X_u \cap X_v| + 1,$$

which implies $|X_u \cap X_v| \geqslant (1-2\varepsilon)\Delta + 1$. Together with Claim 9.5 applied to u, we thus have

$$|N^+(u) \cap X_v| > (1-2\varepsilon)\Delta - 4d. \tag{2}$$

Let us now count the number p of arcs going from $N^+(u) \cap X_v$ to $N^+(u) \setminus X_v$. First, by Claim 9.6, observe that

$$p \leqslant |\partial^+(X_v)| < \frac{8}{\varepsilon} d\Delta. \tag{3}$$

On the other hand, by denseness of u, we know that less than $d\Delta$ arcs are missing in the out-neighbourhood of u. In particular, we have

$$p \geqslant |N^+(u) \cap X_v| \cdot |N^+(u) \setminus X_v| - d\Delta. \tag{4}$$

Combining (2), (3), and (4), straightforward calculations imply $|N^+(u) \setminus X_v| < \frac{9}{\varepsilon(1-2\varepsilon)} d$. Since $d^+(u) \geqslant \Delta - d$ by Claim 9.2, we finally obtain

$$|N^+(u) \cap X_v| > \Delta - \left(1 + \frac{9}{\varepsilon(1-2\varepsilon)}\right) \cdot d > (1-\varepsilon)\Delta,$$

concluding the proof of the Lemma. ◊

□

4 Proof of Theorem 8

As explained in the introduction, this section is devoted to the proof of Theorem 8. Let us first recall it for the reader's convenience.

Theorem 8. *There exists $\Delta_8 \in \mathbb{N}$ such that the following holds. Let $\Delta \geqslant \Delta_8$ be an integer and D a digraph with $\tilde{\Delta}(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$. Then $\vec{\chi}(D) \leqslant \Delta - 1$ unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

We do not explicitly give the value of Δ_8, we simply assume that it is large enough so that all inequalities claimed along the proof actually hold.

We assume the statement does not hold, which means that there exists an integer $\Delta \geqslant \Delta_8$ and a digraph D satisfying $\vec{\chi}(D) \geqslant \Delta \geqslant \tilde{\Delta}(D)$ that does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. Among all such digraphs, we choose D for which $|V(D)| + |A(D)|$ is minimum. We further set $d = \log^3(\Delta)$.

In Subsection 4.1, with Lemma 9 in hand, we show a collection of structural properties on D. In Subsection 4.2, we use the probabilistic method to show that a digraph satisfying these specific properties can be partially coloured with $\Delta - 1$ colours at most, in such a way that the obtained dicolouring can be greedily extended to the whole digraph, thus yielding a contradiction.

4.1 Structure of a minimum counterexample

Remark first that D is Δ-critical, for otherwise we can remove an arc or a vertex without decreasing the dichromatic number, thus yielding a smaller counterexample. We start with two well-known basic properties of critical digraphs.

Lemma 20. *Every vertex $v \in V(D)$ satisfies $\Delta - 1 \leqslant d^-(v), d^+(v) \leqslant \Delta + 1$. Moreover, we have $\Delta \leqslant \Delta_{\max}(D) \leqslant \Delta + 1$.*

Proof. If some vertex $v \in V(D)$ satisfies $\min(d^-(v), d^+(v)) \leqslant \Delta - 2$, we can extend a $(\Delta - 1)$-dicolouring of $D - v$ (which exists as D is Δ-critical) to D by choosing for v a colour that is not appearing in the in- or out-neighbourhood of v, thus contradicting $\vec{\chi}(D) = \Delta$. This shows the lower bound. As for the upper bound, assume for a contradiction that v satisfies $\max(d^-(v), d^+(v)) \geqslant \Delta + 2$. Using the lower bound we just proved, we thus have

$$d^+(v) \cdot d^-(v) \geqslant (\Delta - 1)(\Delta + 2) > \Delta^2 \geqslant \tilde{\Delta}(D)^2.$$

This is a contradiction to the definition of $\tilde{\Delta}(D)$. This shows the first part of the statement.

For the second part, it remains to justify that $\Delta_{\max}(D) \neq \Delta - 1$. Suppose that $\Delta_{\max}(D) = \Delta - 1$, then by the directed Brooks' Theorem (Theorem 19), and because $\vec{\chi}(D) \geqslant \Delta$, some connected component of D must be isomorphic to $\overleftrightarrow{K}_\Delta$, a contradiction. □

Lemma 21. *If uv is a simple arc of D, then v has a simple out-neighbour, and u has a simple in-neighbour.*

Proof. By criticality of D, let φ be a $(\Delta-1)$-dicolouring of $D \setminus \{uv\}$. Since φ is not a dicolouring of D, D coloured with φ contains a monochromatic directed cycle, and among all of them we let $\mathscr{C}$ be an induced one. Clearly, $\mathscr{C}$ contains uv, for otherwise $\mathscr{C}$ is a monochromatic directed cycle of $D \setminus \{uv\}$ coloured with φ. Since $\mathscr{C}$ is induced and uv is simple, the successor of v in $\mathscr{C}$ is indeed one of its simple out-neighbours. Similarly, the predecessor of u in $\mathscr{C}$ is one of its simple in-neighbours. □

By definition, D does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. The following shows that it remains true even when we add one or two arcs between two vertices of D.

Lemma 22. *For every pair of vertices $u, v \in V(D)$, $D \cup \{uv, vu\}$ does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Proof. Throughout the proof, we will often rely on the fact that $\min(d^-(w), d^+(w)) \leqslant \Delta$ for every vertex w of D. Let us see first that $D \cup \{uv, vu\}$ does not contain any induced copy of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. (If $D \cup \{uv, vu\}$ contains a non-induced copy of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, then it contains

a copy of $\overleftrightarrow{K}_\Delta$ and we will deal with this later.) Assume for a contradiction that this is not true. Let T be a set of vertices inducing a directed 3-cycle in $D \cup \{uv, vu\}$ and W a biclique of $D \cup \{uv, vu\}$ of size $\Delta - 2$, such that all arcs between T and W are in $A(D) \cup \{uv, vu\}$. Note that $u, v \in T \cup W$; otherwise, D itself would contain a copy of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, a contradiction. Let φ be a $(\Delta - 1)$-dicolouring of $D - T - W$, which exists due to the Δ-criticality. We are going to extend φ to a $(\Delta - 1)$-dicolouring of D, yielding a contradiction.

Since T induces a directed 3-cycle in $D \cup \{uv, vu\}$, at least one of $\{u, v\}$, say v, belongs to W. Let x and y be two distinct vertices in $T \setminus \{u\}$. Since none of $\{\langle x, y\rangle, \langle u, v\rangle\}$ is a digon of D, and each of these four vertices has at most three out-neighbours outside $T \cup W$, we can extend φ to $V(D) \setminus W \cup \{u, v\}$ in a way that $\varphi(x) = \varphi(y) \neq \varphi(u) = \varphi(v)$. Then, each $w \in W \setminus \{u, v\}$ has at most $\Delta - 2$ forbidden colours, so we can extend φ to $V(D)$ greedily.

Let us now see that $D \cup \{uv, vu\}$ does not contain any copy of $\overleftrightarrow{K}_\Delta$. Suppose for a contradiction that we can find a set W of $\Delta - 2$ vertices such that $X = \{u, v\} \cup W$ is a biclique of $D \cup \{uv, vu\}$ of size Δ.

Claim 22.1. *For every $w \in W$, $d^-(w) = d^+(w) = \Delta$.*

Proof of claim. Since $\tilde{\Delta}(D) \leqslant \Delta$, it is enough to show that $d^-(w), d^+(w) \geqslant \Delta$. Since D is Δ-critical, there is a $(\Delta - 1)$-dicolouring φ of $D - X$. Note that $\langle u, v\rangle$ is not a digon, otherwise X would be a biclique of D, a contradiction. Moreover, by Lemma 20, both u and v have at most three out-neighbours outside W. Therefore, φ can be extended to $V(D) \setminus W$ by assigning the same colour to u and v. Now, if there was a vertex $w \in W$ with $\min(d^-(w), d^+(w)) \leqslant \Delta - 1$, then, due to u and v having the same colour, we could extend φ to a $(\Delta - 1)$-dicolouring of D greedily, by taking care that w is coloured in the last place, a contradiction. $\Diamond$

Let us denote by $w_1, \ldots, w_{\Delta-2}$ the vertices in W, and, for $1 \leqslant i \leqslant \Delta - 2$, let w_i^- and w_i^+ be the unique in-neighbour and out-neighbour of w_i in $V(D) \setminus X$, given by Claim 22.1. Note that w_i^- and w_i^+ are not necessarily distinct, and that some (non-trivial) identities of the kind $w_i^- = w_j^-$, $w_i^+ = w_j^+$, or $w_i^- = w_j^+$ may hold. Generalising the argument used in the proof of Claim 22.1, we actually obtain the following.

Claim 22.2. *Let φ be a $(\Delta - 1)$-dicolouring of $D - X$. Then,*

(i) for every $1 \leqslant i \leqslant \Delta - 2$ there is a monochromatic directed path from w_i^+ to w_i^- in $D - X$ (possibly of length 0), and

(ii) $\varphi(w_1^-) = \varphi(w_1^+) = \ldots = \varphi(w_{\Delta-2}^-) = \varphi(w_{\Delta-2}^+)$.

Proof of claim. Assume that for some $1 \leqslant i \leqslant \Delta - 2$, property *(i)* does not hold. Then we can extend φ to a $(\Delta - 1)$-dicolouring of D as follows. First, we assign the same colour to u and v (we have already justified in the proof of Claim 22.1 that this is possible), and then, we colour the vertices in $W \setminus \{w_i\}$ greedily. Finally, we assign to w_i a colour not assigned to any vertex in $X \setminus \{w_i\}$. This last step does not create any monochromatic directed cycle in $D[X]$, nor, by our assumption, any involving w_i and the vertices outside X. So φ can be extended to a $(\Delta - 1)$-dicolouring of D, a contradiction.

Now let us see *(ii)*. By *(i)* we already know that $\varphi(w_i^-) = \varphi(w_i^+)$ for every $1 \leqslant i \leqslant \Delta - 2$. Assume that $\varphi(w_i^+) \neq \varphi(w_j^+)$ for some $i \neq j$. Then we can extend φ to a $(\Delta - 1)$-dicolouring of D as follows. First, we assign the same colour to u and v, with the additional requirement that this colour is different from $\varphi(w_j^+)$. The justification that this can be done without creating monochromatic directed cycles is the same as before. Then, we assign to w_i the colour $\varphi(w_j^+)$. By assumption, this does not create any monochromatic directed cycle. We can then colour the vertices in $W \setminus \{w_j\}$ greedily, because they have two out-neighbours sharing a colour and one

uncoloured out-neighbour. Finally, w_j has two pairs of out-neighbours sharing a colour, namely, u, v and w_i, w_j^+, so w_j can also be coloured greedily using one of the original $\Delta - 1$ colours, a contradiction. $\Diamond$

The point is that now some structure is forced.

Claim 22.3. *Let $1 \leqslant i, j \leqslant \Delta - 2$.*

(i) If $w_i^- \neq w_j^+$ and $w_j^- \neq w_i^+$, then the digraph $D' = D - X \cup \{w_i^- w_j^+, w_j^- w_i^+\}$ contains a copy of $\overleftrightarrow{K}_\Delta$ or of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.

(ii) If $w_i^- \neq w_j^+$ and $w_j^- = w_i^+$, then the digraph $D - X \cup \{w_i^- w_j^+\}$ contains a copy of $\overleftrightarrow{K}_\Delta$ or of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.

Proof of claim. We prove (i); (ii) can be proved analogously. Let us begin by showing that $\vec{\chi}(D') \geqslant \Delta$. Assume for a contradiction that there is a $(\Delta - 1)$-dicolouring φ of D'. Since φ is also a dicolouring of $D - X$, by Claim 22.2 we can find directed paths from w_i^+ to w_i^- and from w_j^+ to w_j^-, in which all the vertices have been assigned the colour $\varphi(w_i^+)$. Together with the arcs $w_i^- w_j^+$ and $w_j^- w_i^+$, this yields a monochromatic directed cycle in D'. Hence, $\vec{\chi}(D') \geqslant \Delta$.

By construction, observe that both inequalities $d^-_{D'}(x) \leqslant d^-_D(x)$ and $d^+_{D'}(x) \leqslant d^+_D(x)$ hold for all vertices $x \in V(D')$. Therefore, D' satisfies $\vec{\chi}(D') \geqslant \Delta \geqslant \tilde{\Delta}(D')$. The choice of D implies that D' contains a copy of $\overleftrightarrow{K}_\Delta$ or of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. $\Diamond$

As a consequence, we obtain the following.

Claim 22.4. *Let $1 \leqslant i, j \leqslant \Delta - 2$ such that $w_i^- \neq w_j^+$. Then, either $N^\pm(w_i^-) \cap N^\pm(w_j^+)$ contains a biclique of $D - X$ of size $\Delta - 4$, or $w_j^- \neq w_i^+$ and $N^\pm(w_j^-) \cap N^\pm(w_i^+)$ contains such a biclique.*

Proof of claim. Let K be the copy of $\overleftrightarrow{K}_\Delta$ or of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$ found by (i) or (ii) of Claim 22.3. Recall that K cannot be a subdigraph of D, so it uses at least one arc of $\{w_i^- w_j^+, w_j^- w_i^+\}$. Assume that $w_i^- w_j^+ \in A(K)$, the other case being symmetric.

Note that, in K, for each pair of vertices $\{x, y\}$, there exists a biclique of size $\Delta - 3$ in $N^\pm_{D'}(x) \cap N^\pm_{D'}(y)$. In particular K contains a biclique Y_{ij} of size $\Delta - 3$ in $N^\pm_{D'}(w_i^-) \cap N^\pm_{D'}(w_j^+)$. Note that Y_{ij} is not necessarily a biclique of D as, in K, it may contain the arc $w_j^- w_i^+$. However, either $Y_{ij} \setminus \{w_j^-\}$ or $Y_{ij} \setminus \{w_i^+\}$ is a biclique of D in $N^\pm_D(w_i^-) \cap N^\pm_D(w_j^+)$. The result follows. $\Diamond$

With the next claim, we make sure that Claim 22.4 can be used.

Claim 22.5. *There exist $1 \leqslant i, j \leqslant \Delta - 2$ such that $w_i^- \neq w_j^+$.*

Proof of claim. Suppose that the statement is not true. Then, there is a vertex w such that $w_i^- = w_i^+ = w$ for every $1 \leqslant i \leqslant \Delta - 2$. In other words, w forms a digon with every vertex of W, just as u and v. Observe that the subdigraph of D induced by $\{u, v, w\}$ is acyclic. Indeed, if $D[\{u, v, w\}]$ had a digon, the vertices of the digon together with W would form a biclique of D of size Δ. And if, otherwise, $D[\{u, v, w\}]$ was a copy of $\vec{C}_3$, then $D[X \cup \{w\}]$ would be isomorphic to $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, a contradiction.

Now, by the Δ-criticality of D, there is a $(\Delta - 1)$-dicolouring of $D - X - w$. Since the set $\{u, v, w\}$ is acyclic (and because, by Lemma 20, u, v and w have at most 3 out-neighbours outside W each), φ can be extended to a $(\Delta - 1)$-dicolouring of $D - W$ by assigning the same colour to u, v, and w. Then, by Claim 22.1, this can be extended to a $(\Delta - 1)$-dicolouring of D by colouring the vertices in W greedily, a contradiction. $\Diamond$

The final claim shows the existence of vertices with in-degree and out-degree at least $\Delta + 2$. This contradicts Lemma 20, thus completing the proof of Lemma 22.

Claim 22.6. *There exists some $1 \leqslant i \leqslant \Delta - 2$ and two sets of vertices $Z^-, Z^+ \subseteq V(D)$ such that*

(i) $|Z^-| + |Z^+| \geqslant 11$;

(ii) $w_i^- \in Z^-$ *and, unless* Z^+ *is empty,* $w_i^+ \in Z^+$;

(iii) every vertex in $Z^- \cup Z^+$ *is of the form* w_j^- *or* w_j^+ *for some* $1 \leqslant j \leqslant \Delta - 2$.

(iv) for every $z \in Z^- \setminus \{w_i^-\}$, $N^\pm(w_i^-) \cap N^\pm(z)$ *contains a biclique* Y_z^- *of* $D - X$ *of size* $\Delta - 4$;

(v) for every $z \in Z^+ \setminus \{w_i^+\}$, $N^\pm(w_i^+) \cap N^\pm(z)$ *contains a biclique* Y_z^+ *of* $D - X$ *of size* $\Delta - 4$.

Proof of claim. The proof is by induction on $|Z^-| + |Z^+|$. Let us prove first that the statement holds if (i) is replaced by $|Z^-| \geqslant 2$. By Claims 22.4 and 22.5, it is enough to take $Z^- = \{w_i^-, w_j^+\}$ and $Z^+ = \varnothing$ for some adequate $1 \leqslant i, j \leqslant \Delta - 2$.

Let us now assume that the statement holds if (i) is replaced by $|Z^-| + |Z^+| \geqslant t$, with $2 \leqslant t \leqslant 10$, and prove that it is also true with $|Z^-| + |Z^+| \geqslant t + 1$. By Lemma 20, each vertex $z \in Z^- \cup Z^+$ has at most five in-neighbours and five out-neighbours in W. Since Δ is large, there exists some $1 \leqslant k \leqslant \Delta - 2$ with $w_k^-, w_k^+ \notin Z^- \cup Z^+$. We apply Claim 22.4 to w_i^- and w_k^+. There are two possible outcomes. If $N^\pm(w_i^-) \cap N^\pm(w_k^+)$ contains a biclique of $D - X$ of size $\Delta - 4$, then we simply add w_k^+ to Z^-. Else, $N^\pm(w_k^-) \cap N^\pm(w_i^+)$ contains a biclique of $D - X$ of size $\Delta - 4$, and we add w_k^- to Z^+, together with w_i^+ if it was not already in Z^+. ◊

Let $1 \leqslant i \leqslant \Delta - 2$ and $Z^-, Z^+ \subseteq V(D)$ given by Claim 22.6. Let us assume that $|Z^-| \geqslant 6$; if $|Z^+| \geqslant 6$, the proof goes the same way. We can further assume that Z^- has exactly 6 elements; otherwise, we simply truncate it. Observe that, for any $z \in Z^- \setminus \{w_i^-\}$, the biclique Y_z^- given by Claim 22.6 is disjoint from Z^-: indeed, by Claims 22.2 and 22.6, any $(\Delta - 1)$-dicolouring of $D - X$ is constant on Z^-, and z forms a digon with every vertex of Y_z^-. Now, by Lemma 20, for any $z \in Z^- \setminus \{w_i^-\}$, $|N^\pm(w_i^-) \setminus Y_z^-| \leqslant 4$. Therefore,

$$\left| \bigcap_{z \in Z^- \setminus \{w_i^-\}} Y_z^- \right| \geqslant (\Delta - 4) - 4(|Z^-| - 2) > 0.$$

Let thus $y \in \bigcap_{z \in Z^- \setminus \{w_i^-\}} Y_z^-$ and $z \in Z^- \setminus \{w_i^-\}$. Note that y forms a digon with every vertex in $Y_z^- \sqcup Z^- \setminus \{y\}$. Thus $|N^\pm(y)| \geqslant \Delta + 1$, a contradiction to $\tilde{\Delta}(D) \leqslant \Delta$. □

With the same type of arguments, we now forbid copies of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-3}$ in D.

Lemma 23. *The digraph D does not contain any copy of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-3}$.*

Proof. Assume for a contradiction that there exist two disjoint sets $\{x, y, z\}, W \subseteq V(D)$ such that $D[\{x, y, z\}]$ contains a 3-cycle, W is a biclique of size $\Delta - 3$, and D contains all possible digons between $\{x, y, z\}$ and W. We let $X = \{x, y, z\} \cup W$. Note that, by Lemma 22, $D[\{x, y, z\}]$ contains at most one digon. We thus assume without loss of generality that both xy and yz are simple arcs.

Claim 23.1. *For every $w \in W$, $d^-(w) = d^+(w) = \Delta$.*

Proof of claim. The proof is similar to that of Claim 22.1. By Lemma 20, it is enough to show that $d^-(w), d^+(w) \geqslant \Delta$. Let φ be a $(\Delta - 1)$-dicolouring of $D - X$. We extend φ to $V(D) \setminus W$ by assigning the same colour to x and y. Now, if there was a vertex $w \in W$ with $\min(d^-(w), d^+(w)) \leqslant \Delta - 1$, then, due to x and y having the same colour, we could extend φ

to a $(\Delta-1)$-dicolouring of D greedily, by taking care that w is coloured in the last place, a contradiction. $\Diamond$

We label $w_1, \ldots, w_{\Delta-3}$ the vertices of W, and for every $i \in [\Delta-3]$, we denote by w_i^- and w_i^+ the unique in- and out-neighbour of w_i outside X. Following exactly the proof of Claim 22.2, we obtain the following. We give it for completeness.

Claim 23.2. *Let φ be a $(\Delta-1)$-dicolouring of $D-X$. Then,*

(i) for every $1 \leqslant i \leqslant \Delta-3$ there is a monochromatic directed path from w_i^+ to w_i^- in $D-X$ (possibly of length 0), and

(ii) $\varphi(w_1^-) = \varphi(w_1^+) = \ldots = \varphi(w_{\Delta-3}^-) = \varphi(w_{\Delta-3}^+)$.

Proof of claim. If (i) does not hold for some $i \in [\Delta-3]$, then we can extend φ to a $(\Delta-1)$-dicolouring of D as follows. First, we assign a same colour to x and y, and then, we colour the vertices in $\{z\} \cup W \setminus \{w_i\}$ greedily, starting with z. Finally, we assign to w_i a colour not assigned to any vertex in $X \setminus \{w_i\}$.

For (ii), assume that $\varphi(w_i^+) \neq \varphi(w_j^+)$ for some $i \neq j$. Then we can extend φ to a $(\Delta-1)$-dicolouring of D as follows. First, we assign the same colour c_1 to x and y, and a colour c_2 to z, with the additional requirement that both c_1 and c_2 are different from $\varphi(w_j^+)$. Then, we assign to w_i the colour $\varphi(w_j^+)$. We can then colour the vertices in $W \setminus \{w_j\}$ greedily. At the end, w_j has two pairs of out-neighbours sharing a colour, namely, x, y and w_i, w_j^+, so w_j can also be coloured greedily, a contradiction. $\Diamond$

Actually, all vertices w_i^-, w_i^+ must be the same one vertex.

Claim 23.3. $w_1^- = w_1^+ = \ldots = w_{\Delta-3}^- = w_{\Delta-3}^+$

Proof of claim. Assume first that, for some $i \in [\Delta-3]$, $w_i^- \neq w_i^+$. Let D' be the digraph obtained from $D-X$ by further adding the arc $w_i^- w_i^+$. Note that $\tilde{\Delta}(D) \geqslant \tilde{\Delta}(D')$, and that D' does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$ by Lemma 22. Therefore, the minimality of $|V(D)| + |A(D)|$ implies the existence of a $(\Delta-1)$-dicolouring of D', which by construction is a dicolouring of $D-X$ without any monochromatic directed path from w_i^+ to w_i^-, a contradiction to Claim 23.2.

Assume then that, for some i, j, $w_i^- = w_i^+ \neq w_j^- = w_j^+$. Similarly, let D' be the digraph obtained from $D-X$ by further adding the digon $\langle w_i^-, w_j^- \rangle$. Since $w_i^- = w_i^+$ and $w_j^- = w_j^+$, we have $\tilde{\Delta}(D) \geqslant \tilde{\Delta}(D')$, and D' does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$ by Lemma 22. Therefore, the minimality of $|V(D)| + |A(D)|$ implies the existence of a $(\Delta-1)$-dicolouring φ of D', which by construction is a dicolouring of $D-X$ satisfying $\varphi(w_i^-) \neq \varphi(w_j^-)$, a contradiction to Claim 23.2. $\Diamond$

Let $w \in V(D) \setminus X$ be such that $W \subseteq N^{\pm}(w)$. Note that D contains all possible digons between W and $\{x, y, z, w\}$. By Lemma 22, we get that, in $D[\{x, y, z, w\}]$, no vertex is incident to two distinct digons. It is straightforward to check that, in particular, $\overline{D}[\{x, y, z, w\}]$ admits a matching $\{x_1y_1, x_2y_2\}$.

Let φ be a $(\Delta-1)$-dicolouring of $D - (X \cup \{w\})$. We can extend φ to D as follows. We first give a same colour c_1 to both x_1 and y_1 that is not appearing in $N^+(x_1) \cup N^+(y_1)$, and similarly a same colour c_2 to both x_2 and y_2. Now, two colours are repeated in the out- and in-neighbourhoods of every vertex of W, so φ can be greedily extended to D, yielding the contradiction. $\square$

For the next lemma, we first need a few specific definitions. We say that $X \subseteq V(D)$ is a **quasi-biclique** of D if one of the following holds:

(1) X is a biclique;

(2) $D[X] \cup \{xy\} = \overleftrightarrow{K}_{|X|}$ for some $x, y \in X$;

(3) $D[X] = \vec{C}_3 \boxplus \overleftrightarrow{K}_{|X|-3}$;

(4) $D[X] - x = \overleftrightarrow{K}_{|X|-1}$ and $|N^+(x) \cap X| \geqslant \frac{99}{100}\Delta$ hold for some $x \in X$; or

(5) X admits a partition (R, W, K) such that $|R| \leqslant 4$, $|K| > \frac{1}{2}|X|$, $W \cup K$ is a biclique, R dominates K, and vertices in K have in-degree exactly $\Delta + 1$.

For i $\in$ [5], we say that X is a **quasi-biclique of type (i)** if X satisfies property (i) above and none of the properties (j) for j < i.

In what follows, by applying the dense-decomposition lemma to D, we show that D is made of a collection of pairwise-disjoint quasi-bicliques and a set of sparse vertices. Recall that d was set to $\log^3 \Delta$ at the beginning of the section.

Lemma 24. *There is a partition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ of $V(D)$ such that:*

(i) each X_i is a quasi-biclique of D of size $|X_i| > \Delta - 300d$;

(ii) for every $i \in [t]$ and $u \in V(D) \setminus X_i$, $|N^+(u) \cap X_i| < \frac{100}{101}\Delta$.

(iii) vertices in S are d-sparse.

Proof. From now on, we denote $\Delta_{\max}(D)$ by $\Delta_{\max}$ for conciseness. Recall that, by Lemma 20, $\Delta \leqslant \Delta_{\max} \leqslant \Delta + 1$. We apply Lemma 9 with $\varepsilon = \frac{1}{100}$ and $d = \log^3(\Delta)$. We thus obtain a partition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ of $V(D)$ such that:

- for every $i \in [t]$, $|X_i| > \Delta - 300d$;
- for every $i \in [t]$ and every $u \in V(D)$, $u \in X_i$ if and only if $|N^+(u) \cap X_i| \geqslant \frac{99}{100}\Delta_{\max}$; and
- vertices in S are d-sparse.

In particular, the second property implies that vertices $u \in V(D) \setminus X_i$ satisfy $|N^+(u) \cap X_i| < \frac{99}{100}(\Delta + 1) < \frac{100}{101}\Delta$. Clearly, it thus remains to prove that each X_i is a quasi-biclique. We thus fix $i \in [t]$, and denote $D[X_i]$ by D_i. To prove that X_i is a quasi-biclique, our strategy is first to show that the complement $\overline{D_i}$ of D_i has no matching of size 3. We first prove that $\nu\left(\overline{D_i}\right)$ is indeed at most a small fraction of Δ.

Claim 24.1. $\nu\left(\overline{D_i}\right) < \dfrac{1}{50}\Delta_{\max}$.

Proof of claim. Assume for a contradiction that $\nu\left(\overline{D_i}\right) \geqslant \Delta_{\max}/50$, and let $M = \{u_1v_1, \ldots, u_\ell v_\ell\}$ be a matching of $\overline{D_i}$ of size exactly $\ell = \lceil \Delta_{\max}/50 \rceil$. Note that, for every $j \in [\ell]$, there is at most one arc between u_j and v_j in D.

Let φ be a $(\Delta - 1)$-dicolouring of $D - X_i$, which exists as D is Δ-critical. We will show that φ can be extended into a $(\Delta - 1)$-dicolouring of D, thus yielding a contradiction. We first colour the vertices covered by M by sequentially choosing, for every $j \in [\ell]$, a common colour $c_j \in [\Delta - 1]$ for both u_j and v_j that is not already appearing in $N^+(u_j) \cup N^+(v_j)$. Note that this does not create any monochromatic directed cycle as there exists at most one arc between u_j and v_j. Let us show that such a colour c_j exists. By Lemma 9(iii), both u_j and v_j have

at most $\Delta_{\max}/100$ out-neighbours in $V(D) \setminus X_i$. Moreover, at most $j-1$ colours appear in X_i when colouring $\{u_j, v_j\}$ (namely $c_1, \ldots, c_{j-1}$). Hence, the number of colours that appear in $N^+(u_j) \cup N^+(v_j)$ is at most

$$\frac{1}{50}\Delta_{\max} + \ell - 1 \leqslant \frac{1}{25}\Delta_{\max} < \Delta - 1.$$

Once the vertices covered by M are coloured, we colour the remaining vertices of X_i greedily, by choosing for each of them a colour of $[\Delta - 1]$ that is not appearing in its out-neighbourhood. We claim that this is always possible, so assume for a contradiction that, for some $x \in X_i \setminus V(M)$, all $\Delta - 1$ colours appear in $N^+(x)$ when x is considered. Let $\iota(x)$ be the number of indices j such that $\{u_j, v_j\} \subseteq N^+(x)$. Since any two vertices matched in M received the same colour, the number of colours appearing in $N^+(x)$ is at most $d^+(x) - \iota(x)$. We thus have $d^+(x) - \iota(x) \geqslant \Delta - 1$, which together with Lemma 20 implies $\iota(x) \leqslant 2$. Therefore, at least $\ell - 2$ vertices covered by M do not belong to $N^+(x)$. Together with Lemma 9(i), this implies

$$|N^+(x) \cap X_i| \leqslant |X_i| - \ell + 2 < \frac{98}{100}\Delta_{\max} + 4d + 4 < \frac{99}{100}\Delta_{\max}.$$

By Lemma 9(iii), x does not belong to X_i, a contradiction. The claim follows. ◊

We can now prove that indeed, $\nu(\overline{D_i}) \leqslant 2$.

Claim 24.2. $\nu(\overline{D_i}) \leqslant 2$.

Proof of claim. Let $M = \{u_1v_1, \ldots, u_{|M|}v_{|M|}\}$ be a maximum matching of $\overline{D_i}$, and assume for a contradiction that $|M| \geqslant 3$. By Claim 24.1, $|M| < \frac{1}{50}\Delta_{\max}$. We denote by Y the vertices of X_i that are not covered by M. Note that, by maximality of M, Y is a biclique of D. We define Z as the set of vertices in Y that are out-neighbours of all vertices $u_1, u_2, u_3, v_1, v_2, v_3$, so formally:

$$Z = Y \cap \left(\bigcap_{j=1}^{3} N^+(u_j)\right) \cap \left(\bigcap_{j=1}^{3} N^+(v_j)\right).$$

Subclaim 24.2.1. $|Z| \geqslant 3$.

Proof of subclaim. Let u be any vertex covered by M. Recall that, by Lemma 9(iii),

$$|N^+(u) \cap X_i| \geqslant \frac{99}{100}\Delta_{\max}.$$

Together with Claim 24.1, this implies $|N^+(u) \cap Y| \geqslant \frac{95}{100}\Delta_{\max}$. Since, by Lemma 9(i), $|Y| < |X_i| \leqslant \Delta_{\max} + 1 + 4d$, we obtain

$$|Y \setminus N^+(u)| < \frac{5}{100}\Delta_{\max} + 4d + 1 < \frac{6}{100}\Delta_{\max}.$$

Since this holds for every u saturated by M, and in particular for every $u \in \{u_1, u_2, u_3, v_1, v_2, v_3\}$, we have

$$|Y \setminus Z| < \frac{36}{100}\Delta_{\max}. \tag{5}$$

Recall that the size of Y is larger than $|X_i| - \frac{4}{100}\Delta_{\max}$ by Claim 24.1, which together with Lemma 9(i) implies

$$|Y| > \frac{96}{100}\Delta_{\max} - 300d. \tag{6}$$

Combining (5) and (6), straightforward calculations imply $|Z| > \frac{1}{2}\Delta_{\max} \geqslant 3$, as desired. ◿

We let φ be a $(\Delta-1)$-dicolouring of $D-Y$, with the extra property that $\varphi(u_j)=\varphi(v_j)$ holds for every $j \in [|M|]$. We skip the proof that φ exists, it is shown by taking a $(\Delta-1)$-dicolouring of $D-X_i$ and extending it to $D-Y$ by choosing, for every $j \in [|M|]$, a colour that is not appearing in $N^+(u_j) \cup N^+(v_j)$, exactly as in the proof of Claim 24.1.

We now prove that φ can be extended into a $(\Delta-1)$-dicolouring of D, thus yielding the contradiction. We first greedily colour the vertices of $Y-Z$ by choosing for each of them a colour that does not appear in its out-neighbourhood. Let us show that this is possible. Since Y is a biclique, and because $|Z| \geqslant 3$ by Subclaim 24.2.1, note that when we consider $v \in Y-Z$, it has at least three uncoloured neighbours (namely the vertices of Z). Since, by Lemma 20, $d^+(v) \leqslant \Delta+1$, this implies that the number of colours appearing in $N^+(v)$ (when considered) is at most $\Delta-2$.

We then extend the obtained dicolouring to D by choosing for every vertex $z \in Z$ a colour that does not appear in its in-neighbourhood. Again, let us show that this is possible. Since all vertices $u_1, u_2, u_3, v_1, v_2, v_3$ belong to $N^-(z)$ (by definition of Z), and because $\varphi(u_j)=\varphi(v_j)$ for every $j \in [3]$, the number of colours appearing in the in-neighbourhood of z (when considered) is at most $d^-(z)-3$, which is at most $\Delta-2$ by Lemma 20. This yields a contradiction, and concludes the proof of Claim 24.2. $\Diamond$

We will now prove that X_i is a quasi-biclique. We distinguish two cases, depending on the value of $\nu(\overline{D_i})$.

Case 1: $\nu(\overline{D_i}) \leqslant 1$.

If $\nu(\overline{D_i})=0$, then X_i satisfies (1). We thus assume that $\nu(\overline{D_i})=1$ and we let u and v be two vertices with at most one arc between them. Observe first that if $(X_i \setminus \{u,v\}) \subseteq N^{\pm}(u)$, then X_i satisfies (4), and X_i is a quasi-biclique. We can thus assume that there exists $x_u \in X_i \setminus \{u,v\}$ such that there is at most one arc between x_u and u. Similarly, we can assume that there exists $x_v \in X_i \setminus \{u,v\}$ such that there is at most one arc between x_v and v. Since $\nu(\overline{D_i})=1$, we must have $x_u=x_v$, and D contains every possible digon between $\{u,v,x_u\}$ and $X_i \setminus \{u,v,x_u\}$. Note that $D[\{u,v,x_u\}]$ does not contain any digon by construction. Hence, if it contains a directed cycle, it has size exactly 3, and X_i satisfies (3).

Assume now that $D[\{u,v,x_u\}]$ is acyclic and let us reach a contradiction. Let φ be any $(\Delta-1)$-dicolouring of $D-X_i$. We can extend φ to $\{u,v,x_u\}$ in such a way that $\varphi(u)=\varphi(v)=\varphi(x_u)$. Again, this is possible because $D[\{u,v,x_u\}]$ is acyclic, and because D contains all digons between $\{u,v,x_u\}$ and $X_i-\{u,v,x_u\}$. Now, φ can be extended to the remaining vertices of X_i because each such vertex has one colour used three times in both its in- and out-neighbourhoods.

Case 2: $\nu(\overline{D_i})=2$.

Let $\{u_1v_1, u_2v_2\}$ be a maximum matching of $\overline{D_i}$. Let $R=\{u_1,v_1,u_2,v_2\}$ and K be the set of vertices in $X_i \setminus R$ that are in the out-neighbourhood of all vertices in R, so formally

$$K=(X_i \setminus R) \cap \left(\bigcap_{u \in R} N^+(u) \right).$$

By definition, note that K is disjoint from R. Let $W=X_i \setminus (R \cup K)$. We are now going to show that the partition (R,W,K) of X_i satisfies (5). By construction, we have:

- $|R|=4$,
- $W \cup K$ is a biclique (as M is maximum), and

- R dominates K.

It remains to show that $K > \frac{1}{2}|X_i|$ and that every vertex in K has in-degree $\Delta+1$. We first show the former. The proof is similar to that of Subclaim 24.2.1. Let y be any vertex in R. By Lemma 9(iii), $|N^+(y) \cap X_i| \geqslant \frac{99}{100}\Delta_{\max}$. Since, by Lemma 9(i), $|X_i| < \Delta_{\max} + 1 + 4d$, we have $|(X_i \setminus N^+(y)) \cup R| < \frac{1}{100}\Delta_{\max} + 4 + 4d < \frac{2}{100}\Delta_{\max}$. Since this holds for every $y \in R$, we have

$$|X_i \setminus K| < \frac{8}{100}\Delta_{\max}. \tag{7}$$

Recall that $\Delta_{\max} - 300d < |X_i| < \Delta_{\max} + 4d + 1$ by Lemma 9(i). Hence (7) implies $|K| > \frac{1}{2}|X_i|$, as desired.

To prove that every vertex $x \in K$ has in-degree at least $\Delta + 1$, assume for a contradiction that $d^-(x) \leqslant \Delta$ for some $x \in K$. Let φ be a $(\Delta - 1)$-dicolouring of $D - (K \cup W)$, with the extra property that $\varphi(u_1) = \varphi(v_1)$ and $\varphi(u_2) = \varphi(v_2)$.

Let y be any vertex in K distinct from x. Since two colours (namely $\varphi(u_1)$ and $\varphi(u_2)$) are repeated in the in-neighbourhood of both x and y, and because $d^-(x) \leqslant \Delta$, one can extend φ to D by first colouring the vertices of $(K \cup W) \setminus \{x, y\}$ (choosing for each of them a colour that is not appearing either in its in- or out-neighbourhood), and then colouring y and x in this order. This yields the desired contradiction, shows that X_i is a biclique of type (5), and concludes the proof of Lemma 24. □

From now on, we fix a decomposition $X_1 \sqcup \ldots \sqcup X_t \sqcup S$ of $V(D)$ satisfying the properties of Lemma 24. For the next lemma, we need a few additional technical definitions. Let X be a quasi-biclique of D, and let r be any positive real number. We define $\mathscr{Z}_r(X)$ as the set of vertices in $V(D) \setminus X$ with less than r in- and out-neighbours in X, that is

$$\mathscr{Z}_r(X) = \left\{ z \in V(D) \setminus X : \max\left(|N^-(z) \cap X|, |N^+(z) \cap X|\right) < r \right\}.$$

The **saving-part** $\mathscr{S}(X)$ of X is defined as:

- X itself if X is of type (1);
- $X \setminus \{x, y\}$ if X is of type (2);
- $X \setminus V(\vec{C}_3)$ if X is of type (3);
- $X \setminus \{x\}$ if X is of type (4); and
- K if X is of type (5).

We say that a vertex $x \in \mathscr{S}(X)$ is an r**-saviour** of X if one of the following holds:

(a) $d^+(x) = \Delta - 1$ and $\mathscr{Z}_r(X) \cap N^+(x) \neq \varnothing$;

(b) $d^-(x) = \Delta - 1$ and $\mathscr{Z}_r(X) \cap N^-(x) \neq \varnothing$;

(c) $d^+(x) \leqslant \Delta$ and $|\mathscr{Z}_r(X) \cap N^+(x)| \geqslant 2$; or

(d) $d^-(x) \leqslant \Delta$ and $|\mathscr{Z}_r(X) \cap N^-(x)| \geqslant 2$.

Intuitively speaking, if r is small enough compared to Δ, when applying the pseudo-random colouring process of the next section, a saviour will have a quite high probability to be uncoloured and to have either two repeated colours in its out- or in-neighbourhood (if (c) or (d) holds) or

only one repeated colour but one missing arc (if (a) or (b) holds). If this property holds for at least three saviours of X, and if these saviours are in the out-neighbourhood of every vertex of X, we will be able to extend a partial dicolouring of D to X.

The following two lemmas say that, in a minimum counterexample to Theorem 8, large quasi-bicliques indeed have many saviours. From now on, we fix $r = \log^4 \Delta$.

Lemma 25. *For every $i \in [t]$, if X_i is of type (5) with partition (R_i, K_i, W_i), then elements of K_i are r-saviours.*

Proof. We fix $i \in [t]$ such that X_i is of type (5). For the sake of conciseness, we denote X_i, R_i, K_i, and W_i by X, R, K, and W respectively. We further denote $\mathscr{S}(X)$ by $\mathscr{S}$ and denote $\mathscr{Z}_r(X)$ by $\mathscr{Z}_r$. By definition, recall that $|R| \leqslant 4$, R dominates K, $K \cup W$ is a biclique, $|K| > \frac{1}{2}|X|$, and vertices in K have in-degree $\Delta + 1$. We claim that all vertices in K satisfy (a), thus implying the statement.

Assume for a contradiction that $s \in K$ does not satisfy (a). By definition of Δ and because $d^-(s) = \Delta + 1$, we have $d^+(s) = \Delta - 1$. Since $d^-(s) \neq d^+(s)$, s is adjacent to a simple arc, and by Lemma 21 we know that s has a simple out-neighbour y. Since R dominates K, and $K \cup W$ is a biclique, we thus necessarily have $y \in V(D) \setminus X$. As s does not satisfy (a), $y \notin \mathscr{Z}_r$, that is, y has at least r out-neighbours or r in-neighbours in X. Let u_y be any vertex in $(K \cup W) \setminus N^+(y)$ distinct from s. Note that u_y exists because $y \notin X$, so $|N^+(y) \cap X| < \frac{100}{101}\Delta$, while $|K \cup W| = |X| - |R| > \Delta - 300d - 4$.

Let φ be a $(\Delta - 1)$-dicolouring of $D - (X \cup \{y\})$. We claim that there exists a colour c that is not appearing either in $N^-(y) \cup N^-(u_y)$ or in $N^+(y) \cup N^+(u_y)$. To see this, assume that y has r out-neighbours in X. Then y has r uncoloured out-neighbours, and u_y has at least $|K \cup W| - 1 \geqslant |X| - 5$ uncoloured out-neighbours. The total number of colours appearing in $N^+(y) \cup N^+(u_y)$ is thus at most

$$(\Delta + 1) - r + (\Delta + 1) - (\Delta - 301d) = \Delta - \log^4(\Delta) + 301 \log^3(\Delta) + 2,$$

which is less than $\Delta - 1$ when Δ is large enough. Similarly, if y has r in-neighbours in X, then at most $\Delta + 1 - r$ colours appear in its in-neighbourhood, while at most $301d$ colours appear in the in-neighbourhood of u_y.

We thus extend φ by first choosing colour c for both y and u_y. Then, we greedily colour the vertices of R and W in this order. This is possible because K is large, uncoloured, and in the out-neighbourhood of all vertices of X. We then colour the vertices of K, keeping s for the end. Since the vertices in $K \setminus \{s\}$ have out-degree $\Delta - 1$ and an uncoloured out-neighbour, this is always possible for them. For s, this is possible because $d^+(s) = \Delta - 1$ and one colour (namely c) is repeated in its out-neighbourhood. This yields a contradiction. □

Lemma 26. *For every $i \in [t]$, if X_i is of type (1)–(4), it has at least $\frac{1}{2}|X_i|$ r-saviours.*

Proof. We fix $i \in [t]$ and assume that X_i is of type (1)–(4). Using ideas analogous to the one used in the proof of Lemma 25, we first show a collection of structural results around non-r-saviour vertices. We then leverage these to show that many vertices must indeed be r-saviours.

For the sake of conciseness, again, we denote X_i by X and $\mathscr{S}(X)$ by $\mathscr{S}$. Similarly, for every r', we denote $\mathscr{Z}_{r'}(X)$ by $\mathscr{Z}_{r'}$. We further denote by $\mathscr{R}$ the set of r-saviours of X. Finally, we let Y be $V(D) \setminus X$, unless X is of type (4) in which case Y is $(V(D) \setminus X) \cup \{x\}$, where x is the special vertex witnessing that X is of type (4).

Along the proof, we will use many times that $|\mathscr{S}| \geqslant |X| - 3$ and that every vertex $s \in \mathscr{S}$ is linked with digons to all vertices in $X \setminus (Y \cup \{s\})$. Note that these two properties hold because X is of type (1)–(4). We further have the following.

Claim 26.1. *Every vertex $s \in \mathscr{S}$ satisfies $|N(s) \cap (X \setminus Y)| \leqslant \Delta - 2$.*

Proof of claim. If X is of type (1) or (4), this is because D does not contain any biclique of size Δ. Else, if X is of type (2), this is a consequence of Lemma 22. Finally, if X is of type (3), this is a consequence of Lemma 23. $\Diamond$

Intuitively, and similarly to the proof of Lemma 25, when some vertex $y \in Y$ has many neighbours in $(X \setminus Y)$, we mainly want to argue that we can extend a $(\Delta - 1)$-colouring of $D - (X \cup \{y\})$ by first colouring y and a vertex $u_y \in X \setminus Y$ with a same colour, and then greedily colour the remaining vertices in X in a specific order. The first step consists of showing that actually, we can find such a vertex u_y that is not linked with a digon to y.

Claim 26.2. *Let $s \in \mathscr{S}$ and $y \in Y$ be a neighbour of s. There exists a vertex $u_y \in X \setminus Y$, distinct from s, such that $\langle y, u_y \rangle$ is not a digon of D. Moreover, if $y_1, y_2 \in Y$ are two distinct neighbours of s, then there exist distinct vertices $u_1, u_2 \in X \setminus Y$, both distinct from s, such that none of $\{\langle y_1, u_1 \rangle, \langle y_2, u_2 \rangle\}$ are digons of D.*

Proof of claim. We first prove the first part of the claim. Assume that this is not the case, so y is linked with a digon to all vertices in $X \setminus (Y \cup \{s\})$. Since vertices outside X have less than $\frac{100}{101}\Delta$ out-neighbours in X, and because $|X| > \Delta - 300d$, we have $y \in X$. Since y is chosen in Y, it means that $Y \cap X \neq \varnothing$, hence X is of type (4), and y is the special vertex such that $X \setminus \{y\}$ is a biclique of D. Since y is a neighbour of s, adding at most one arc between y and s to $D[X]$ yields a complete digraph. Hence X is either of type (1) or (2), a contradiction (recall that X being of type (i) implies that X is not of type (j) for any j < i).

To show the second part, note that at most one of y_1, y_2 belongs to X, so assume by symmetry $y_2 \notin X$. We apply the first part of the statement to y_1 and thus find u_1. Since $y_2 \notin X$, we can let u_2 be any vertex of $X \setminus (N^+(y_2) \cup Y \cup \{s, u_1\})$, which is non-empty as y_2 has less than $\frac{100}{101}\Delta$ out-neighbours in X, and $|X| > \Delta - 300d$. $\Diamond$

From now on, let us show some structure around vertices in $\mathscr{S} \setminus \mathscr{R}$, *i.e.* non r-saviours of $\mathscr{S}$.

Claim 26.3. *Every vertex $s \in \mathscr{S} \setminus \mathscr{R}$ satisfies $d^-(s) = d^+(s) = \Delta$.*

Proof of claim. Assume for a contradiction that $\min(d^+(s), d^-(s)) = \Delta - 1$. The proof is the same as that of Lemma 25, but we give it for completeness.

If $d^+(s) = \Delta - 1$, we let y be any out-neighbour of s in Y, otherwise $d^-(s) = \Delta - 1$ and we let y be any in-neighbour of s in Y. In both cases, y exists by Claim 26.1. Since $s \notin \mathscr{R}$, we know that $y \notin \mathscr{Z}_r$, for otherwise s would satisfy either (a) or (b). By definition of $\mathscr{Z}_r$, either $y \in X$, $|N^-(y) \cap X| \geqslant r$, or $|N^+(y) \cap X| \geqslant r$. If $y \in X$, then X being a quasi-biclique implies $|N^+(y) \cap X| \geqslant r$, so we can ignore this case. Let us thus assume that $|N^+(y) \cap X| \geqslant r$ or $|N^-(y) \cap X| \geqslant r$.

Let u_y be any vertex in $X \setminus (\{s\} \cup Y)$ such that $\langle y, u_y \rangle$ is not a digon of D, which exists by Claim 26.2. Let v be any vertex in $(N(y) \cap \mathscr{S}) \setminus \{u_y, s\}$, which exists as $|\mathscr{S}| \geqslant |X| - 3$.

Let φ be a $(\Delta - 1)$-dicolouring of $D - (X \cup \{y\})$. We greedily extend φ to D, thus yielding the contradiction. We first give a same colour c to both y and u_y that is not appearing in $N^+(y) \cup N^+(u_y)$ or $N^-(y) \cup N^-(u_y)$. This is possible because y has at least r uncoloured in- or out-neighbours and u_y is linked with digons to at least $|X| - 2 > \Delta - 301d$ uncoloured vertices. We then greedily colour the vertices of X, starting with the possible vertex in $Y \cap X$, and keeping v and s for the end (in this order). Since both v and s are linked with digons to all vertices in $X \setminus Y \setminus \{v, s\}$, it is possible to choose for each of them. For v, if $d^+(v) = \Delta - 1$ or $d^-(v) = \Delta - 1$, then we can pick a colour that is not appearing in $N^+(v)$ or in $N^-(v)$ as $\langle v, s \rangle$ is a digon and s is uncoloured. Else, $d^-(v) = d^+(v) = \Delta$, s is an uncoloured in-neighbour of v, and v has a

repeated colour (namely c) in its in- or out-neighbourhood. We can thus choose a colour that is not appearing in $N^-(v)$ or in $N^+(v)$. We finally consider s, for which either $d^+(s) = \Delta - 1$ and one colour (namely c) is repeated in $N^+(s)$, or $d^-(s) = \Delta - 1$ and c repeated in $N^-(s)$. In both cases, we can extend φ to D, a contradiction. $\Diamond$

For vertices s with in-degree and out-degree Δ, the argument used above does not work anymore, as we can only save one colour for s. However, if two neighbours y_1, y_2 outside $X \setminus Y$ indeed have a lot of neighbours in X, then we can manage to save two colours for s, and the argument works.

Claim 26.4. *For every vertex $s \in \mathscr{S}$ with* $\max(d^+(s), d^-(s)) \leqslant \Delta$,

$$|(N^+(s) \cap Y) \setminus \mathscr{Z}_{r-3}| \leqslant 1 \quad \textit{and} \quad |(N^-(s) \cap Y) \setminus \mathscr{Z}_{r-3}| \leqslant 1.$$

Proof of claim. Assume for a contradiction that some $s \in \mathscr{S}$ with $d^+(s) \leqslant \Delta$ has two distinct out-neighbours $y_1, y_2 \in Y$, both of them having at least $r - 3$ in- or out-neighbours in X. The case of s having $d^-(s) \leqslant \Delta$ and two in-neighbours $y_1, y_2 \in Y$, none of them belonging to $\mathscr{Z}_{r-3}$, is proved analogously.

Let φ be any $(\Delta - 1)$-dicolouring of $D - (X \cup \{y_1, y_2\})$. We will show that φ extends to D, thus yielding the contradiction. By Claim 26.2, we let u_1 and u_2 be two distinct vertices of $X \setminus Y$, both distinct from s, such that none of $\{\langle y_1, u_1\rangle, \langle y_2, u_2\rangle\}$ are digons of D. We finally let v be any vertex in $N(y_1) \cap \mathscr{S}$ distinct from u_1, u_2, and s. Note that we excluded the case of X being of type (5), so $|\mathscr{S}| \geqslant |X| - 3$, and v exists as $|N(y_1) \cap (\mathscr{S} \setminus \{u_1, u_2, s\})| \geqslant r - 9 > 0$. By definition, observe that $\langle s, v\rangle$ is a digon of D, and that D contains all possible digons between $\{s, v\}$ and $\{u_1, u_2\}$.

We first extend φ by choosing for y_1 and u_1 a common colour that is not appearing in $N^+(y_1) \cup N^+(u_1)$ if y_1 has $r-3$ out-neighbours in X, or in $N^-(y_1) \cup N^-(u_1)$ otherwise. Similarly, we choose for y_2 and u_2 a common colour that is not appearing either in $N^+(y_2) \cup N^+(u_2)$ or $N^-(y_2) \cup N^-(u_2)$. To see that this is possible, note that both y_1 and y_2 have at least $r - 4$ uncoloured in- or out-neighbours, and that both u_1 and u_2 have at least $\Delta - 301d$ uncoloured in- and out-neighbours (the ones in $X \setminus \{u_1\}$). The number of forbidden colours is thus at most

$$(\Delta + 1) - (r - 4) + (\Delta + 1) - (\Delta - 301d) = \Delta - \log^4(\Delta) + 301 \log^3(\Delta) + 6,$$

which is smaller than $\Delta - 1$ when Δ is large enough. We finally greedily colour the vertices in X that are not already coloured, starting with the possible vertex in Y (if it is not y_1 or y_2), and finishing with v and s in this order. The possible vertex in Y can be coloured because it has at least $\frac{99}{100}\Delta - 2$ uncoloured out-neighbours. Vertices in $X \setminus (Y \cup \{v, s\})$ can be coloured because they are linked with digons to both v and s, which are uncoloured.

At the end, v has a colour repeated in its in- or out-neighbourhood (the colour of u_1 and y_1) and one uncoloured neighbour s to which v is linked with a digon. Therefore, a colour is missing either in $N^-(v)$ or in $N^+(v)$. Finally, two colours are repeated in the out-neighbourhood of s (namely the colour of y_1 and u_1, and the colour of y_2 and u_2). Since we assumed $d^+(s) \leqslant \Delta$, we can actually colour s, yielding the contradiction. $\Diamond$

Recall that $|\mathscr{S}| \geqslant |X| - 3$ and that every vertex $s \in \mathscr{S}$ is linked with digons to all vertices in $X \setminus Y$. Therefore, since $|X| > \Delta - 300d$, the statement follows from Claims 26.3 and 26.4 when every vertex s of $\mathscr{S}$ satisfies $|N^{\pm}(s) \cap (X \setminus Y)| < \Delta - 2$ (note that $\mathscr{Z}_{r-3} \subseteq \mathscr{Z}_r$). Since vertices in $\mathscr{S}$ have at most $\Delta - 2$ neighbours in $X \setminus Y$ by Claim 26.1, we are done unless one of the following holds:

1. $D[X \setminus Y]$ is a biclique of size $\Delta - 1$ (if X is of type (1) or (4));

2. $D[X] \cup \{xy\} = \overleftrightarrow{K}_{\Delta-1}$ for some $x, y \in X$ (if X is of type (2)); or

3. $D[X] = \vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-4}$ (if X is of type (3)).

From now on, we thus assume that we are in one of the three cases above. Furthermore, combining Claims 26.1, 26.3, and 26.4, we obtain that every vertex $s \in \mathscr{S} \setminus \mathscr{R}$ has exactly two in-neighbours y_s^-, z_s^- and exactly two out-neighbours y_s^+, z_s^+ in Y (we may have $y_s^- = y_s^+$ and/or $z_s^- = z_s^+$). Furthermore, as $s \notin \mathscr{R}$ and by Claim 26.4, exactly one of $\{y_s^-, z_s^-\}$ belongs to $\mathscr{Z}_{r-3}$, and the other one has at least r in- or out-neighbours in X. Similarly, exactly one of $\{y_s^+, z_s^+\}$ belongs to $\mathscr{Z}_{r-3}$, and the other one has at least r in- or out-neighbours in X.

Henceforth, for every vertex $s \in \mathscr{S} \setminus \mathscr{R}$, we thus denote by z_s^- the unique vertex in $N^-(s) \cap \mathscr{Z}_{r-3}$ and by y_s^- the other vertex in $N^-(s) \cap Y$. Similarly, we denote by z_s^+ the unique vertex in $N^+(s) \cap \mathscr{Z}_{r-3}$ and by y_s^+ the other vertex in $N^+(s) \cap Y$. It turns out that z_s^- and z_s^+ are necessarily the same vertex.

Claim 26.5. *For every vertex $s \in \mathscr{S} \setminus \mathscr{R}$, $z_s^- = z_s^+$.*

Proof of claim. Assume for a contradiction that, for some $s \in \mathscr{S} \setminus \mathscr{R}$, $z_s^- s$ and $s z_s^+$ are two simple arcs. Let D' be the digraph obtained from D by removing $X \cup \{y_s^-, y_s^+\}$ and adding the arc $z_s^- z_s^+$ (note that we may have $y_s^- = y_s^+$). Note that $\tilde{\Delta}(D') \leqslant \tilde{\Delta}(D)$, and that by Lemma 22, D' does not contain any copy of $\overleftrightarrow{K}_\Delta$ nor $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. The minimality of D implies the existence of a $(\Delta-1)$-dicolouring φ of D'.

Hence, φ is a dicolouring of $D - (X \cup \{y_s^-, y_s^+\})$. Let us show that it extends to D, thus yielding the contradiction. Let u_y be any vertex in $\mathscr{S}$ such that $\langle y_s^-, u_y \rangle$ is not a digon of D, the existence of which is guaranteed by Claim 26.2. We first extend φ by colouring y_s^- and u_y with a common colour c that is not appearing either in $N^+(y_s^-) \cup N^+(u_y)$ or in $N^-(y_s^-) \cup N^-(u_y)$. If $y_s^- \neq y_s^+$, we further colour y_s^+ with a colour that is not appearing in $N^+(y_s^+) \cup \{z_s^-\}$ or in $N^-(y_s^-) \cup N^-(u_y)$.

Next, we greedily extend the obtained dicolouring to $D - \{s, v\}$ by choosing for every vertex a colour that is not appearing either in its in- or out-neighbourhood, where v is an arbitrary fixed vertex in $N(y_s^-) \cap (\mathscr{S} \setminus \{s, u_y\})$. Since s is uncoloured, $\langle s, v \rangle$ is a digon, and one colour is repeated in either $N^-(v)$ or $N^+(v)$, we can further extend this dicolouring to $D - s$. Finally, if one colour is not appearing in $N^-(s)$, we choose this colour for s, and we are done. If not, every colour of $[\Delta - 1]$ is appearing exactly once in $N^-(s)$, except the colour of u_y and y_s^- that is appearing twice exactly. We thus choose colour $\varphi(z_s^-)$ for s. We claim that the obtained colouring ψ is a dicolouring of D, so assume for a contradiction that D, coloured with ψ, contains a monochromatic directed cycle $\mathscr{C}$.

Since X is of type (1) to (4), we have $(X \setminus (Y \cup \{s\})) \subseteq N^{\pm}(s)$. Moreover, we explicitly fixed $\psi(y_s^+)$ to a colour distinct from $\psi(z_s^-)$, so $V(\mathscr{C}) \setminus V(D')$ is exactly $\{s\}$. In particular, the successor of s in $\mathscr{C}$ is z_s^+. Since ψ is equal to φ on $V(D')$, this implies that D', coloured with φ, contains a monochromatic path from z_s^+ to z_s^-. Since D' contains the arc $z_s^- z_s^+$, this is a contradiction. $\Diamond$

Henceforth, for better readability, given a vertex $s \in \mathscr{S} \setminus \mathscr{R}$, we thus let $z_s = z_s^- = z_s^+$.

Claim 26.6. *For every vertex $s \in \mathscr{S} \setminus \mathscr{R}$, $|N^+(z_s) \cap X| \leqslant 3$ and $|N^-(z_s) \cap X| \leqslant 3$.*

Proof of claim. Assume for a contradiction that, for some $s \in \mathscr{S} \setminus \mathscr{R}$, $|N^+(z_s) \cap X| \geqslant 4$, the other case being proved symmetrically.

Let u_y be any vertex in $X \setminus Y$, distinct from s, such that $\langle y_s^-, u_y \rangle$ is not a digon of D, the existence of which is guaranteed by Claim 26.2. If possible, we further choose u_y such that $u_y \notin N(y_s^-)$. We then let u_z be any vertex in $N(y_s^-) \cap \mathscr{S}$ distinct from vertices of $N^+(z_s) \cup \{u_y\}$

(note that it is also distinct from s as $s \in N^+(z_s)$). Let us justify that u_z exists. Recall that $z_s \in \mathscr{Z}_{r-3}$, and $|\mathscr{S} \setminus X| \leqslant 3$, hence

$$\Big|(N(y_s^-) \cap \mathscr{S}) \setminus (N^+(z_s) \cup \{u_y\})\Big| > \Big|N(y_s^-) \cap X\Big| - r - \Big|N(y_s^-) \cap \{u_y\}\Big|.$$

If u_y is not a neighbour of y_s^-, this is non-negative as $|N(y_s^-) \cap X| \geqslant r$. If u_y is a neighbour of y_s^- then, by choice of u_y, $(X \setminus Y) \subseteq N(y_s^-)$, which implies $|N(y_s^-) \cap X| \geqslant |X| - 1 \geqslant \Delta - 2$ and, again, $|(N(y_s^-) \cap \mathscr{S}) \setminus (N^+(z_s) \cup \{u_y\})| > 0$. This shows the existence of u_z.

We finally let v be any vertex in $N(y_s^-) \cap \mathscr{S}$ that is distinct from s, u_y, and u_z. Let φ be a $(\Delta - 1)$-dicolouring of $D - (X \cup \{y_s^-, z_s\})$. Let us show that it extends to D, thus yielding the contradiction.

We first give to z_s and u_z a common colour c that is not appearing in $N^+(z_s)$, and that is neither appearing in one of $N^-(u_z), N^+(u_z)$. Note that this does not create any monochromatic directed cycle because $zu_z \notin A(D)$. To see that such a colour c exists, note that at most $\Delta - 3$ colours appear in $N^+(z_s)$, and at most one colour appears either in $N^-(u_z)$ or in $N^+(u_z)$. For u_z, this is because $u_z \in \mathscr{S} \cap N(y_s^-)$, so u_z is linked with digons to at least $\Delta - 2$ vertices in X, and it is adjacent to y_s^-, which is uncoloured.

We then give to y_s^- and u_y a common colour that is not appearing either in $N^-(u_y) \cup N^-(y_s^-)$ or in $N^+(u_y) \cup N^+(y_s^-)$, which is possible because y_s^- has r in- or out-neighbours in X. We finally greedily colour the vertices in X that are not already coloured, starting with the possible vertex in Y, and finishing with v and s in this order. The possible vertex in Y can be coloured because it has at least $\frac{99}{100}\Delta - 2$ uncoloured out-neighbours. Vertices in $X \setminus (Y \cup \{v, s\})$ can be coloured because they are linked with digons to both v and s, which are uncoloured. At the end, v has a colour repeated in its in- or out-neighbourhood (the colour of y_s^- and u_y) and one uncoloured neighbour s to which v is linked with a digon. Hence, v can be coloured. Finally, two colours are repeated in the in-neighbourhood of s (namely the colour of y_s^- and u_y, and the one of z_s and u_z). Since $d^-(s) = \Delta$ by Claim 26.3, we can actually colour s, yielding the contradiction. $\Diamond$

Claim 26.7. *For every vertex $s \in \mathscr{S} \setminus \mathscr{R}$, we have $|N(y_s^-) \cap (\mathscr{S} \setminus \mathscr{R})| \leqslant 10$.*

Proof of claim. Assume for a contradiction that, for some $s \in \mathscr{S} \setminus \mathscr{R}$, $|N(y_s^-) \cap (\mathscr{S} \setminus \mathscr{R})| \geqslant 11$. Therefore, either $N^- = N^-(y_s^-) \cap (\mathscr{S} \setminus \mathscr{R})$ or $N^+ = N^+(y_s^-) \cap (\mathscr{S} \setminus \mathscr{R})$ has order at least 6. We let $N = N^-$ if $|N^-| > |N^+|$, and $N = N^+$ otherwise.

Let u_y be any vertex in $X \setminus Y$, distinct from s, such that $\langle y_s^-, u_y \rangle$ is not a digon of D, the existence of which is guaranteed by Claim 26.2. We further let s' be any vertex in

$$N \setminus \Big(N^+(z_s) \cup \{u_y, s\}\Big),$$

which is non-empty by Claim 26.6. We finally let v be any vertex in $N(y_s^-) \cap (X \setminus Y)$ distinct from s, u_y, and s'.

Note that s' is chosen in $(\mathscr{S} \setminus \mathscr{R})$, which allows us to consider vertex $z_{s'}$. Thus, let D' be the digraph obtained from D by first removing $X \cup \{y_s^-\}$, and further adding the digon $\langle z_s, z_{s'} \rangle$. Note that z_s and $z_{s'}$ are indeed distinct because $s' \notin N^+(z_s)$, and $z_{s'}$ is an in-neighbour of s' by definition. Moreover, $z_{s'}$ is distinct from y_s^- by Claim 26.6 applied to s'. By Claim 26.5, both z_s and $z_{s'}$ are linked with a digon to a vertex in X (namely s and s'), so in particular $\tilde{\Delta}(D') \leqslant \tilde{\Delta}(D)$. Furthermore, by Lemma 22, D' does not contain copy of $\overleftrightarrow{K}_\Delta$ nor $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. The minimality of D thus implies the existence of a $(\Delta - 1)$-dicolouring φ of D'. Let us show that it extends to D, thus yielding the contradiction.

We first give to s' colour $\varphi(z_s)$. This is possible because $s' \in (\mathscr{S} \setminus \mathscr{R}) \cap N(y_s^-)$, so exactly one colour appears in both the in- and out-neighbourhood of s', namely the one of $z_{s'}$, and $\varphi(z_{s'}) \neq \varphi(z_s)$ because $\langle z_s, z_{s'} \rangle$ is a digon of D'.

We then give to y_s^- and u_y a common colour that is not appearing either in $N^+(u_y) \cup N^+(y_s^-)$ or in $N^-(u_y) \cup N^-(y_s^-)$. We finally greedily colour the vertices in X that are not already coloured, starting with the possible vertex in Y, and finishing with v and s in this order. Again, this is always possible using the fact that we gave the same colour to both z_s and s', and the same to both y_s^- and u_y. $\Diamond$

With Claim 26.7 in hand, we are now able to prove that almost all vertices in $\mathscr{S}$ are in $\mathscr{R}$. To do this, let us count the following sum in two different ways:

$$S = \sum_{s \in \mathscr{S} \setminus \mathscr{R}} |N(y_s^-) \cap \mathscr{R}|.$$

Let s be any vertex in $\mathscr{S} \setminus \mathscr{R}$. Recall that $|X \setminus \mathscr{S}| \leqslant 3$. Hence, together with Claim 26.7 we thus have

$$|N(y_s^-) \cap \mathscr{R}| \geqslant |N(y_s^-) \cap X| - 13 \geqslant r - 13.$$

On the other hand, recall that vertices $s \in \mathscr{S}$ (and, a fortiori, vertices in $\mathscr{R}$) satisfy $|N^{\pm}(s) \cap (X \setminus Y)| \geqslant \Delta - 2$, because we are in one of the following cases:

1. $D[X \setminus Y]$ is a biclique of size $\Delta - 1$ (if X is of type (1) or (4));
2. $D[X] \cup \{xy\} = \overleftrightarrow{K}_{\Delta-1}$ for some $x, y \in X$ (if X is of type (2)); or
3. $D[X] = \vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-4}$ (if X is of type (3)).

This implies in particular that vertices $s \in \mathscr{S}$ have at most four neighbours in Y, *i.e.* $|N(s) \cap Y| \leqslant 4$. Since, for every vertex $s \in \mathscr{S} \setminus \mathscr{R}$, y_s^- belongs to Y, we have $S \leqslant 4|\mathscr{R}|$. Together with the inequality above, we thus have

$$4|\mathscr{R}| \geqslant S \geqslant |\mathscr{S} \setminus \mathscr{R}| \cdot (r - 13),$$

which implies

$$|\mathscr{R}| \cdot \left(\frac{4}{r - 13} + 1 \right) \geqslant |\mathscr{S}| \geqslant |X| - 3.$$

Since $r = \log^4(\Delta)$ and $|X| > \Delta - 300d$, and because Δ is arbitrarily large, we obtain $|\mathscr{R}| > \frac{1}{2}|X|$, concluding the proof of Lemma 26. $\square$

Now that we know that each X_i has a large number of saviours, we can derive the following. For every $i \in [t]$, we denote by $X_i^\star$ the set of vertices of X_i that are in the closed out-neighbourhood of all vertices of X_i, so formally

$$X_i^\star = X_i \cap \left(\bigcap_{x \in X_i} N^+[x] \right).$$

Lemma 27. *For every $i \in [t]$, there exists a collection $\mathscr{T}_i$ of $\lceil \frac{1}{50r} \Delta \rceil$ k_i-tuples of vertices of D, such that every vertex appears in at most one tuple and only once within that tuple, and such that one of the following holds.*

(A) *$k_i = 2$, and every $(x, u) \in \mathscr{T}_i$ satisfies $x \in X_i^\star$, $d^+(x) = \Delta - 1$, and $u \in N^+(x) \cap \mathscr{Z}_r(X_i)$.*

(B) $k_i = 2$, *and every* $(x, u) \in \mathscr{T}_i$ *satisfies* $x \in X_i^\star$, $d^-(x) = \Delta - 1$, *and* $u \in N^-(x) \cap \mathscr{Z}_r(X_i)$.

(C) $k_i = 3$, *and every* $(x, u, v) \in \mathscr{T}_i$ *satisfies* $x \in X_i^\star$, $d^+(x) \leqslant \Delta$, *and* $u, v \in N^+(x) \cap \mathscr{Z}_r(X_i)$.

(D) $k_i = 3$, *and every* $(x, u, v) \in \mathscr{T}_i$ *satisfies* $x \in X_i^\star$, $d^-(x) \leqslant \Delta$, *and* $u, v \in N^-(x) \cap \mathscr{Z}_r(X_i)$.

Proof. Let us fix $i \in [t]$ and let $\mathscr{S}_i^\star = X_i^\star \cap \mathscr{S}(X_i)$. By Lemma 24, X_i is a quasi-biclique. If it is a quasi-biclique of type (1), (2) or (3), then $\mathscr{S}_i^\star = \mathscr{S}(X_i)$ and $|\mathscr{S}_i^\star| \geqslant |X_i| - 3$ by definition of $\mathscr{S}(X_i)$. If it is of type (4), then $|\mathscr{S}_i^\star| \geqslant \frac{99}{100}\Delta$ by definition.

Therefore, in cases (1)–(4), since $\Delta - 300d < |X_i| \leqslant \Delta + 1$ (the upper bound coming from the fact that D does not contain a biclique of size Δ), and because Δ is large enough, we have $|\mathscr{S}_i^\star| \geqslant \frac{98}{100}|X_i|$, and in particular $|\mathscr{S}_i \setminus \mathscr{S}_i^\star| \leqslant \frac{2}{100}|X_i|$. By Lemma 26, $\mathscr{S}_i$ contains at least $\frac{1}{2}|X_i|$ r-saviours of X_i. Using the remark above, $\mathscr{S}_i^\star$ contains at least $\frac{48}{100}|X_i|$ r-saviours.

If X_i is of type (5) and (R_i, K_i, W_i) is the given partition of X_i, then by definition $K_i = \mathscr{S}_i^\star$, and $|K_i| > \frac{1}{2}|X_i|$. Therefore, by Lemma 25, $\mathscr{S}_i^\star$ contains more than $\frac{48}{100}|X_i|$ r-saviours again.

Since each r-saviour satisfies (a), (b), (c), or (d), by the pigeonhole principle, let $\gamma \in \{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$ be such that $\mathscr{S}_i^\star$ contains at least $\frac{12}{100}|X_i|$ r-saviours, all satisfying property (γ), and let Y be the set of such r-saviours. We now greedily construct $\mathscr{T}_i$ by repeating the following process as long as Y is non-empty:

1. choose any vertex $x \in Y$;
2. choose any vertex u in $N^+(x) \cap \mathscr{Z}_r(X_i)$ if $\gamma \in \{\mathrm{a}, \mathrm{c}\}$ and in $N^-(x) \cap \mathscr{Z}_r(X_i)$ otherwise;
3. choose any vertex $v \neq u$ in $N^+(x) \cap \mathscr{Z}_r(X_i)$ if $\gamma = \mathrm{c}$ and in $N^-(x) \cap \mathscr{Z}_r(X_i)$ if $\gamma = \mathrm{d}$;
4. if $\gamma \in \{\mathrm{a}, \mathrm{b}\}$, add (x, u) to $\mathscr{T}_i$ and remove $N(u)$ from Y;
5. else if $\gamma \in \{\mathrm{c}, \mathrm{d}\}$, add (x, u, v) to $\mathscr{T}_i$ and remove $N(u) \cup N(v)$ from Y.

Note that steps 2 and 3 above are always possible by definition of x being an r-saviour. By construction, the obtained $\mathscr{T}_i$ thus satisfies one of (A), (B), (C) or (D). Furthermore, steps 4 and 5 guarantee that the elements of $\mathscr{T}_i$ are pairwise disjoint. Finally, at each step, the size of Y decreases by at most $4r$. Hence,

$$|\mathscr{T}_i| \geqslant \frac{3}{100r}|X_i| \geqslant \frac{1}{50r}\Delta,$$

where in the second inequality we used $|X_i| \geqslant \Delta - 300d$ and the fact that Δ is large enough. The result follows. □

From now on, for every $i \in [t]$, we fix a collection $\mathscr{T}_i$ as in the statement of Lemma 27.

Remark 28. *Let us note, for future use, that* $|X_i^\star| \geqslant \frac{49}{100}\Delta$ *for every* $i \in [t]$, *as is implied by the proof of Lemma 27.*

4.2 The probabilistic analysis

Let φ be a partial $(\Delta-1)$-dicolouring of D. For an integer $i \in [t]$, we say that φ is i**-extendable** if there exist three distinct uncoloured vertices $x \in X_i^\star$, each of which satisfying one of the following:

(*i*) $d^+(x) = \Delta - 1$ and one colour is repeated in $N^+(x)$,

(*ii*) $d^-(x) = \Delta - 1$ and one colour is repeated in $N^-(x)$,

(*iii*) $d^+(x) \leqslant \Delta$ and two colours are repeated in $N^+(x)$, or

(*iv*) $d^-(x) \leqslant \Delta$ and two colours are repeated in $N^-(x)$.

We say that φ is **extendable** if it is i-extendable for every $i \in [t]$, and for every d-sparse vertex $s \in S$, three colours are repeated in $N^+(s)$.

Let us briefly justify that an extendable partial dicolouring of D can actually be extended to D. We colour the uncoloured vertices of $X_1, \ldots, X_t, S$ in this order. When colouring the vertices of X_i, we keep the three special vertices described above for the end. Since they are uncoloured and belong to $X_i^\star$, for each other vertex of X_i, there is a colour not appearing in its out-neighbourhood. When we end up with these three vertices, clearly we can choose a colour that does not appear in either the in- or out-neighbourhood. At the end, we can colour each vertex $s \in S$ with a colour that does not appear in its out-neighbourhood, as three colours are repeated in it.

The following lemma thus implies $\vec{\chi}(D) \leqslant \Delta - 1$, yielding a contradiction and concluding the proof of Theorem 8.

Lemma 29. *The digraph D admits an extendable partial $(\Delta - 1)$-dicolouring.*

Proof. Let ψ be a random colouring of the vertices of D, where each vertex is assigned a colour from $[\Delta - 1]$ independently and uniformly at random. Let V_ψ be the set of vertices x such that $\psi(x) = \psi(x^-) = \psi(x^+)$ for some $x^- \in N^-(x)$ and some $x^+ \in N^+(x)$. We consider the random partial dicolouring φ obtained from ψ by uncolouring simultaneously all the vertices in V_ψ. By construction, φ is indeed a partial $(\Delta - 1)$-dicolouring of D. We are going to show, using the Lovász Local Lemma, that φ is extendable with positive probability.

Let us define the events that we want to avoid. For each d-sparse vertex $s \in S$, let A_s be the event that fewer than three colours are repeated in $N^+(s)$. And, for each $i \in [t]$, let B_i be the event that φ is not i-extendable. We note that A_s depends only on the colour choices for the vertices within distance 2 of s in the underlying undirected graph $\mathrm{UG}(D)$ of D. Likewise, B_i depends only on the colour choices for the vertices within distance 2 of X_i in $\mathrm{UG}(D)$. Therefore, each of these events is mutually independent of all the others, except at most $(2\Delta_{\max})^5$ of them. By the Lovász Local Lemma (Lemma 16) and Lemma 20, it will be enough to show that each of these events holds with probability at most Δ^{-6}. This is the purpose of Claims 29.1 and 29.2.

Claim 29.1. *For any vertex $s \in S$, A_s holds with probability at most $e^{-\log^2 \Delta}$.*

Proof of claim. We let M_s be the random variable counting the number of colours c such that:

- c is assigned to two distinct out-neighbours x and y of s not inducing a digon, and
- for every distinct $x, y \in N^+(s)$ not inducing a digon, if x and y are assigned colour c, then both of them retain the colour.

It suffices to show that $M_s < 3$ holds with probability at most $e^{-\log^2 \Delta}$. To do so, we first estimate $\mathbb{E}(M_s)$ and then show that M_s is concentrated around its expectation. In what follows, we denote by U_s the set of unordered pairs of distinct out-neighbours of s not inducing a digon.

Subclaim 29.1.1. $\mathbb{E}(M_s) \geqslant \dfrac{|U_s|}{e^6(\Delta - 1)}$.

Proof of subclaim. Let $\tilde{M}_s$ be the number of colours c such that:

- c is assigned to exactly two out-neighbours x and y of s,

- $\langle x, y\rangle$ is not a digon, and
- both x and y retain colour c.

Observe that $M_s \geqslant \tilde{M}_s$. Given a colour $c \in [\Delta - 1]$ and two distinct out-neighbours x and y of s such that $\langle x, y\rangle$ is not a digon, let us consider the event $E_{\{x,y\},c}$ that x and y are assigned colour c, no other vertex in $N^+(s)$ is assigned colour c, and both x and y retain the colour. Let $\mathscr{W}_{x,y} = N^+(s) \cup N(x) \cup N(y) \setminus \{x, y\}$. Since $\langle x, y\rangle$ is not a digon, we have

$$\begin{aligned}\mathbb{P}(E_{\{x,y\},c}) &\geqslant \mathbb{P}\Big(\text{for every } w \in \mathscr{W}_{x,y},\ \psi(w) \neq c = \psi(x) = \psi(y)\Big)\\ &\geqslant \left(\frac{1}{\Delta-1}\right)^2 \left(\frac{\Delta-2}{\Delta-1}\right)^{5\Delta_{\max}} \geqslant \frac{1}{e^6(\Delta-1)^2},\end{aligned}$$

where in the last inequality we used that Δ is large enough, $\lim_{\Delta\to+\infty}\left(\frac{\Delta-2}{\Delta-1}\right)^{5\Delta} = \frac{1}{e^5} > \frac{1}{e^6}$, and that $\Delta \leqslant \Delta_{\max} \leqslant \Delta + 1$. By linearity of the expectation, we conclude that

$$\mathbb{E}(M_s) \geqslant \mathbb{E}(\tilde{M}_s) \geqslant (\Delta - 1) \cdot |U_s| \cdot \frac{1}{e^6(\Delta-1)^2} = \frac{|U_s|}{e^6(\Delta-1)}.$$

◿

Subclaim 29.1.2. *For any* $\lambda > 504\sqrt{\frac{|U_s|}{\Delta-1}} + 2752$, *we have*

$$\mathbb{P}(|M_s - \mathbb{E}(M_s)| > \lambda) \leqslant 8\exp\left(\frac{-\lambda^2}{512\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right).$$

Proof of subclaim. We consider two auxiliary random variables: $M_s^\star$, which counts the number of colours c assigned to both x and y for some $\{x, y\} \in U_s$, and M_s^{del}, which counts the number of colours c such that, for some $\{x, y\} \in U_s$, c is assigned to x and y, but not retained by both of them. Observe that $M_s = M_s^\star - M_s^{\text{del}}$. Let us show that these auxiliary random variables are concentrated, hence implying that M_s is concentrated as well.

Given $\{x, y\} \in U_s$ and a colour $c \in [\Delta - 1]$, let $E^\star_{\{x,y\},c}$ be the event that x and y are assigned colour c. We have that

$$\mathbb{P}(E^\star_{\{x,y\},c}) = \frac{1}{(\Delta-1)^2}.$$

Thus, by the union bound,

$$\mathbb{E}(M_s^\star) \leqslant (\Delta - 1) \cdot |U_s| \cdot \frac{1}{(\Delta-1)^2} = \frac{|U_s|}{\Delta-1}. \tag{8}$$

We note that changing the outcome of any colour assignment affects $M_s^\star$ by at most 1. Moreover, whenever $M_s^\star \geqslant k$ for some integer k, this can be certified by revealing $2k$ colour assignments: for each colour c counted by $M_s^\star$, it suffices to exhibit two vertices x and y with $\{x, y\} \in U_s$ that have been assigned colour c. By Talagrand's inequality (Lemma 17) and (8),

$$\mathbb{P}(|M_s^\star - \mathbb{E}(M_s^\star)| > \lambda) \leqslant 4\exp\left(\frac{-\lambda^2}{32 \cdot 2 \cdot (\mathbb{E}(M_s^\star) + \lambda)}\right) \leqslant 4\exp\left(\frac{-\lambda^2}{64\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right) \tag{9}$$

for any $\lambda > 126\sqrt{2\frac{|U_s|}{\Delta-1}} + 688 \geqslant 126\sqrt{2\mathbb{E}(M_s^\star)} + 344 \cdot 2$.

Similarly, M_s^{del} is affected by at most 1 when the outcome of any colour assignment is changed. Moreover, whenever $M_s^{\mathrm{del}} \geqslant k$, this can be certified by revealing at most $4k$ colour assignments: for each one of the k colours, it suffices to exhibit two vertices x and y with $\{x, y\} \in U_s$, and one in-neighbour x^- and one out-neighbour x^+ of x, such that x, y, x^- and x^+ have all been assigned that colour. By Talagrand's inequality (Lemma 17), (8), and the fact that $M_s^{\mathrm{del}} \leqslant M_s^\star$, we have

$$\mathbb{P}(|M_s^{\mathrm{del}} - \mathbb{E}(M_s^{\mathrm{del}})| > \lambda) \leqslant 4\exp\left(\frac{-\lambda^2}{32 \cdot 4 \cdot (\mathbb{E}(M_s^{\mathrm{del}}) + \lambda)}\right) \leqslant 4\exp\left(\frac{-\lambda^2}{128\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right) \tag{10}$$

for any $\lambda > 252\sqrt{\frac{|U_s|}{\Delta-1}} + 1376 \geqslant 126\sqrt{4\mathbb{E}(M_s^{\mathrm{del}})} + 344 \cdot 4$.

Recall that $M_s = M_s^\star - M_s^{\mathrm{del}}$. Hence, by linearity of the expectation and the triangle inequality, we have $|M_s - \mathbb{E}(M_s)| \leqslant |M_s^\star - \mathbb{E}(M_s^\star)| + |M_s^{\mathrm{del}} - \mathbb{E}(M_s^{\mathrm{del}})|$. Combining (9) and (10), we thus obtain

$$\begin{aligned}
\mathbb{P}(|M_s - \mathbb{E}(M_s)| > \lambda) &\leqslant \mathbb{P}(|M_s^\star - \mathbb{E}(M_s^\star)| + |M_s^{\mathrm{del}} - \mathbb{E}(M_s^{\mathrm{del}})| > \lambda) \\
&\leqslant \mathbb{P}(|M_s^\star - \mathbb{E}(M_s^\star)| > \tfrac{\lambda}{2}) + \mathbb{P}(|M_s^{\mathrm{del}} - \mathbb{E}(M_s^{\mathrm{del}})| > \tfrac{\lambda}{2}) \\
&\leqslant 4\exp\left(\frac{-\lambda^2}{256\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right) + 4\exp\left(\frac{-\lambda^2}{512\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right) \\
&\leqslant 8\exp\left(\frac{-\lambda^2}{512\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right)
\end{aligned}$$

for any $\lambda > 504\sqrt{\frac{|U_s|}{\Delta-1}} + 2752$. ◿

To end the proof of the claim, we note that, since s is d-sparse, out of the $d^+(s)(d^+(s) - 1)$ possible arcs between the vertices of $N^+(s)$, at least

$$d^+(s)(d^+(s) - 1) - (\Delta_{\max}(\Delta_{\max} - 1) - d\Delta_{\max}) \geqslant (\Delta - 1)(\Delta - 2) - (\Delta + 1)\Delta + d(\Delta - 1) \geqslant \frac{d\Delta}{2}$$

of them are not present in D. This implies that $|U_s| \geqslant \frac{d\Delta}{4}$. In particular, if we take $\lambda = \frac{|U_s|}{e^6(\Delta-1)} - 3$, the condition of Subclaim 29.1.2 holds. By Subclaims 29.1.1 and 29.1.2, we obtain

$$\mathbb{P}(A_s) \leqslant \mathbb{P}(M_s < 3) \leqslant \mathbb{P}(\mathbb{E}(M_s) - M_s > \lambda) \leqslant 8\exp\left(\frac{-\lambda^2}{512\left(\frac{|U_s|}{\Delta-1} + \lambda\right)}\right) \leqslant e^{-\log^2 \Delta},$$

as desired. ◊

Claim 29.2. *For any $i \in [t]$, B_i holds with probability at most $e^{\frac{-\Delta}{r^3}}$.*

Proof of claim. Let $\mathscr{T}_i$ be a set given by Lemma 27 (of size exactly $\lceil \frac{\Delta}{50r} \rceil$). For the sake of conciseness, we write the proof assuming that $\mathscr{T}_i$ satisfies (C); the other cases can be argued analogously. Let T_i be the set of vertices appearing in some triple of $\mathscr{T}_i$. Let M_i be the random variable counting the number of triples $(x, u, v) \in \mathscr{T}_i$ such that:

(1) vertex x is uncoloured;

(2) both u and v retain their colours, which both appear on some vertex of $X_i^\star$; and

(3) for every vertex of the triple, its colour is not assigned to any other vertex of T_i.

We note that, by (1) and (2), φ is i-extendable with probability at least $\mathbb{P}(M_i \geqslant 3)$. (In cases (B) and (D), we take into account that $X_i^\star$ is a biclique.) Subclaim 29.2.1 shows that M_i is large on average, and Subclaim 29.2.2 that it is concentrated around its expected value. The claim is then obtained by combining these two facts. The purpose of condition (3) is to prepare the ground for using Azuma's inequality.

Subclaim 29.2.1. $\mathbb{E}(M_i) \geqslant \dfrac{\Delta}{e^{37}r}$.

Proof of subclaim. Given a triple $(x, u, v) \in \mathscr{T}_i$, three distinct vertices $x' \in X_i^\star \setminus T_i$, $u' \in X_i^\star \setminus (T_i \cup N(u))$, and $v' \in X_i^\star \setminus (T_i \cup N(v))$, and three distinct colours $c_1, c_2, c_3 \in [\Delta - 1]$, let $E_{x,u,v,x',u',v',c_1,c_2,c_3}$ be the event that

- c_1 is assigned to x and x', but not to any vertex in $X_i^\star \cup T_i \setminus \{x, x'\}$;
- c_2 is assigned to u and u', but not to any vertex in $X_i^\star \cup T_i \cup N(u) \cup N(u') \setminus \{u, u'\}$;
- c_3 is assigned to v and v', but not to any vertex in $X_i^\star \cup T_i \cup N(v) \cup N(v') \setminus \{v, v'\}$.

Since $\langle x, x'\rangle$ is a digon, these conditions ensure that the triple (x, u, v) is counted by M_i. Since $|T_i| = 3\lceil \frac{\Delta}{50r} \rceil$, we have that

$$\mathbb{P}(E_{x,u,v,x',u',v',c_1,c_2,c_3}) \geqslant \left(\frac{1}{\Delta - 1}\right)^6 \left(\frac{\Delta - 4}{\Delta - 1}\right)^{|X_i^\star| + |T_i| + 8\Delta} \geqslant \left(\frac{1}{\Delta - 1}\right)^6 e^{-30}.$$

By Remark 28 and the fact that $u, v \in \mathscr{Z}_r(X_i)$, there are at least $\left(\frac{48}{100}\Delta\right)^3$ ways of choosing vertices x', u', and v'; moreover, there are $(\Delta-1)(\Delta-2)(\Delta-3)$ possible ways of choosing colours c_1, c_2, and c_3. Since for every fixed $(x, u, v) \in \mathscr{T}_i$, the events of the form $E_{x,u,v,x',u',v',c_1,c_2,c_3}$ are pairwise disjoint, by linearity of the expectation, we have

$$\mathbb{E}(M_i) \geqslant |\mathscr{T}_i| \left(\frac{48}{100}\Delta\right)^3 (\Delta - 1)(\Delta - 2)(\Delta - 3) \left(\frac{1}{\Delta - 1}\right)^6 e^{-30} \geqslant \frac{\Delta}{e^{37}r}.$$

◿

Subclaim 29.2.2. *For any* $\lambda \geqslant 0$, $\mathbb{P}(|M_i - \mathbb{E}(M_i)| > \lambda) \leqslant 2\exp\left(\frac{-\lambda^2}{44\Delta}\right)$.

Proof of subclaim. We will be interested in how much M_i can vary when the colour assignment of an arbitrary vertex is altered. For that, we first fix some notation. Let $\tau, \tau' : V(D) \to [\Delta - 1]$ be two arbitrary colour assignments that differ everywhere. We consider the event E that $\psi = \tau$ and, for every vertex w of D, the event E_w that $\psi|_{V(D)\setminus\{w\}} = \tau|_{V(D)\setminus\{w\}}$ and $\psi(w) = \tau'(w)$. We let $\mathscr{M}$ and $\mathscr{M}_w$ be the sets of triples of $\mathscr{T}_i$ satisfying (1), (2) and (3) when E and E_w hold, respectively. In particular, $|\mathscr{M}|$ and $|\mathscr{M}_w|$ are the values $M_i(E)$ and $M_i(E_w)$.

We are going to consider the random outcomes defining ψ in a specific order, depending on the given $i \in [t]$. We fix a labelling $w_1, \ldots, w_n$ of the vertices of D such that $w_1, \ldots, w_q \in V(D) \setminus (X_i^\star \cup T_i)$, and $w_{q+1}, \ldots w_n \in X_i^\star \cup T_i$ for some q. For $1 \leqslant j \leqslant n$, we denote by E_j and E'_j the events that $\psi(w_1) = \tau(w_1), \ldots, \psi(w_j) = \tau(w_j)$ and that $\psi(w_1) = \tau(w_1), \ldots, \psi(w_{j-1}) = \tau(w_{j-1})$ and $\psi(w_j) = \tau'(w_j)$.

Let us now discuss how much M_i can vary when changing the random assignment of an arbitrary vertex; in other words, let us fix $1 \leqslant j \leqslant n$ and examine the quantity $\big||\mathscr{M}| - |\mathscr{M}_{w_j}|\big|$. We note that a triple $(x, u, v) \in \mathscr{T}_i$ with $\{\tau(x), \tau(u), \tau(v)\} \cap \{\tau(w_j), \tau'(w_j)\} = \varnothing$ belongs to $\mathscr{M}$

if and only if it belongs to $\mathscr{M}_{w_j}$. On the other hand, by (3), both $\mathscr{M}$ and $\mathscr{M}_{w_j}$ contain at most two triples $(x,u,v) \in \mathscr{T}_i$ with $\{\tau(x),\tau(u),\tau(v)\} \cap \{\tau(w_j),\tau'(w_j)\} \neq \varnothing$. Therefore,

$$\left|M_i(E) - M_i(E_{w_j})\right| \leqslant 2. \tag{11}$$

For $1 \leqslant j \leqslant q$, the previous bound (11) can be improved to 0 in some situations. Let us denote by $\mathcal{N}_j$ the set of neighbours of w_j in $X_i^\star \cup T_i$, and let F_j be the event that some vertex in $\mathcal{N}_j$ is assigned colour $\tau(w_j)$ or $\tau'(w_j)$. Let us discuss when do $M_i(E)$ and $M_i(E_{w_j})$ differ. If $\mathscr{M} \neq \mathscr{M}_{w_j}$, then, since w_j is not in $X_i^\star \cup T_i$, changing the outcome of $\psi(w_j)$ from $\tau(w_j)$ to $\tau'(w_j)$ has to affect the instruction of uncolouring a vertex y from $X_i^\star \cup T_i$. In particular, y must be adjacent to w_j, and its assigned colour $\tau(y)$ under E and E_{w_j} must be $\tau(w_j)$ or $\tau'(w_j)$. Hence, $E, E_{w_j} \subseteq F_j$. Therefore, for every $1 \leqslant j \leqslant q$, changing the outcome of $\psi(w_j)$ does not affect M_i on $\overline{F_j}$. We deduce that

$$\mathbb{E}\left(M_i \mid E_j \cap \overline{F_j}\right) = \mathbb{E}\left(M_i \mid E'_j \cap \overline{F_j}\right). \tag{12}$$

We also need some control over the probability of F_j, for every $1 \leqslant j \leqslant q$. We note that, by the definition of the ordering $w_1, \ldots, w_n$, F_j is independent of both E_j and E'_j. Therefore, by the union bound,

$$\mathbb{P}(F_j \mid E_j) = \mathbb{P}(F_j \mid E'_j) = \mathbb{P}(F_j) \leqslant \frac{2|\mathcal{N}_j|}{\Delta - 1} \tag{13}$$

for every $1 \leqslant j \leqslant q$. We now bound

$$\sum_{j=1}^{n} \left|\mathbb{E}(M_i \mid E_j) - \mathbb{E}(M_i \mid E'_j)\right|^2,$$

which will allow us to apply Azuma's inequality. We split the sum into two chunks. For the terms $q+1 \leqslant j \leqslant n$, (11) yields

$$\sum_{j=q+1}^{n} \left|\mathbb{E}(M_i \mid E_j) - \mathbb{E}(M_i \mid E'_j)\right|^2 \leqslant 2^2 |X_i^\star \cup T_i| \leqslant 5\Delta.$$

For $1 \leqslant j \leqslant q$, (11), (12), and (13) yield

$$\begin{aligned}
&\sum_{j=1}^{q} \left|\mathbb{E}(M_i \mid E_j) - \mathbb{E}(M_i \mid E'_j)\right|^2 \\
&\leqslant \sum_{j=1}^{q} 2\left|\mathbb{E}(M_i \mid E_j) - \mathbb{E}(M_i \mid E'_j)\right| \\
&= 2\sum_{j=1}^{q} \Big|\mathbb{E}\left(M_i \mid E_j \cap F_j\right)\mathbb{P}\left(F_j \mid E_j\right) + \mathbb{E}\left(M_i \mid E_j \cap \overline{F_j}\right)\mathbb{P}\left(\overline{F_j} \mid E_j\right) \\
&\qquad - \mathbb{E}\left(M_i \mid E'_j \cap F_j\right)\mathbb{P}\left(F_j \mid E'_j\right) - \mathbb{E}\left(M_i \mid E'_j \cap \overline{F_j}\right)\mathbb{P}\left(\overline{F_j} \mid E'_j\right)\Big| \\
&= 2\sum_{j=1}^{q} \left|\mathbb{E}(M_i \mid E_j \cap F_j)\mathbb{P}(F_j \mid E_j) - \mathbb{E}(M_i \mid E'_j \cap F_j)\mathbb{P}(F_j \mid E'_j)\right| \\
&\leqslant 2\sum_{j=1}^{q} \left|\mathbb{E}(M_i \mid E_j \cap F_j) - \mathbb{E}(M_i \mid E'_j \cap F_j)\right| \frac{2|\mathcal{N}_j|}{\Delta - 1} \\
&\leqslant \frac{8}{\Delta - 1}\sum_{j=1}^{q} |\mathcal{N}_j| \leqslant \frac{16}{\Delta - 1}\Delta_{\max}|X_i^\star \cup T_i| \leqslant 17\Delta.
\end{aligned}$$

Altogether,

$$\sum_{j=1}^{n} \left|\mathbb{E}(M_i \mid E_j) - \mathbb{E}(M_i \mid E'_j)\right|^2 \leqslant 22\Delta.$$

The statement now follows from Azuma's inequality (Lemma 18). ◿

To end the proof of the Claim 29.2, we combine Subclaims 29.2.1 and 29.2.2 to obtain

$$\mathbb{P}(B_i) \leqslant \mathbb{P}(M_i < 3) \leqslant \mathbb{P}\left(\mathbb{E}(M_i) - M_i > \frac{\Delta}{e^{37}r} - 3\right) \leqslant 2\exp\left(\frac{-\left(\frac{\Delta}{e^{37}r} - 3\right)^2}{44\Delta}\right) \leqslant e^{\frac{-\Delta}{r^3}},$$

as desired. ◊

It follows from Claims 29.1 and 29.2 that each of the bad events occurs with probability at most Δ^{-6}. As discussed above, this allows us to conclude the proof of Lemma 29 with the Lovász Local Lemma (Lemma 16), and thus the proof of Theorem 8. □

5 Deriving the analogous result for Δ^+

As explained in the introduction, we now derive Theorem 12 from Theorem 8. Let us first restate it here for convenience.

Theorem 12. *There exists $\Delta_{12} \in \mathbb{N}$ such that the following holds. Let $\Delta \geqslant \Delta_{12}$ be an integer and D a digraph with $\Delta^+(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$. Then $\vec{\chi}(D) \leqslant \Delta - 1$ unless D contains $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Proof. We show the result for $\Delta_{12} = \max(\Delta_8, 556)$. Let us thus fix $\Delta \geqslant \Delta_{12}$, and assume for a contradiction that there exists a digraph D with $\overleftrightarrow{\omega}(D) \leqslant \Delta - 1$, $\Delta^+(D) \leqslant \Delta$, and $\vec{\chi}(D) \geqslant \Delta$, and such that D does not contain $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. Among all such digraphs, we choose D such that:

(1) $|V(D)|$ is minimum, and

(2) with respect to (1), the number of simple arcs of D is minimum.

We start with the following easy property of D that we will need later.

Claim 12.1. *D does not contain any induced copy of $\vec{P}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Proof of claim. Assume for a contradiction that $X \subseteq V(D)$ induces a copy of $\vec{P}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. Let $u, v, w \in X$ be the three vertices inducing the copy of $\vec{P}_3$.

By minimality of D, let φ be a $(\Delta - 1)$-dicolouring of $D - X$. We extend φ to D by first giving a common colour to u, v, w. This is possible because u, v, w have at least $\Delta - 2$ uncoloured out-neighbours, so one colour is not appearing in $N^+(u) \cup N^+(v) \cup N^+(w)$.

We then greedily extend φ to $X \setminus \{u, v, w\}$ by choosing for every such vertex x a colour that is not appearing in its out-neighbourhood. This is always possible because $\{u, v, w\} \subseteq N^+(x)$ and $\varphi(u) = \varphi(v) = \varphi(w)$. ◊

Similarly to the proof of Theorem 8, by (1), every vertex v satisfies $d^+(v), d^-(v) \geqslant \Delta - 1$, but this time we do not have any upper bound on $d^-(v)$. However, we show that there exists some vertex with in-degree exactly $\Delta - 1$ and out-degree at least Δ, for otherwise we are directly done by Theorem 8.

Claim 12.2. *There exists $v \in V(D)$ satisfying $d^-(v) = \Delta - 1 < d^+(v)$.*

Proof of claim. Let us prove that there exists v with $d^-(v) < d^+(v)$. The fact that $d^-(v) = \Delta - 1$ then follows as inequality $\Delta - 1 \leqslant d^+(v) \leqslant \Delta$ holds for every vertex v. Assume for a contradiction that $d^-(v) \geqslant d^+(v)$ for every vertex $v \in V(D)$. Since

$$\sum_{v \in V(D)} d^-(v) = |A(D)| = \sum_{v \in V(D)} d^+(v),$$

we thus have $d^+(v) = d^-(v)$ for every vertex $v \in V(D)$. In particular, $\tilde{\Delta}(D) = \Delta^+(D) \leqslant \Delta$, and Theorem 8 implies $\vec{\chi}(D) \leqslant \Delta - 1$, a contradiction. ◊

Let $\mathcal{V}_{\Delta-1}$ be the set of vertices with in-degree $\Delta - 1$ and out-degree at least Δ. The whole proof consists in identifying a vertex $v \in \mathcal{V}_{\Delta-1}$ with a specific structure, and then justifying that a $(\Delta - 1)$-dicolouring of $D - N^-[v]$ can be extended to D, thus yielding the contradiction. Let us first prove the following two properties of the vertices of $\mathcal{V}_{\Delta-1}$.

Claim 12.3. *Every vertex $v \in \mathcal{V}_{\Delta-1}$ has a simple in-neighbour u and a simple out-neighbour w.*

Proof of claim. Clearly v has a simple out-neighbour w as $d^+(v) > d^-(v)$. To see that v has a simple in-neighbour, let $D' = D \setminus \{vw\}$. By choice of D, D' admits a $(\Delta - 1)$-dicolouring φ. Clearly, D coloured with φ contains a monochromatic induced directed cycle $\mathscr{C}$, which necessarily contains vw. Since $\mathscr{C}$ is induced, the predecessor of v is a simple in-neighbour of v. ◊

Claim 12.4. *For every $v \in \mathcal{V}_{\Delta-1}$, $N^-(v)$ is a biclique of D.*

Proof of claim. Let us fix $v \in \mathcal{V}_{\Delta-1}$, and let D' be the digraph obtained from D by first removing all arcs of

$$\{vw : w \in N_D^+(v) \setminus N_D^-(v)\},$$

and then adding all arcs of

$$\{vu : u \in N_D^-(v) \setminus N_D^+(v)\}.$$

Note that the $d_{D'}^+(x) \leqslant \Delta$ for all vertices $x \neq v$ because we only added arcs from v to $V(D) \setminus \{v\}$. We further have $d_{D'}^+(v) = d_D^-(v) = \Delta - 1$, so $\Delta^+(D') \leqslant \Delta$. Claim 12.3 implies that D' contains less simple arcs than D. By choice of D, either $\vec{\chi}(D') \leqslant \Delta - 1$ or D' contains a copy of $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.

The former case does not hold, as every dicolouring of D' is a dicolouring of D. Indeed, assume that φ is a dicolouring of D', but D coloured with φ contains a monochromatic directed cycle $\mathscr{C}$. Clearly, $\mathscr{C}$ contains v, for otherwise it is a cycle of D'. Let u be the predecessor of v in $\mathscr{C}$, then $\langle u, v \rangle$ is a monochromatic digon of D', a contradiction.

Therefore D' contains a copy K of $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. We can further assume that K is an induced copy of $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, as adding any arc to $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$ yields a copy of $\overleftrightarrow{K}_\Delta$.

Clearly, $v \in V(K)$, otherwise K is a subdigraph of D. Since $d_{D'}^-(v) = d_{D'}^+(v) = \Delta - 1$, and because $N_{D'}^-(v) = N_{D'}^+(v)$, K is necessarily a copy of $\overleftrightarrow{K}_\Delta$, and $V(K) \setminus \{v\} = N^-(v)$ induces a biclique on D, as desired. ◊

The following will allow us to find our specific vertex $v \in \mathcal{V}_{\Delta-1}$.

Claim 12.5. *There exists $v \in \mathcal{V}_{\Delta-1}$ such that every simple in-neighbour u of v satisfies $d^-(u) \leqslant 2\Delta - 4$.*

Proof of claim. Assume that the statement does not hold, that is, every vertex $v \in \mathcal{V}_{\Delta-1}$ has a simple in-neighbour u with in-degree at least $2\Delta - 3$. We use a simple discharging argument to

reach a contradiction. Let X be the set of vertices with out-degree $\Delta - 1$, and for every vertex v, let its initial charge $\mathrm{ch}(v)$ be exactly $d^-(v)$. Then the total charge in D is:

$$\sum_{v \in V(D)} \mathrm{ch}(v) = |A(D)| = \sum_{v \in V(D)} d^+(v) = \Delta \cdot |V(D)| - |X|. \tag{14}$$

We now apply the following rule:

For every $v \in \mathcal{V}_{\Delta-1}$ and $u \in N^-(v) \setminus N^+(v)$ with $d^-(u) \geqslant 2\Delta - 3$, v receives 2 from u. (R)

Let $\mathrm{ch}^\star(v)$ denote the charge of a vertex $v \in V(D)$ after applying (R). We have the following lower bounds on $\mathrm{ch}^\star(v)$.

- If $v \in \mathcal{V}_{\Delta-1}$, then $\mathrm{ch}^\star(v) \geqslant \Delta + 1$.

  By assumption, v has a simple neighbour u with in-degree at least $2\Delta - 3$. Therefore, u sends 2 to v. Moreover, v does not send any charge as it has in-degree $\Delta - 1$.

- If $\Delta \leqslant d^-(v) \leqslant 2\Delta - 4$, then $\mathrm{ch}^\star(v) \geqslant \Delta$.

  By definition, the initial charge of v is at least Δ, and v does not send any charge.

- If $d^-(v) \geqslant 2\Delta - 3$, then $\mathrm{ch}^\star(v) \geqslant \Delta$.

  This is clear if v does not send any charge. If it does send charge, then it has a simple out-neighbour $w \in \mathcal{V}_{\Delta-1}$. By Claim 12.4, v belongs to a biclique of size $\Delta - 1$. Hence v has at most two simple out-neighbours, so it sends charge at most twice. Its final charge is then at least $2\Delta - 3 - 4 \geqslant \Delta$.

- If none of the cases above applies, then $d^-(v) = d^+(v) = \Delta - 1$. In particular, $v \in X$ and $\mathrm{ch}^\star(v) = \Delta - 1$.

Therefore, the total charge in D is

$$\sum_{v \in V(D)} \mathrm{ch}^\star(v) \geqslant \Delta \cdot |V(D)| - |X| + |\mathcal{V}_{\Delta-1}| > \Delta \cdot |V(D)| - |X|,$$

where in the last inequality we used that $\mathcal{V}_{\Delta-1}$ is non-empty by Claim 12.2. By (14), we thus have $\sum_v \mathrm{ch}^\star(v) > \sum_v \mathrm{ch}(v)$, a contradiction. ◊

From now on, we thus fix a vertex $v \in \mathcal{V}_{\Delta-1}$ such that each of its simple in-neighbours has in-degree at most $2\Delta - 4$. We further fix such a vertex $u \in N^-(v) \setminus N^+(v)$, we denote $X = N^-[v]$, and we arbitrarily label $x_1, \ldots, x_{\Delta-2}$ the vertices of $X \setminus \{u, v\}$.

In what remains, using arguments similar to the ones of Section 4.1, especially Lemma 22, we justify that a $(\Delta-1)$-dicolouring of $D - X$ can be extended to D, thus yielding the contradiction. However, the fact that vertices may have arbitrarily large in-degree requires a more technical analysis. We start with the following analogue of Claim 22.2.

Claim 12.6. *For every $i \in [\Delta - 2]$, there exists a unique vertex $w_i \in N^+(x_i) \setminus X$. Moreover, for every $(\Delta - 1)$-dicolouring φ of $D - X$,*

(i) for every $i \in [\Delta - 2]$, $D - X$ coloured with φ contains a monochromatic path from w_i to $N^-(x_i)$; and

(ii) for all $i, j \in [\Delta - 2]$, $\varphi(w_i) = \varphi(w_j)$.

Proof of claim. Let φ be a $(\Delta-1)$-dicolouring of $D-X$. We first extend φ to $D-(X\setminus\{u,v\})$ by choosing for u and v a colour that is not appearing in $N^-(u)\cup N^-(v)$. This is possible because uv is simple, no in-neighbour of v is coloured, and u has at most $(2\Delta-4)-(\Delta-2)=\Delta-2$ in-neighbours in $V(D)\setminus X$.

Now, one colour (namely $\varphi(u)=\varphi(v)$) is repeated in the out-neighbourhood of every vertex $x\in X\setminus\{u,v\}$. Then, for every i, x_i has out-degree exactly Δ, and admits an out-neighbour w_i and an in-neighbour y_i in $V(D)\setminus X$ such that $D-X$, coloured with φ, contains a monochromatic path P_i from w_i to y_i (we may have $w_i=y_i$). Indeed, if this is not the case, we can greedily colour the vertices of $X\setminus\{u,v\}$, keeping x_i for the end, by choosing for every vertex a colour that is not appearing in its out-neighbourhood. The uniqueness of w_i comes from the fact that x_i has exactly $\Delta-1$ out-neighbours in X. This shows *(i)*.

To show *(ii)*, assume for a contradiction that, for some i,j, $\varphi(w_i)\neq\varphi(w_j)$. If $\varphi(w_j)=\varphi(u)$, then we can greedily extend φ, keeping x_j for the end, as one colour (namely $\varphi(u)$) appears three times in $N^+(x_j)$. Hence assume that $\varphi(w_j)\neq\varphi(w_i)$ and $\varphi(w_j)\neq\varphi(u)$. We can thus extend φ by first colouring w_i with $\varphi(w_j)$, and then keeping x_j for the end, as two colours are now repeated in $N^+(x_j)$. $\Diamond$

From now on, we will consider two distinct partitions $(I^{\mathrm{dig}},I^{\mathrm{s}})$ and $(I^{\pm},I^{\leqslant 6},I^{>6})$ of $[\Delta-2]$. We define $(I^{\mathrm{dig}},I^{\mathrm{s}})$ as follows:

$$I^{\mathrm{dig}}=\Big\{i\in[\Delta-2]:\langle x_i,w_i\rangle \text{ is a digon of } D\Big\}\quad\text{and}\quad I^{\mathrm{s}}=\Big\{i\in[\Delta-2]:w_i\in N^+(x_i)\setminus N^-(x_i)\Big\},$$

and $(I^{\pm},I^{\leqslant 6},I^{>6})$ as follows:

$$\begin{aligned}
I^{\pm}&=\{i\in[\Delta-2]:N^+(w_i)\cap X\neq\varnothing\},\\
I^{\leqslant 6}&=\{i\in[\Delta-2]:w_i \text{ has at most six simple out-neighbours}\}\setminus I^{\pm},\quad\text{and}\\
I^{>6}&=\{i\in[\Delta-2]:w_i \text{ has at least seven simple out-neighbours}\}\setminus I^{\pm}.
\end{aligned}$$

By definition, observe that $I^{\mathrm{dig}}\subseteq I^{\pm}$ and $(I^{\leqslant 6}\cup I^{>6})\subseteq I^{\mathrm{s}}$. Also, note that one set of $(I^{\pm},I^{\leqslant 6},I^{>6})$ is large, as it has size at least $(\Delta-2)/3$. Using Claim 12.6, our goal is now to show that each of the sets $(I^{\pm},I^{\leqslant 6},I^{>6})$ has bounded size, thus yielding a contradiction. We first exclude the case of some vertex w being adjacent with digons to all vertices of $X\setminus\{u,v\}$.

Claim 12.7. *For every $w\in V(D)\setminus X$, there exists $x\in X\setminus\{u,v\}$ such that $\langle x,w\rangle\not\subseteq A(D)$.*

Proof of claim. Assume for a contradiction that $(X\setminus\{u,v\})\cup\{w\}$ is a biclique, for some $w\in V(D)\setminus X$. Recall that $X\setminus\{v\}$ is also a biclique. We now distinguish two cases, depending on whether u is the unique simple in-neighbour of v or not, see Figure 2 for an illustration.

Assume first that u is the unique simple in-neighbour of v, see Figure 2(a). In this case, $D[\{u,v,w\}]$ is acyclic, for otherwise D contains either $\overleftrightarrow{K}_{\Delta}$ or $\vec{C}_3\boxplus\overleftrightarrow{K}_{\Delta-2}$. Let φ be a $(\Delta-1)$-dicolouring of $D-(X\cup\{w\})$. Since u,v,w all have $\Delta-2$ uncoloured out-neighbours, we can extend φ by first giving to u,v,w the same colour. We can then extend φ to D by choosing for each vertex $x\in X\setminus\{u,v\}$ a colour that is not appearing in $N^+(x)$, which is possible because one colour (namely $\varphi(v)$) is appearing three times, a contradiction.

Assume then that v has a simple in-neighbour u' distinct from u, see Figure 2(b). Again, let φ be a $(\Delta-1)$-dicolouring of $D-(X\cup\{w\})$. We now extend φ by choosing a common colour for both u' and v that is not appearing in $N^-(u')\cup N^-(v)$. This is possible because no in-neighbour of v is coloured, and by choice of v, $d^-(u')\leqslant 2\Delta-4$ so $|N^-(u')\setminus(X\cup\{w\})|\leqslant\Delta-3$. We then extend φ by giving a common colour to u and w. Again, this is possible as $\langle u,w\rangle$ is not a digon (otherwise D contains $\overleftrightarrow{K}_{\Delta}$), and both u and w have $\Delta-3$ uncoloured out-neighbours. We can

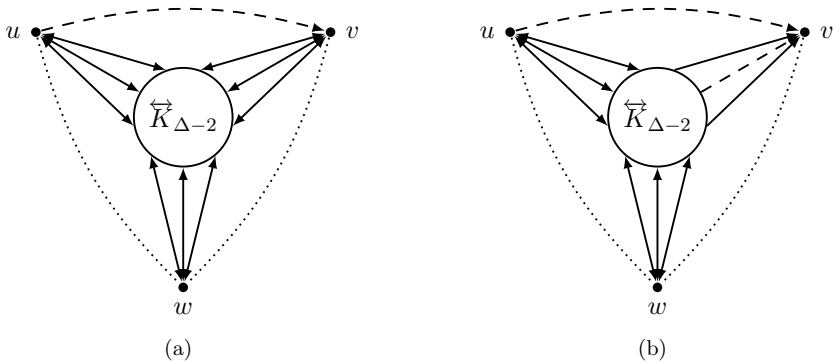

Figure 2: The two distinguished cases in the proof of Claim 12.7. When two vertices are linked with a dotted edge, it illustrates that there is at most one arc between them. A dashed arc is a simple arc.

finally extend φ to D by choosing for each vertex $x \in X \setminus \{u, u', v\}$ a colour that is not appearing in $N^+(x)$, which is possible because two colours are repeated in $N^+(x)$, a contradiction. $\Diamond$

We now need a few extra definitions. For every $i \in I^{\mathrm{s}}$, we let N_i be $N^-(x_i) \setminus X$. We further let D_i be the digraph obtained from $D - X$ by adding all arcs from N_i to w_i (note that $w_i \notin N_i$ by definition of I^{s}).

Claim 12.8. *For every $i \in I^{\mathrm{s}}$, D_i contains a copy of $\overleftrightarrow{K}_{\Delta}$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.*

Proof of claim. Assume that this is not the case. By construction, $\Delta^+(D_i) \leqslant \Delta$, so the minimality of D implies the existence of a $(\Delta - 1)$-dicolouring φ of D_i. Then, by Claim 12.6, since φ is a particular dicolouring of $D - X$, there exists a monochromatic path P_i from w_i to some $y_i \in N_i$. The concatenation of P_i and the arc $y_i w_i$ is a monochromatic directed cycle of D_i coloured with φ, a contradiction. $\Diamond$

Claim 12.8 allows us to do the following. For every $i \in I^{\mathrm{s}}$, we fix an induced copy K_i of either $\overleftrightarrow{K}_{\Delta}$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$ in D_i, and we fix $\tilde{N}_i$ to:

$$\tilde{N}_i = (V(K_i) \cap N^-(x_i)) \setminus N^-(w_i).$$

The key point is that $\tilde{N}_i$ can play the role of N_i, that is $\tilde{N}_i \subseteq N^-(x_i) \setminus X$ and in every $(\Delta - 1)$-dicolouring of $D - X$, there is a monochromatic path from w_i to $\tilde{N}_i$, but we further have the extra property that $\tilde{N}_i$ is almost the set of simple out-neighbours of w_i. We show these properties in the next two claims.

Claim 12.9. *For every $(\Delta-1)$-dicolouring φ of $D-X$ and every $i \in I^{\mathrm{s}}$, there is a monochromatic path from w_i to $\tilde{N}_i$.*

Proof of claim. Let φ_i be the restriction of φ to $V(K_i)$. Clearly, φ_i is not a dicolouring of K_i, as $\vec{\chi}(K_i) = \Delta$. Let $\mathscr{C}$ be a monochromatic directed cycle in K_i coloured with φ_i. Since the arcs of $A(K_i) \setminus A(D)$ all have the same head (namely w_i), and because $\mathscr{C}$ is not a cycle of D, there exists a unique arc xy of $\mathscr{C}$ in $A(K_i) \setminus A(D)$. Since $xy \in A(D_i) \setminus A(D)$, we have $y = w_i$, and $x \in N^-(x_i) \setminus N^-(w_i)$. The claim follows. $\Diamond$

Claim 12.10. *Let $i \in I^{\mathrm{s}}$, and $S_i^+ = N^+(w_i) \setminus N^-(w_i)$ be the set of simple out-neighbours of w_i. Then*

$$|S_i^+ \setminus \tilde{N}_i| \leqslant 2 \quad \textit{and} \quad |\tilde{N}_i \setminus S_i^+| \leqslant 1.$$

Moreover, $\tilde{N}_i \cap S_i^+ \neq \varnothing$.

Proof of claim. By definition, K_i is obtained from $D[V(K_i)]$ by adding all arcs from $\tilde{N}_i$ to w_i. Observe also that, in K_i, w_i is linked with digons to at least $\Delta - 2$ vertices. Let us call $Z_i = N^{\pm}_{K_i}(w_i)$ the set of these vertices. Suppose for a contradiction that we can find three simple out-neighbours of w_i outside $\tilde{N}_i$. Note that all of them lie outside Z_i, regardless of whether they are in $V(K_i)$ or not. Therefore, w_i has at least $3 + (\Delta - 2) = \Delta + 1$ out-neighbours in D_i, a contradiction, because all of them are also out-neighbours in D. The first inequality follows.

For the second inequality, recall that all vertices in $\tilde{N}_i$ are in-neighbours of w_i in K_i because K_i is an induced copy in D_i, and vertices in $\tilde{N}_i$ belong to $N^-(x_i)$. However, they are not in-neighbours of w_i in D by definition of $\tilde{N}_i$. Since, in K_i, w_i has at most one simple in-neighbour, it means that all vertices in $\tilde{N}_i$, except possibly one, are simple out-neighbours of w_i in D. The second inequality follows.

Assume now, for a contradiction, that $\tilde{N}_i$ contains no element of S_i^+. By the second inequality, $|\tilde{N}_i| = 1$ ($\tilde{N}_i$ is not empty otherwise K_i would be a subdigraph of D). Let us denote by y the unique vertex in $\tilde{N}_i$. Recall that $K_i \subseteq D \cup \{yw_i\}$, and K_i is either $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. Since y is not a simple out-neighbour of w_i in D, then yw_i is a simple arc in K_i, which implies that K_i is exactly $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, and that $D[V(K_i)]$ is exactly $\vec{P}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. This is a contradiction to Claim 12.1. ◊

The last important property is that the sets $(\tilde{N}_i)_{i \in I^{\mathrm{s}}}$ have limited overlap, as we show now.

Claim 12.11. *For all distinct $i, j, k \in I^{\mathrm{s}}$, we have $\tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k = \varnothing$.*

Proof of claim. Assume for a contradiction that $y \in \tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k$. This implies in particular $|N^+(y) \cap X| \geqslant 3$ (as $\{x_i, x_j, x_k\} \subseteq N^+(y)$).

We further know that y, in K_i, has at least $\Delta - 1$ out-neighbours. By definition of K_i, all of these except w_i are out-neighbours of y in D; we thus have $|N^+_D(y) \cap V(K_i)| \geqslant \Delta - 2$. Since $V(K_i) \cap X = \varnothing$, we get $d^+(y) \geqslant \Delta + 1$, a contradiction. ◊

We now have enough material to show that each of the sets $(I^{\pm}, I^{\leqslant 6}, I^{>6})$ has bounded size. We start with the following, which intuitively implies that if $I \subseteq [\Delta - 2]$ is large, then so is $W_I = \{w_i : i \in I\}$.

Claim 12.12. *For every $w \in V(D) \setminus X$, $|N^-(w) \cap \{x_1, \ldots, x_{\Delta-2}\}| \leqslant 14$.*

Proof of claim. Let I_w be the set of integers $i \in [\Delta - 2]$ such that $w_i = w$, and assume for a contradiction that $|I_w| \geqslant 15$.

We first assume that $|I_w \cap I^{\mathrm{dig}}| \geqslant 3$, so w is linked with digons to at least three vertices in $\{x_1, \ldots, x_{\Delta-2}\}$. Let us fix $j \in I_w \cap I^{\mathrm{dig}}$. By Claim 12.7, there exists $i \in [\Delta - 2]$ such that $\langle w, x_i \rangle$ is not a digon of D. We claim that w is not an out-neighbour of x_i. Indeed, if it is the case, then $w = w_i$ is a simple out-neighbour of x_i, and w has at least $\Delta - 1$ out-neighbours in $V(K_i) \subseteq V(D) \setminus X$. Since w has three out-neighbours in X, this is a contradiction.

Therefore, $w \neq w_i$. Let φ be a $(\Delta - 1)$-dicolouring of $D - (X \cup \{w\})$. Since w has three out-neighbours in X, we can extend φ to $D - X$ by choosing for w a colour that is not appearing in $N^+(w)$ and that is further distinct from $\varphi(w_i)$. We obtain a $(\Delta - 1)$-dicolouring φ of $D - X$ with $\varphi(w_i) \neq \varphi(w_j)$, a contradiction to Claim 12.6(*ii*).

Henceforth, we assume that $I_w \cap I^{\mathrm{s}}$ has size at least $15-2=13$. Let S_w^+ be the set of simple out-neighbours of w. We conclude by distinguishing two cases, depending on the size of S_w^+. In both cases, we get a contradiction by showing that some $y \in S_w^+$ has out-degree at least $\Delta+1$.

Assume first that $|S_w^+| \geqslant 7$, and consider three distinct indices $i,j,k \in I_w \cap I^{\mathrm{s}}$. By the first inequality of Claim 12.10, we have

$$|S_w^+ \cap \tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k| \geqslant |S_w^+| - |(S_w^+ \setminus \tilde{N}_i) \cup (S_w^+ \setminus \tilde{N}_j) \cup (S_w^+ \setminus \tilde{N}_k)| \geqslant 7-6=1.$$

We thus have a simple out-neighbour y of w which belongs to $\tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k$. This is a contradiction to Claim 12.11.

Assume then that $|S_w^+| \leqslant 6$. Recall that, for every $i \in I_w \cap I^{\mathrm{s}}$, $\tilde{N}_i$ contains a simple out-neighbour of w by the last part of Claim 12.10. Hence, since $|I_w \cap I^{\mathrm{s}}| \geqslant 13$, and because w has at most six simple out-neighbours, by the pigeonhole principle, some simple out-neighbour y of w belongs to $\tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k$ for three distinct $i,j,k \in I_w \cap I^{\mathrm{s}}$. Again, this is a contradiction to Claim 12.11. $\Diamond$

In the next three claims, we finally show that each of the sets $(I^{\pm}, I^{\leqslant 6}, I^{>6})$ has bounded size. We start with $I^{\pm}$.

Claim 12.13. $|I^{\pm}| \leqslant 70$.

Proof of claim. Assume for a contradiction that $|I^{\pm}| \geqslant 71$. By Claim 12.12, there exists $I \subseteq I^{\pm}$ such that $|I| = \lceil 71/14 \rceil = 6$ and vertices $(w_i)_{i \in I}$ are pairwise distinct. For every distinct $i,j \in I$, we define D_{ij} as the digraph obtained from $D-X$ by further adding the digon $\langle w_i, w_j \rangle$, see Figure 3(a) for an illustration.

Since i,j belong to $I^{\pm}$, we have $\Delta^+(D_{ij}) \leqslant \Delta$. Furthermore, $\vec{\chi}(D_{ij}) \geqslant \Delta$, as any $(\Delta-1)$-dicolouring φ of D_{ij} would yield a $(\Delta-1)$-dicolouring of $D-X$ satisfying $\varphi(w_i) \neq \varphi(w_j)$, a contradiction to Claim 12.6.

Therefore, the minimality of D implies that each D_{ij} contains an induced copy of either $\overleftrightarrow{K}_{\Delta}$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. In both cases, we conclude on the existence of a biclique $\mathscr{L}_{ij}$ in $D-X$ of size $\Delta-4$ such that

$$\mathscr{L}_{ij} \subseteq N^{\pm}(w_i) \cap N^{\pm}(w_j).$$

Let us fix two distinct $i,j \in I$. Observe that, for any $k \in I \setminus \{i,j\}$, $|\mathscr{L}_{ij} \setminus \mathscr{L}_{ik}| \leqslant 3$, for otherwise w_i has at least Δ out-neighbours in $V(D) \setminus X$ and one out-neighbour in X, a contradiction. Therefore,

$$\left| \bigcap_{k \in I \setminus \{i\}} \mathscr{L}_{ik} \right| \geqslant |\mathscr{L}_{ij}| - \sum_{k \in I \setminus \{i,j\}} |\mathscr{L}_{ij} \setminus \mathscr{L}_{ik}| \geqslant (\Delta-4) - 3(|I|-2) > 0.$$

Let us thus fix $y \in \bigcap_{k \in I \setminus \{i\}} \mathscr{L}_{ik}$. Then y is an in-neighbour of each w_k, $k \in I$. Furthermore, y has $\Delta-5$ out-neighbours in $\mathscr{L}_{ij}$. Note that $\{w_k : k \in I\}$ is disjoint from $\mathscr{L}_{ij}$, as if $w_k \in \mathscr{L}_{ij}$ then $\langle w_i, w_k \rangle$ is a digon of $D-X$, and D_{ik} is a subdigraph of D, a contradiction. Hence, $d^+(y) \geqslant |I| + \Delta - 5 = \Delta+1$, a contradiction. $\Diamond$

We now move to the cases of $I^{\leqslant 6}$ and $I^{>6}$. For two integers $i,j \in I^{\leqslant 6} \cup I^{>6}$ such that $w_i \neq w_j$, we define D_{ij} as the digraph obtained from $D-X$ by adding all arcs from $\tilde{N}_i$ to w_j and all arcs from $\tilde{N}_j$ to w_i. Observe that $w_j \notin \tilde{N}_i$, for otherwise w_j would have an out-neighbour in X (namely x_i), which by definition would imply $j \in I^{\pm}$. See Figure 3(b) for an illustration.

For all $i,j \in I^{\leqslant 6} \cup I^{>6}$ with $w_i \neq w_j$, observe that D_{ij} satisfies $\Delta^+(D_{ij}) \leqslant \Delta$ by construction (this is because $\tilde{N}_i \subseteq N^-(x_i)$ and $\tilde{N}_j \subseteq N^-(x_j)$). Moreover, $\vec{\chi}(D_{ij}) \geqslant \Delta$. To see this, assume

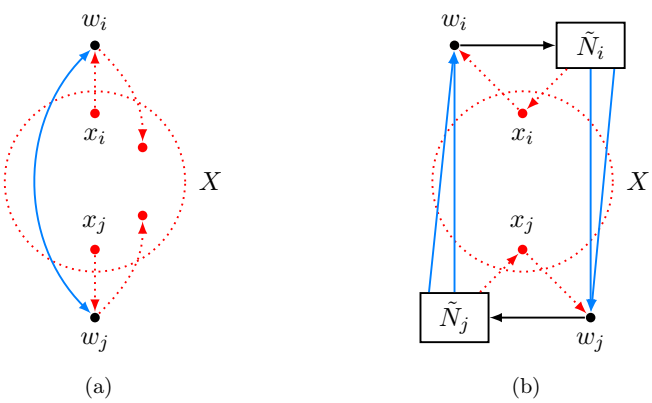

Figure 3: The construction of D_{ij} when $i, j \in I^{\pm}$ (Figure (a)), and $i, j \in I^{\leqslant 6} \cup I^{>6}$ (Figure (b)). In both figures, D_{ij} is built from D by removing the dotted red part and adding the solid blue one. In Figure (b), w_i does not have any out-neighbour in X nor any in-neighbour in $\tilde{N}_i$. Moreover, all but at most one vertex of $\tilde{N}_i$ are out-neighbours of w_i. The same properties hold for w_j and $\tilde{N}_j$.

for a contradiction that D_{ij} admits a $(\Delta - 1)$-dicolouring φ. Then φ is a particular $(\Delta - 1)$-dicolouring of $D - X$, so by Claim 12.6 we have $\varphi(w_i) = \varphi(w_j)$. Moreover, by Claim 12.9 (recall that $I^{\leqslant 6} \cup I^{>6} \subseteq I^{\mathrm{s}}$), D_{ij}, coloured with φ, contains a monochromatic directed path P_i from w_i to some $y_i \in \tilde{N}_i$ and one P_j from w_j to $y_j \in \tilde{N}_j$. Hence the concatenation of P_i, $y_i w_j$, P_j, and $y_j w_i$ is a monochromatic directed cycle in D_{ij} coloured with φ, a contradiction.

The choice of D thus implies that, for all $i, j \in I^{\leqslant 6} \cup I^{>6}$ with $w_i \neq w_j$, D_{ij} contains an induced copy K_{ij} of either $\overleftrightarrow{K}_{\Delta}$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$.

We now bound the size of $I^{\geqslant 6}$. The proof is analogous to that of Claim 12.13, but requires more technical arguments. Intuitively, for integers $i, j \in I^{\leqslant 6}$, the digraph D_{ij} is obtained from $D - X$ by adding only a constant number of arcs. Using this, we will find a large biclique in $D - X$ that is linked with digons to many vertices in $\bigcup_{i \in I^{\geqslant 6}} \tilde{N}_i$, hence implying that some vertex in this biclique has too large out-degree.

Claim 12.14. $|I^{\leqslant 6}| \leqslant 441$.

Proof of claim. Assume for a contradiction that $|I^{\leqslant 6}| \geqslant 442$. We claim that there exists $I \subseteq I^{\leqslant 6}$, $|I| = 22$, such that:

1. vertices $(w_i)_{i \in I}$ are pairwise distinct,
2. sets $(\tilde{N}_i)_{i \in I}$ are pairwise disjoint.

To see the existence of I, we build an auxiliary undirected graph $G^{\leqslant 6}$ with vertex-set $V(G^{\leqslant 6}) = I^{\leqslant 6}$ and edge-set $E(G^{\leqslant 6}) = E_1 \cup E_2$, where

$$E_1 = \{\{i, j\} : w_i = w_j\} \quad \text{and} \quad E_2 = \{\{i, j\} : \tilde{N}_i \cap \tilde{N}_j \neq \varnothing\}.$$

Our goal is thus to show that $G^{\leqslant 6}$ admits an independent set I of size 22. For this, fix an integer $i \in I^{\leqslant 6}$, which is also a vertex of $G^{\leqslant 6}$. By Claim 12.12, i has at most 13 neighbours in $G^{\leqslant 6}$ via edges in E_1. Similarly, by Claim 12.11, each vertex in $\tilde{N}_i$ belongs to $\tilde{N}_j$ for at most one $j \neq i$. Furthermore, note that $|\tilde{N}_i| \leqslant 7$ by Claim 12.10 together with the fact that w_i has at

most 6 simple out-neighbours. Hence, i has at most 7 neighbours in $G^{\leqslant 6}$ via edges in E_2. This shows $\Delta(G^{\leqslant 6}) \leqslant 20$. Hence, $G^{\leqslant 6}$ has chromatic number at most $\chi(G^{\leqslant 6}) \leqslant 21$. Take an optimal proper colouring of $G^{\leqslant 6}$, its largest colour class is an independent set of $G^{\leqslant 6}$ of size at least $\lceil 442/21 \rceil = 22$. This shows the existence of I.

Recall that, for every distinct $i, j \in I$, D_{ij} contains an induced copy K_{ij} of either $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. We fix such a K_{ij} for every pair $i, j \in I$. As K_{ij} is not a subgraph of D, it contains an arc from $\tilde{N}_i$ to w_j or an arc from $\tilde{N}_j$ to w_i that is not an arc of D (it might contain both).

We now build an auxiliary tournament T^I as follows. We take $V(T^I) = I$, and we orient each edge ij from i to j if K_{ij} contains an arc from $\tilde{N}_i$ to w_j that is not an arc of D, and from j to i otherwise (if K_{ij} contains both kind of arcs, we orient ij arbitrarily). As T^I has order 22, a simple counting argument implies that it contains a vertex i of in-degree at least 11. We fix such a vertex i and a set $J \subseteq I \setminus \{i\}$ of exactly 11 in-neighbours of i in T^I. Hence, by definition, for every $j \in J$, K_{ij} contains an arc from some $y_j \in \tilde{N}_j$ to w_i that is not an arc of D. We fix such a y_j for every $j \in J$.

For every $j \in J$, recall that K_{ij} contains a biclique $\mathscr{L}'_{ij}$ of size $\Delta - 4$, whose vertices are all linked with digons to both y_j and w_i. Note that the digons of $D_{ij}[\mathscr{L}'_{ij} \cup \{y_j, w_i\}]$ do not necessarily all belong to D, as some of them may contain arcs from $\tilde{N}_j$ to w_i or arcs from $\tilde{N}_i$ to w_j. In fact, $w_j \notin \mathscr{L}'_{ij}$, for otherwise w_j and w_i are linked by a digon in D_{ij}, and so they are in D since $w_j \notin \tilde{N}_i$ and $w_i \notin \tilde{N}_j$ ($w_j \in \tilde{N}_i$ would imply $j \in I^{\pm}$ as $\tilde{N}_i \subseteq N^-(x_i)$). Again, w_i and w_j cannot be linked with a digon in D by Claim 12.6.

Therefore, in D, $\mathscr{L}_{ij} = \mathscr{L}'_{ij} \setminus \tilde{N}_j$ is a biclique of D of size at least $\Delta - 10$ (as $|\tilde{N}_j| \leqslant 7$ and $y_j \notin \mathscr{L}'_{ij}$), and all of them are linked with digons to both w_i and y_j. From now on, let us fix $j \in J$. Observe that, for any $k \in J \setminus \{j\}$, $|\mathscr{L}_{ij} \setminus \mathscr{L}_{ik}| \leqslant 10$, for otherwise w_i has at least $\Delta + 1$ out-neighbours in $\mathscr{L}_{ij} \cup \mathscr{L}_{ik}$, a contradiction. Therefore,

$$\left| \bigcap_{k \in J} \mathscr{L}_{ik} \right| \geqslant |\mathscr{L}_{ij}| - \sum_{k \in J \setminus \{j\}} |\mathscr{L}_{ij} \setminus \mathscr{L}_{ik}| \geqslant (\Delta - 10) - 10(|J| - 1) = \Delta - 110 > 0.$$

Let us thus fix $z \in \bigcap_{k \in J} \mathscr{L}_{ik}$. Then z is an in-neighbour of w_i, and an in-neighbour of y_k for every $k \in J$. Note that the y_k's are pairwise distinct as we chose I such that $\tilde{N}_k$ and $\tilde{N}_{k'}$ are disjoint for all distinct $k, k' \in I$.

Finally, z has at least $\Delta - 11$ out-neighbours in $\mathscr{L}_{ij}$. Observe that $\{y_k : k \in J\}$ is disjoint from $\mathscr{L}_{ij}$, as for every $k \in J$, $y_k w_i$ is an arc of $A(D_{ik}) \setminus A(D)$. Altogether, we thus have

$$d^+(z) \geqslant 1 + |J| + (\Delta - 11) = \Delta + 1,$$

a contradiction. $\Diamond$

We finally bound the size of $I^{>6}$. This time, since the size of $\tilde{N}_i$ might be arbitrarily large, we are not able to find a large biclique with many out-neighbours outside, like we did in the two previous claims. However, because each $\tilde{N}_i$ is indeed large, we will manage to find distinct $i, j, k \in I^{>6}$ for which $\tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k \neq \varnothing$, thus contradicting Claim 12.11.

Claim 12.15. $|I^{>6}| \leqslant 42$.

Proof of claim. Assume for a contradiction that $|I^{\leqslant 6}| \geqslant 43$. Again, by Claim 12.12, there exists $I \subseteq I^{>6}$, $|I| = \lceil 43/14 \rceil = 4$, such that vertices $(w_i)_{i \in I}$ are pairwise distinct.

Again, recall that for every distinct $i, j \in I$, D_{ij} contains an induced copy K_{ij} of either $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. We fix such a K_{ij} for every pair $i, j \in I$. As K_{ij} is not a subgraph of D, it contains at least one vertex of $\{w_i, w_j\}$. Similarly to the previous case, we thus build an

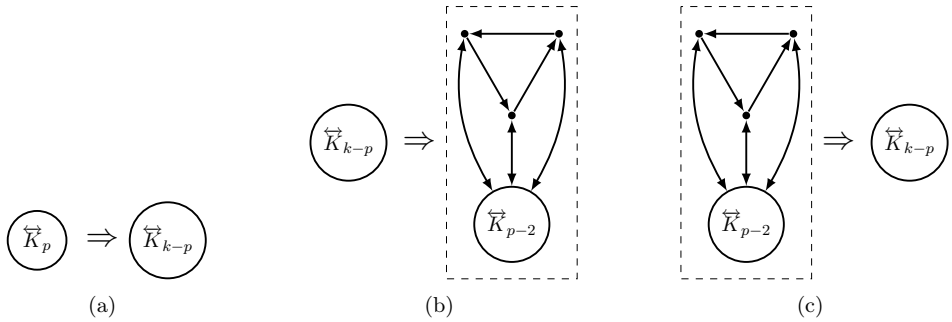

Figure 4: The three possible types of a k-obstruction. In Figure (a), p is any value between 0 and k. In Figures (b) and (c), p is any value between 2 and k.

auxiliary tournament T^I with $V(T^I) = I$, and we orient each edge ij from i to j if K_{ij} contains w_j, and from j to i otherwise (again, we orient ij arbitrarily if it contains both).

As T^I has order 4, it contains a vertex i of in-degree at least 2. We fix such a vertex i and two distinct in-neighbours j, k of i in T^I. By definition, both K_{ij} and K_{ik} contain w_i.

Let us denote by S_i^+ the set of simple out-neighbours of w_i. Observe that, in K_{ij}, w_i is linked with digons to at least $\Delta - 2$ vertices in $V(K_{ij})$, let us denote them by Y_j. By construction, every vertex $y \in Y_j$ is an out-neighbour of w_i in D. Therefore, Y_j contains all but at most two simple out-neighbours of w_i. These vertices from $Y_j \cap S_i^+$ necessarily belong to $\tilde{N}_j$, as they are simple out-neighbours of w_i in D, but linked with digons to w_i in D_{ij}. Hence $|S_i^+ \setminus \tilde{N}_j| \leqslant 2$. Similarly, $|S_i^+ \setminus \tilde{N}_k| \leqslant 2$, and by Claim 12.10, $|S_i^+ \setminus \tilde{N}_i| \leqslant 2$. Therefore, since $|S_i^+| \geqslant 7$ by definition of $I^{>6}$, we have

$$|S_i^+ \cap \tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k| \geqslant |S_i^+| - |(S_i^+ \setminus \tilde{N}_i) \cup (S_i^+ \setminus \tilde{N}_j) \cup (S_i^+ \setminus \tilde{N}_k)| \geqslant 1,$$

and in particular $\tilde{N}_i \cap \tilde{N}_j \cap \tilde{N}_k \neq \varnothing$, a contradiction to Claim 12.11. $\Diamond$

By Claims 12.13, 12.14 and 12.15, and the fact that $(I^{\pm}, I^{\leqslant 6}, I^{>6})$ is a partition of $[\Delta - 2]$, we have $\Delta \leqslant 2 + 70 + 441 + 42 = 555$, a contradiction. $\square$

6 A sufficient condition for $\Delta_{\min}$

Let k be an integer. A **k-obstruction** is a digraph H of order $\ell \in \{k, k+1\}$ admitting a partition (A, B) of its vertex-set, such that H contains every arc from A to B, and such that one of the following holds:

(i) $\ell = k$ and both A and B are bicliques of H,

(ii) $\ell = k+1$, A is a biclique of H, and $H[B]$ is isomorphic to $\vec{C}_3 \boxplus \overleftrightarrow{K}_{|B|-3}$, or

(iii) $\ell = k+1$, $H[A]$ is isomorphic to $\vec{C}_3 \boxplus \overleftrightarrow{K}_{|A|-3}$, and B is a biclique of H.

See Figure 4 for an illustration. Note that A or B might be empty. Observe also that if a symmetric digraph D contains a k-obstruction, then $\overleftrightarrow{\omega}(D) \geqslant k$. Picasarri-Arrieta [36] proved that every digraph D satisfying $\vec{\chi}(D) = \Delta_{\min}(D) + 1$ contains a $(\Delta_{\min}(D) + 1)$-obstruction satisfying *(i)*, thereby generalising Brooks' Theorem. Similarly, the following is a third independent generalisation of Theorem 4 to the directed setting.

Corollary 30. *The following holds for every integer $\Delta \geqslant \Delta_{12}$. Every digraph D with $\Delta_{\min}(D) \leqslant \Delta$ satisfying $\vec{\chi}(D) \geqslant \Delta$ contains a Δ-obstruction.*

Proof. Let D be any digraph with $\Delta_{\min}(D) \leqslant \Delta$ and $\vec{\chi}(D) \geqslant \Delta$. Let (A, B) be the bipartition of $V(D)$ defined as:

$$A = \{v \in V(D) : d^+(v) \leqslant \Delta_{\min}(D)\} \quad \text{and} \quad B = \{v \in V(D) : d^+(v) > \Delta_{\min}(D)\}.$$

Let $\tilde{D}$ be the digraph obtained from D by removing all arcs from B to A, and replacing all arcs from A to B by digons. Let then $\widehat{D}$ be the digraph obtained from $\tilde{D}$ by reversing every arc of $D[B]$. It was proved in [25, Proposition 7.3] that $\Delta^+(\widehat{D}) \leqslant \Delta_{\min}(D) \leqslant \Delta$ and $\vec{\chi}(\widehat{D}) \geqslant \vec{\chi}(D) \geqslant \Delta$, we omit the proof.

By Theorem 12, $\widehat{D}$ contains an induced copy K of $\overleftrightarrow{K}_\Delta$ or $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$. By construction, there is no simple arc between A and B in $\widehat{D}$. Hence, if K is isomorphic to $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\Delta-2}$, its vertices inducing $\vec{C}_3$ all belong to the same side of the bipartition (A, B).

Since, by construction, all arcs of $\widehat{D}$ going from A to B belong to D, we conclude that $D[V(K)]$ is a Δ-obstruction, as desired. □

Noticing that every k-obstruction contains a biclique of size $\lceil \frac{1}{2}(k-1) \rceil$, we obtain the following.

Corollary 13. *For every integer $\Delta \geqslant \Delta_{12}$ and digraph D with $\Delta_{\min}(D) \leqslant \Delta$ and $\overleftrightarrow{\omega}(D) < \frac{1}{2}(\Delta - 1)$, $\vec{\chi}(D) \leqslant \Delta - 1$.*

Similarly, noticing that the underlying graph of any k-obstruction contains a clique of size k, we derive the following, which itself generalises Theorem 4.

Corollary 14. *The following holds for every integer $\Delta \geqslant \Delta_{12}$. Let D be a digraph with underlying graph G. If $\Delta_{\min}(D) \leqslant \Delta$ and $\omega(G) \leqslant \Delta - 1$, then $\vec{\chi}(D) \leqslant \Delta - 1$.*

As mentioned in the introduction, the condition $\overleftrightarrow{\omega}(D) < \frac{1}{2}(\Delta - 1)$ above is best possible, as we have the following hardness result.

Proposition 15. *For every integer $k \geqslant 2$, deciding $\vec{\chi}(D) \leqslant k$ remains NP-complete when restricted to digraphs D with $\Delta_{\min}(D) = k + 1$ and $\overleftrightarrow{\omega}(D) = \lceil \frac{k}{2} \rceil$.*

Proof. The problem is clearly in NP, so let us show that it is NP-hard. The reduction is similar to that of [36, Theorem 9].

We reduce from the problem of deciding whether a given digraph D has dichromatic number at most k, which has been shown to be NP-hard for every fixed integer $k \geqslant 2$ in [15]. Let $D = (V, A)$ be an instance of this problem for a fixed $k \geqslant 2$.

We build $D' = (V', A')$ from D as follows. For each vertex $u \in V$, we associate a copy H_u^- of $\vec{C}_3 \boxplus \overleftrightarrow{K}_{\lceil (k-2)/2 \rceil}$, a copy H_u^+ of $\overleftrightarrow{K}_{\lfloor k/2 \rfloor}$, and we put all arcs from $V(H_u^-)$ to $V(H_u^+)$. For each arc $uv \in A$, we associate all possible arcs xy in A', such that $x \in V(H_u^+)$ and $y \in V(H_v^-)$.

For every $u \in V$, vertices in H_u^- have out-degree at most $k + 1$, and vertices in H_u^+ have in-degree at most $k + 1$. It is also straightforward that $\overleftrightarrow{\omega}(D') = \lceil k/2 \rceil$. Let us prove that $\vec{\chi}(D) \leqslant k$ if and only if $\vec{\chi}(D') \leqslant k$ to get the result.

Assume first that $\vec{\chi}(D) \leqslant k$, and let φ be a k-dicolouring of D. Let φ' be the k-dicolouring of D' defined as follows. For each vertex $u \in V$, choose arbitrarily $x_u^- \in V(H_u^-)$ and $x_u^+ \in V(H_u^+)$, and set $\varphi'(x_u^-) = \varphi'(x_u^+) = \varphi(u)$. Then choose arbitrarily two distinct vertices $y_u, z_u \in V(H_u^-) \setminus \{x_u^-\}$ that are not linked with a digon (which is possible as H_u^- contains an induced directed triangle), and colour them with a common colour distinct from $\varphi(u)$. Finally, choose a distinct colour for every other vertex of $V(H_u^-) \cup V(H_u^+)$.

Observe that φ' is be a k-dicolouring of D'. Indeed, by construction, if D', coloured with φ', contains a monochromatic directed cycle, then it is of the form $x^-_{u_1}, x^+_{u_1}, x^-_{u_2}, x^+_{u_2}, \ldots, x^-_{u_\ell}, x^+_{u_\ell}, x^-_{u_1}$. Hence, $u_1, u_2, \ldots, u_\ell, u_1$ is a monochromatic directed cycle of D coloured with φ, a contradiction.

Assume now that $\vec{\chi}(D') \leqslant k$. Let $\varphi' : V' \to \{1, \ldots, k\}$ be a k-dicolouring of D'. Let φ be the k-dicolouring of D defined as follows. For each vertex $u \in V$, observe that there exist $x^-_u \in V(H^-_u)$ and $x^+_u \in V(H^+_u)$ such that $\varphi'(x^-_u) = \varphi'(x^+_u)$. We set $\varphi(u) = \varphi'(x^-_u)$. Observe that φ is a k-dicolouring of D. Indeed, if D, coloured with φ, contains a monochromatic directed cycle $u_1, u_2, \ldots, u_\ell, u_1$, then $x^-_{u_1}, x^+_{u_1}, x^-_{u_2}, x^+_{u_2}, \ldots, x^-_{u_\ell}, x^+_{u_\ell}, x^-_{u_1}$ is a monochromatic directed cycle in D' coloured with φ', a contradiction. $\square$

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A Proof of Lemma 17

As explained in Section 2, we show in this appendix how Lemma 17 is derived from the original result of Talagrand. The proof is identical to that of [34], applied to the particular case of X being integer valued. Let us first formally state Talagrand's original result.

Let $T_1, \ldots, T_n$ be n independent trials and $\mathcal{A}$ be the set of all sequences of possible outcomes of these. For any $A \subseteq \mathcal{A}$ and any $\ell \geqslant 0$, we let A_ℓ be the set of elements $b \in \mathcal{A}$ which are at distance at most ℓ from A in the following sense. We have $b = (b_1, \ldots, b_n) \in A_\ell$ if and only if, for any sequence of real numbers $\lambda_1, \ldots, \lambda_n$, there exists $a = (a_1, \ldots, a_n) \in A$ such that

$$\sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i \leqslant \ell \cdot \Big(\sum_{i \in [n]} \lambda_i^2 \Big)^{1/2}.$$

Theorem 31 (Talagrand [40])**.** *For any n independent trials $T_1, \ldots, T_n$, any $A \subseteq \mathcal{A}$, and any real $\ell \geqslant 0$, we have*

$$\mathbb{P}(A) \cdot \mathbb{P}(\overline{A_\ell}) \leqslant e^{-\ell^2/4}.$$

We now prove Lemma 17, which we first recall here for convenience.

Lemma 17 (Talagrand's Inequality)**.** *Let X be a random variable valued in $\mathbb{N}$, determined by n independent trials and satisfying the following for some integers $c, r \geqslant 1$:*

1. *changing the outcome of any one trial can affect X by at most c, and*
2. *for every $s \in \mathbb{N}$, if $X \geqslant s$ then there is a set of at most rs trials whose outcomes certify that $X \geqslant s$.*

Then

$$\mathbb{P}\left(|X - \mathbb{E}(X)| > t\right) \leqslant 4 \exp\left(\frac{-t^2}{32c^2 r(\mathbb{E}(X) + t)} \right)$$

for any real number $t > 126c\sqrt{r\mathbb{E}(X)} + 344c^2 r$.

Proof. We proceed in two steps. We first show that X is concentrated around its median, and then show that its median is close to its expectation. Let μ be any integer such that $\mathbb{P}(X \leqslant \mu) \geqslant \frac{1}{2}$ and $\mathbb{P}(X \geqslant \mu) \geqslant \frac{1}{2}$.

Claim 17.1. *For every $t \geqslant 0$, $\mathbb{P}(X < \mu - t) \leqslant 2 \exp\left(\frac{-t^2}{4c^2 r\mu}\right)$.*

Proof of claim. Let $A = \{a \in \mathcal{A} : X(a) \geqslant \mu\}$, $B = \{b \in \mathcal{A} : X(b) < \mu - t\}$, and $\ell = \frac{t}{c\sqrt{r\mu}}$. We first justify that $A \subseteq \overline{B_\ell}$. To see this, let us fix any $a = (a_1, \ldots, a_n) \in A$. We will exhibit a specific sequence of real numbers $\lambda_1, \ldots, \lambda_n$ such that, for every $b = (b_1, \ldots, b_n) \in B$,

$$\sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i > \ell \cdot \Big(\sum_{i \in [n]} \lambda_i^2 \Big)^{1/2},$$

hence implying $A \subseteq \overline{B_\ell}$.

By definition of A, $X(a) \geqslant \mu$, and since μ is an integer, by assumption there exists a set I of at most $r\mu$ indices such that the values of $(a_i)_{i \in I}$ certify $X(a) \geqslant \mu$. For each $i \in [n]$, we set $\lambda_i = c$ if $i \in I$, and $\lambda_i = 0$ otherwise. Let us fix any $b \in B$, and let $b' \in \mathcal{A}$ be the sequence

equal to a on I and equal to b on $[n] \setminus I$. By construction and by the first assumption of the statement, we have

$$X(b) \geqslant X(b') - c \cdot |\{i \in I : a_i \neq b_i\}| \geqslant \mu - \sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i.$$

In the second inequality, we used that the outcomes of b' on I certify $X(b') \geqslant \mu$. By definition of B, we thus have

$$\sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i > t \geqslant \ell \cdot \Big(\sum_{i \in [n]} \lambda_i^2 \Big)^{1/2},$$

where the second inequality follows from

$$\sum_{i \in [n]} \lambda_i^2 = c^2 \cdot |I| \leqslant c^2 \cdot r \cdot \mu = \frac{t^2}{\ell^2}.$$

This shows $A \subseteq \overline{B_\ell}$. By definition of A and μ, we thus have $\mathbb{P}(\overline{B_\ell}) \geqslant \mathbb{P}(A) \geqslant \frac{1}{2}$. By Theorem 31, we thus have

$$\mathbb{P}(X < \mu - t) = \mathbb{P}(B) \leqslant 2e^{-\ell^2/4} = 2 \exp\left(\frac{-t^2}{4c^2 r \mu} \right),$$

as desired. ◊

Claim 17.2. *For every $t \geqslant 0$, $\mathbb{P}(X > \mu + t) \leqslant 2 \exp\left(\frac{-t^2}{4c^2 r(\mu+t+1)} \right)$.*

Proof of claim. The proof is analogous to that of Claim 17.1, with some additional rounding arguments. Therefore, let $A = \{a \in \mathcal{A} : X(a) \leqslant \mu\}$, $B = \{b \in \mathcal{A} : X(b) > \mu + t\}$, $t' = \lfloor t \rfloor + 1$, and $\ell = \frac{t}{c\sqrt{r(\mu+t+1)}}$. Since X is integer-valued, note that B is also $\{b \in \mathcal{A} : X(b) \geqslant \mu + t'\}$.

We now justify that $B \subseteq \overline{A_\ell}$. Let us fix $b = (b_1, \ldots, b_n) \in B$. Since $X(b) \geqslant \mu + t'$, and because $\mu + t'$ is an integer, by assumption there exists a set $I \subseteq [n]$ of at most $r(\mu + t')$ indices such that $(b_i)_{i \in I}$ guarantees $X(b) \geqslant \mu + t'$. For each $i \in [n]$, we define $\lambda_i = c$ if $i \in I$ and $\lambda_i = 0$ otherwise.

Let $a = (a_1, \ldots, a_n)$ be any element of A, and let $a' \in \mathcal{A}$ be the sequence equal to b on I and to a on $[n] \setminus I$. We have

$$X(a) \geqslant X(a') - c \cdot |\{i \in I : a_i \neq b_i\}| \geqslant \mu + t' - \sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i.$$

By definition of A, this implies

$$\sum_{\substack{i \in [n] \\ a_i \neq b_i}} \lambda_i \geqslant t' > \ell \cdot \Big(\sum_{i \in [n]} \lambda_i^2 \Big)^{1/2},$$

where the second inequality follows from

$$\ell^2 \cdot \sum_{i \in [n]} \lambda_i^2 = \ell^2 \cdot c^2 \cdot |I| \leqslant \ell^2 \cdot c^2 \cdot r \cdot (\mu + t') < t'^2.$$

This shows that $B \subseteq \overline{A_\ell}$. Recall that $\mathbb{P}(A) \geqslant \frac{1}{2}$ by definition of μ, so by Theorem 31 we obtain

$$\mathbb{P}(X > \mu + t) = \mathbb{P}(B) \leqslant \mathbb{P}(\overline{A_\ell}) \leqslant 2e^{-\ell^2/4} \leqslant 2 \exp\left(\frac{-t^2}{4c^2 r(\mu + t + 1)} \right),$$

as desired. $\Diamond$

By the union bound, it follows from Claims 17.1 and 17.2 that, for every $t \geqslant 0$,

$$\mathbb{P}(|X-\mu| > t) \leqslant 4\exp\left(\frac{-t^2}{4c^2 r(\mu+t+1)}\right). \tag{15}$$

It remains to justify that μ is close to $\mathbb{E}(X)$.

Claim 17.3. $|\mathbb{E}(X)-\mu| \leqslant 63c\sqrt{r\mathbb{E}(X)} + 172c^2 r$.

Proof of claim. We first assume that $\mu \neq 0$. For every $i \in \mathbb{N}$, we let $I_i \subseteq \mathbb{R}$ be the interval

$$I_i = \left(i \cdot c\sqrt{r\mu},\ (i+1)\cdot c\sqrt{r\mu}\right].$$

Observe that $|\mathbb{E}(X)-\mu| = |\mathbb{E}(X-\mu)|$, and that

$$\begin{aligned}
|\mathbb{E}(X-\mu)| &\leqslant \sum_{i\in\mathbb{N}} (i+1)\cdot c\sqrt{r\mu}\cdot \mathbb{P}(|X-\mu| \in I_i) \\
&= c\sqrt{r\mu}\sum_{i\in\mathbb{N}} \mathbb{P}\left(|X-\mu| \in \bigcup_{j\geqslant i} I_j\right) \\
&= c\sqrt{r\mu}\sum_{i\in\mathbb{N}} \mathbb{P}\left(|X-\mu| > i\cdot c\sqrt{r\mu}\right) \\
&\leqslant 4c\sqrt{r\mu}\sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2\mu}{4(\mu + ic\sqrt{r\mu}+1)}\right),
\end{aligned}$$

where the last inequality follows from (15). This implies

$$\begin{aligned}
\frac{1}{4c\sqrt{r\mu}}|\mathbb{E}(X-\mu)| &\leqslant \sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2\mu}{8\cdot\max(\mu, ic\sqrt{r\mu}+1)}\right) \\
&\leqslant \sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2}{8}\right) + \sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2\mu}{8(ic\sqrt{r\mu}+1)}\right) \\
&\leqslant \sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2}{8}\right) + \sum_{i\in\mathbb{N}} \exp\left(\frac{-i\sqrt{\mu}}{16c\sqrt{r}}\right).
\end{aligned}$$

Observe first that $\sum_{i\in\mathbb{N}} \exp\left(\frac{-i^2}{8}\right) \leqslant \sum_{i\in\mathbb{N}} \exp\left(\frac{-i}{8}\right) = \frac{1}{1-e^{-1/8}} \leqslant 9$.

It remains to bound the second summation. Let $y = \frac{\sqrt{\mu}}{16c\sqrt{r}}$, and observe that

$$S = \sum_{i\in\mathbb{N}} \exp\left(\frac{-i\sqrt{\mu}}{16c\sqrt{r}}\right) = \frac{1}{1-e^{-y}}.$$

Therefore, if $y \geqslant 1$ then $S \leqslant (1-e^{-1})^{-1} < 2$. Else if $y \leqslant 1$ then $e^{-y} \leqslant 1 - \frac{1}{2}y$, and $S \leqslant \frac{2}{y}$. We conclude that

$$|\mathbb{E}(X-\mu)| \leqslant 44c\sqrt{r\mu} + 128c^2 r. \tag{16}$$

Since X is non-negative, $\mu \neq 0$, and $\mathbb{P}(X \geqslant \mu) \geqslant \frac{1}{2}$, by Markov's Inequality we have $\mathbb{E}(X) \geqslant \frac{1}{2}\mu$, which implies

$$|\mathbb{E}(X)-\mu| = |\mathbb{E}(X-\mu)| \leqslant 63c\sqrt{r\mathbb{E}(X)} + 128c^2 r.$$

This proves the claim when $\mu \neq 0$. Henceforth, we assume that $\mu = 0$, that is $\mathbb{P}(X = 0) \geqslant \frac{1}{2}$. Consider the random variable $Y = X + 1$. Observe that all assumptions on X hold for Y, and that $\mathbb{P}(Y \leqslant 1) \geqslant \frac{1}{2}$, $\mathbb{P}(Y \geqslant 1) = 1$. Hence inequality (16) holds for Y and gives

$$|\mathbb{E}(Y-1)| \leqslant 44c\sqrt{r} + 128c^2 r \leqslant 172c^2 r,$$

and the result follows as $\mathbb{E}(X) = \mathbb{E}(Y-1)$. $\Diamond$

We are now ready to prove the statement. Let us fix any real number $t > 126c\sqrt{r\mathbb{E}(X)} + 344c^2 r$. We have

$$\begin{aligned}
\mathbb{P}(|X - \mathbb{E}(X)| > t) &\leqslant \mathbb{P}\left(|X - \mathbb{E}(X)| > \frac{1}{2}t + 63c\sqrt{r\mathbb{E}(X)} + 172c^2 r\right) && \text{by choice of } t \\
&\leqslant \mathbb{P}(|X - \mu| > \frac{1}{2}t) && \text{by Claim 17.3} \\
&\leqslant 4\exp\left(\frac{-t^2}{16c^2 r(\mu + 2t)}\right) && \text{by (15) and } t \geqslant 1 \\
&\leqslant 4\exp\left(\frac{-t^2}{32c^2 r(\mathbb{E}(X) + t)}\right) && \text{by Markov's Inequality.}
\end{aligned}$$

The result follows. $\square$